Exploring Cell Types with the Allen Institute

MIT, Boston

August 17th, 2026



about us/



Jeremy Miller

Senior Scientist

Allen Institute



Elysha Fiabane

Product Manager

Allen Institute



Satrajit Ghosh

Director of the Open Data in Neuroscience Initiative, McGovern Institute

MIT



Dorota Jarecka

Research Scientist

MIT

agenda/morning

- **9:05-9:30 |** Overview of the Allen Institute – Jeremy
- **9:30-10:00 |** Cell typing at the Allen Institute – Jeremy
- **10:00-10:10 |** Breakout session preview
- **10:10-10:25 |** Break
- **10:25-10:50 |** Overview of the Brain Knowledge Platform – Elysha
- **10:50-11:15 |** Allen Brain Cell Atlas Demo – Elysha
- **11:15-12:00 |** User exploration of Allen Institute tools
- **12:00-1:00 | Lunch**

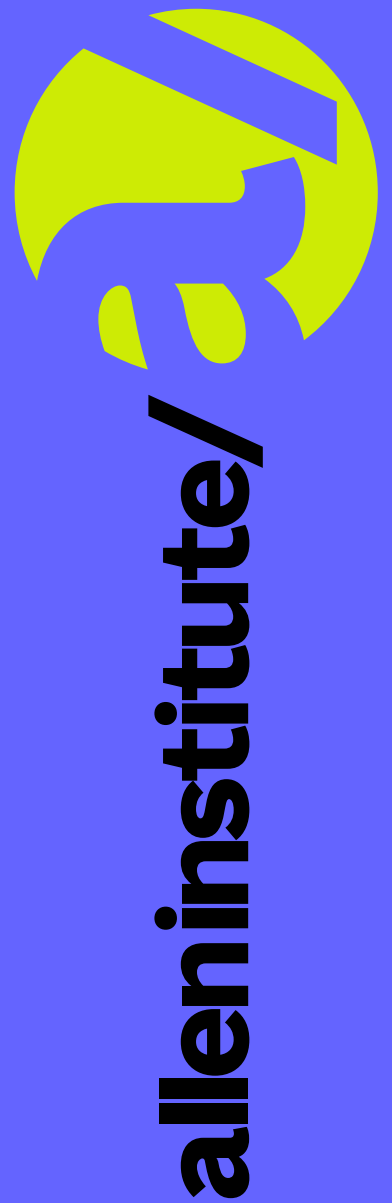
agenda/afternoon

- **12:00-1:00 | Lunch**
- **1:00-3:00 |** Breakout sessions
  - Knowledgebases and data repositories across the BRAIN initiative – Satra & Dorota
  - Computational access to Allen Institute data and knowledge – Elysha & Jeremy
- **3:00-3:15 |** Break
- **3:15-4:00 |** Coming soon / next steps / feedback

  - - - - -
- **4:00-5:00 |** User exploration of Allen Institute tools
- **5:00-6:00 |** Networking reception

about the allen institute/

Jeremy Miller



allen Institute/

- **Founding date**: September 16, 2003
- **Founders**: Paul G. Allen; Jody Allen
- **Employees**: ~900
- **Non-Profit**
  - No university affiliation
  - No degrees, Grad Students, Post-Docs
  - No traditional labs with single PI
- **Foundational Science**

understanding life
advancing health



allen Institute funding & structure/

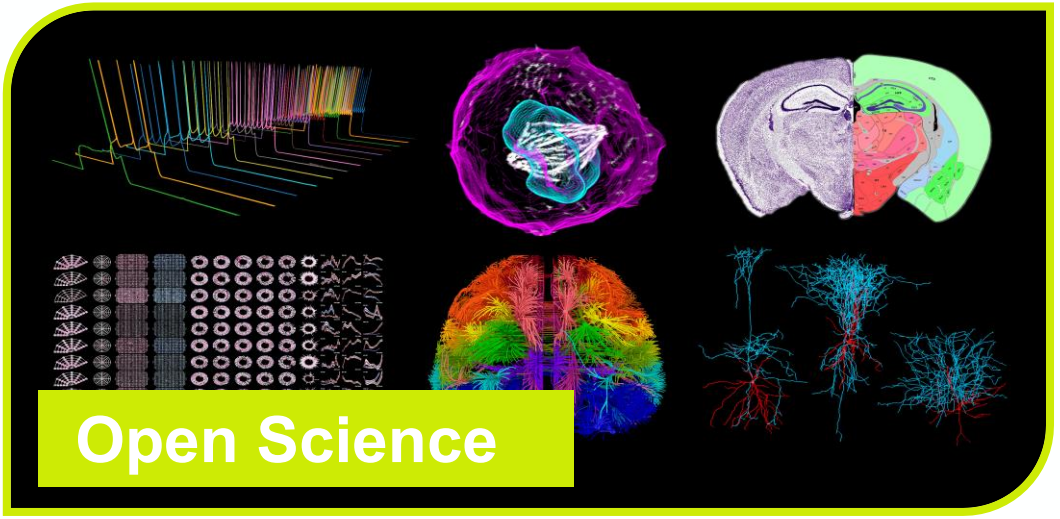
- Majority funded by private funding, particularly ”Fund for Science & Technology” – part of Paul G. Allen’s estates
- We also have public funding, which is primarily NIH Grants.
  - Note: no K series (training). Mainly U & R series
- We are privileged in our funding support so we want to ensure all our resources can be used by the community

- Typical Academic structure: 1 PI with lab of 3-10, all focus on project related to PI’s interest or field
- Allen Institute structure: multiple teams who all work on same big project. There is no single PI.
  - Projects are determined by multiple scientific leaders, along with input from our scientific advisory council

our core principles/









big science/

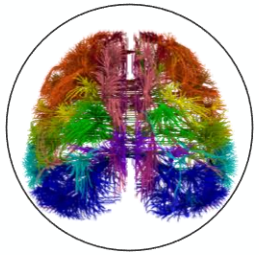
Brain science is a major focus

Allen Institute for Brain Science

Human Brain Atlas



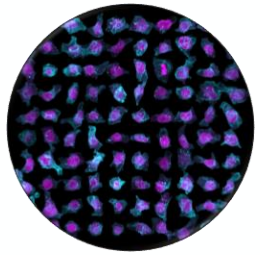
Connectivity Atlas



The Paul G. Allen Frontiers Group



Allen Cell Explorer



Alzheimer's Brain Cell Atlas



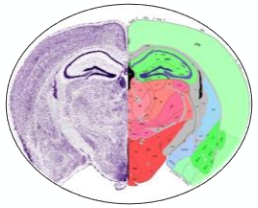
Human Immune System Explorer



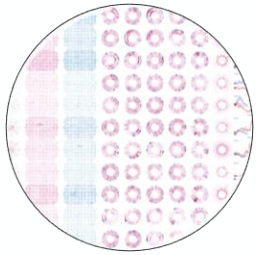
Seattle Hub for Synthetic Biology



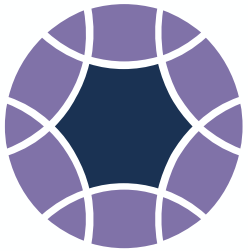
2003 2026



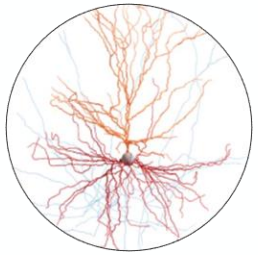
Mouse Brain Atlas



MindScope



Allen Institute for Cell Science



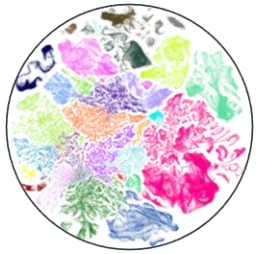
Cell Types Database & Brain Observatory



Allen Institute for Immunology



Allen Institute for Neural Dynamics



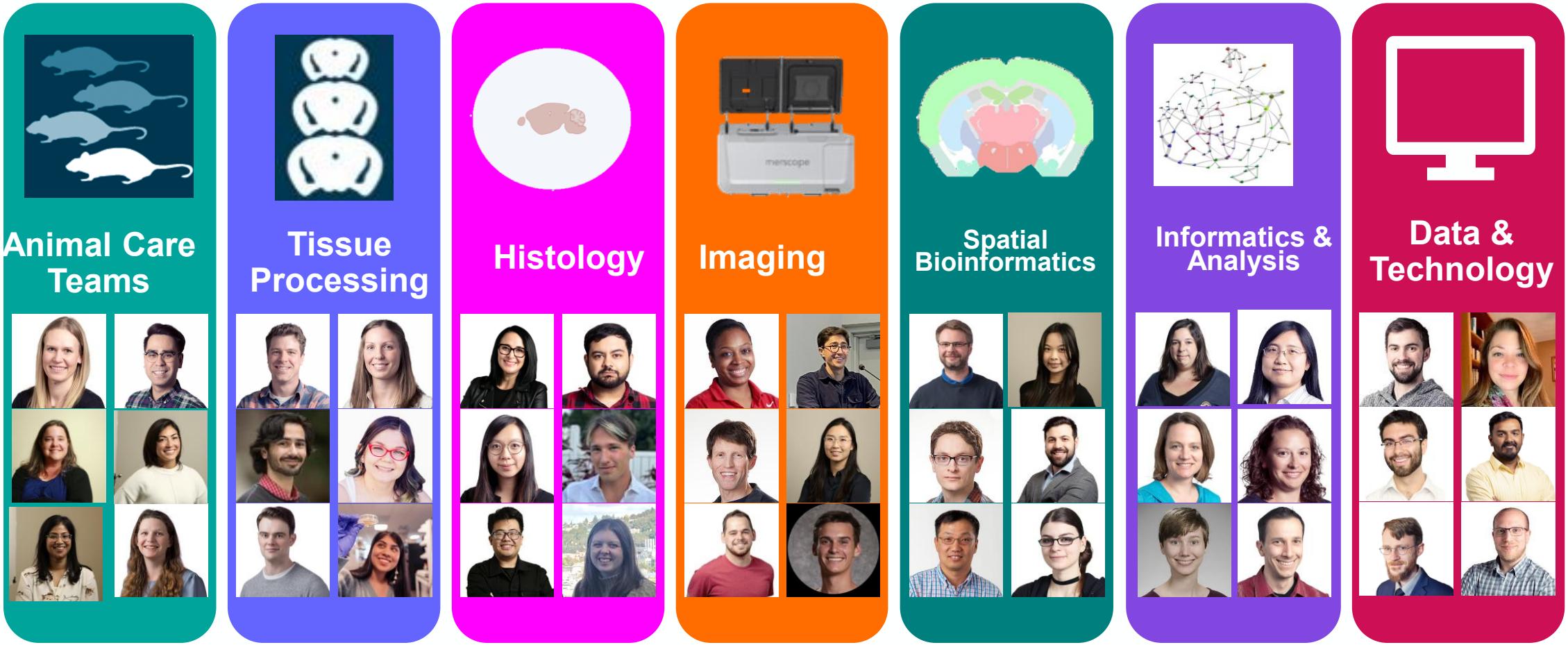
Allen Brain Cell Atlas



Brain Health Accelerator

team science/ example pipeline: spatial transcriptomics/

MANY more folks and teams not pictured!



Additional teams:

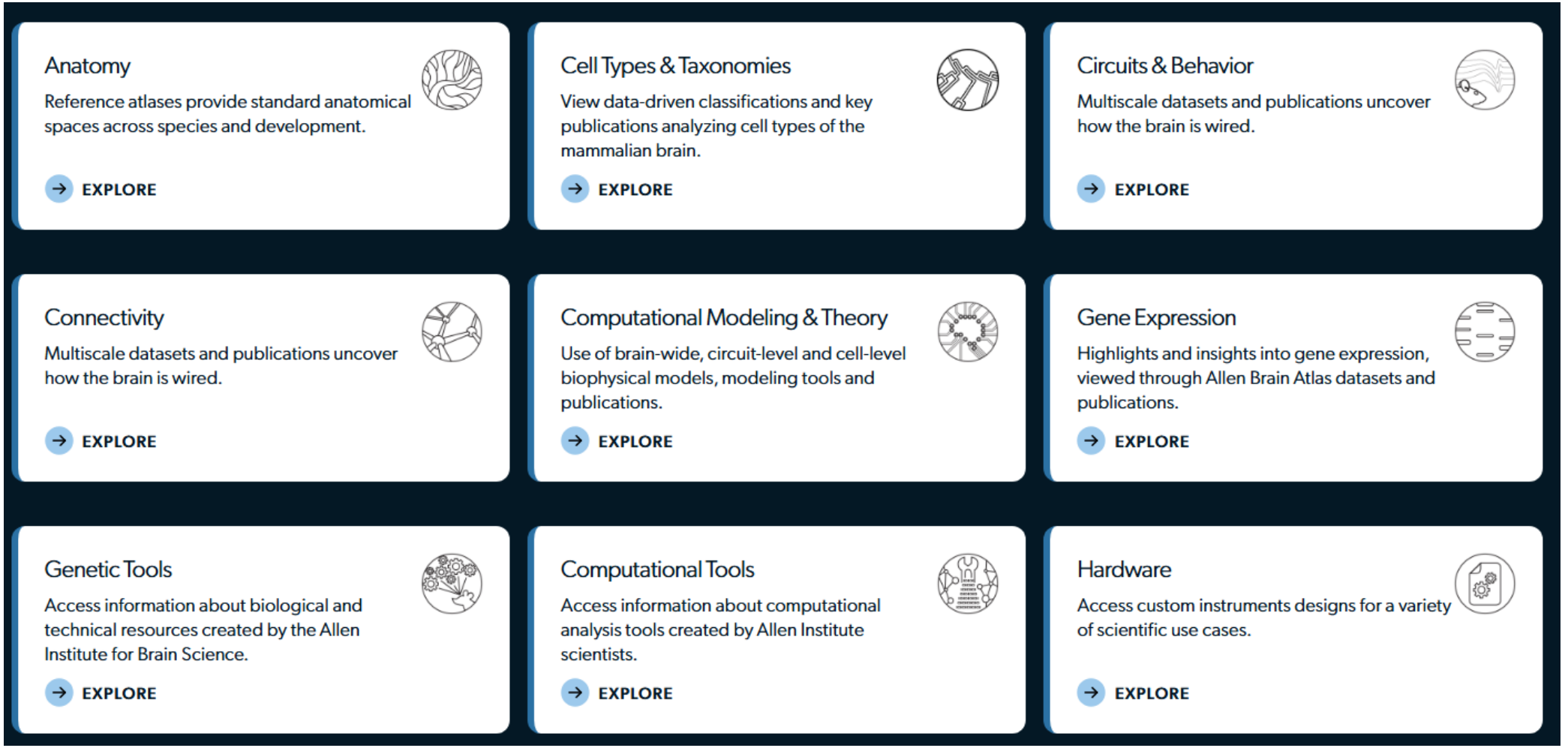
- Project Management
- Scientific Support
- Research Compliance
- Finance
- Shipping & Receiving

open science/

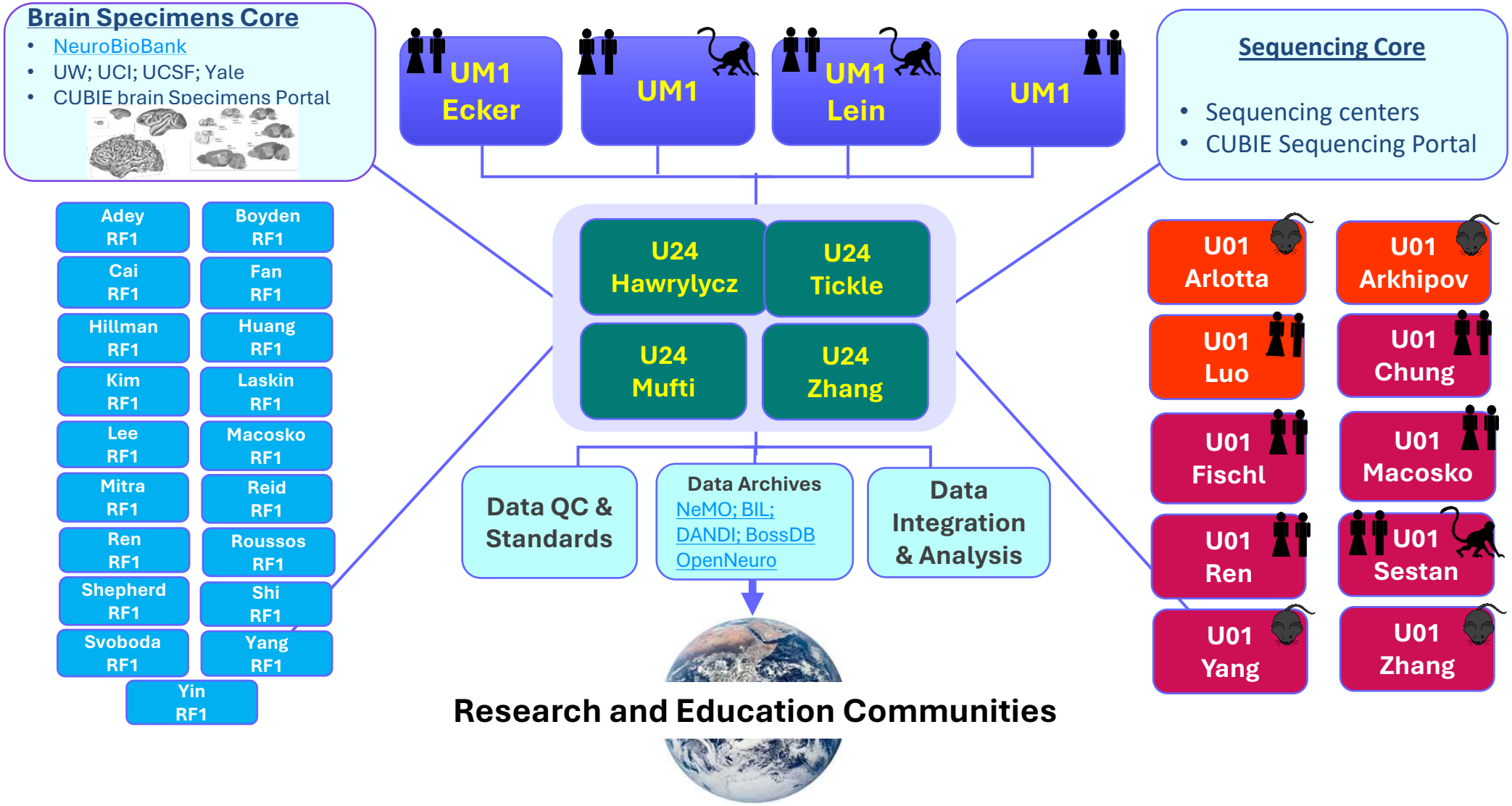
- The goal of open science is to make scientific research data, tools, and knowledge available and accessible to the public

- This is more than just publication!
  - ✓ Having data available and fully public
  - ✓ Publishing protocols (find us on protocols.io)
  - ✓ Creating open-access tools for data use and access
  - ✓ Providing resources & trainings to help use the data & tools
  - ✓ Following FAIR (Findable, Accessible, Interoperable, Reproducible) Principles

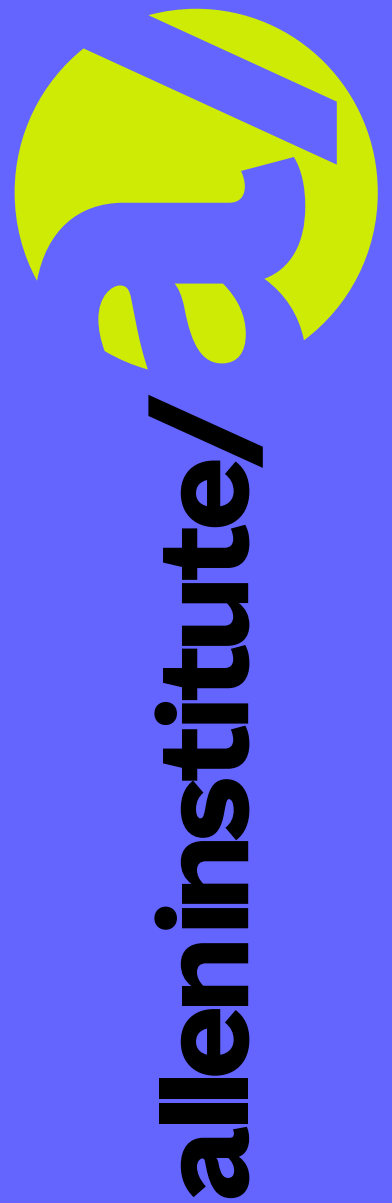
all FREE on Allen Brain Map (brain-map.org)/



many NIH-funded brain atlasing projects focused on different aspects of brain cell typing/

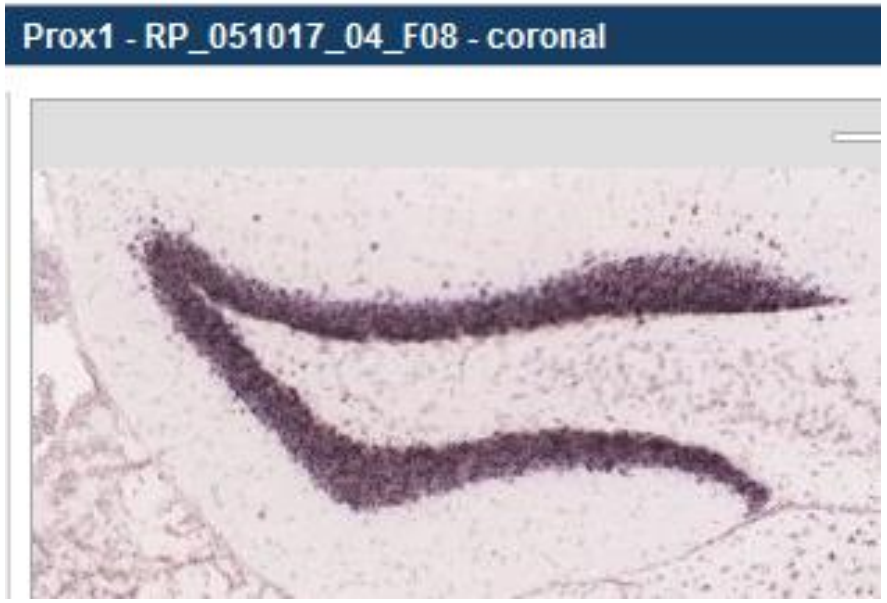


laying the groundwork/ gene expression atlases and reference spaces



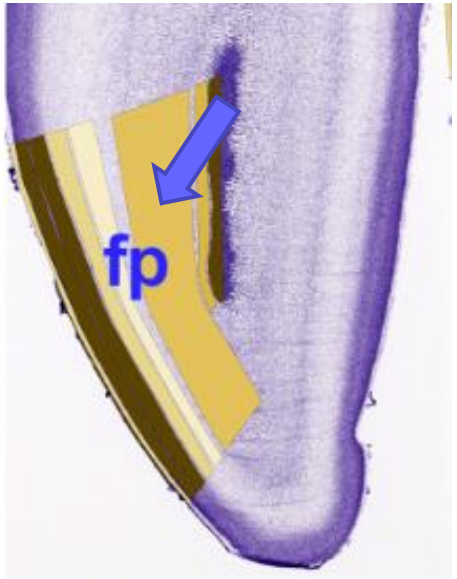
key methods for gene expression atlases/

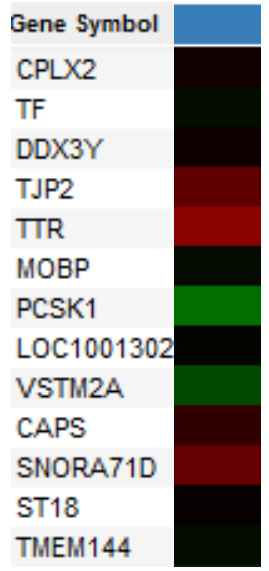
In situ hybridization



- 1 gene per experiment
- Subcellular resolution
- ~40-all genes per study

Microarray & Bulk RNA-seq





- All genes per experiment
- One value per dissection
- 100-3000+ samples per study

key methods for gene expression atlases/

In situ hybridization



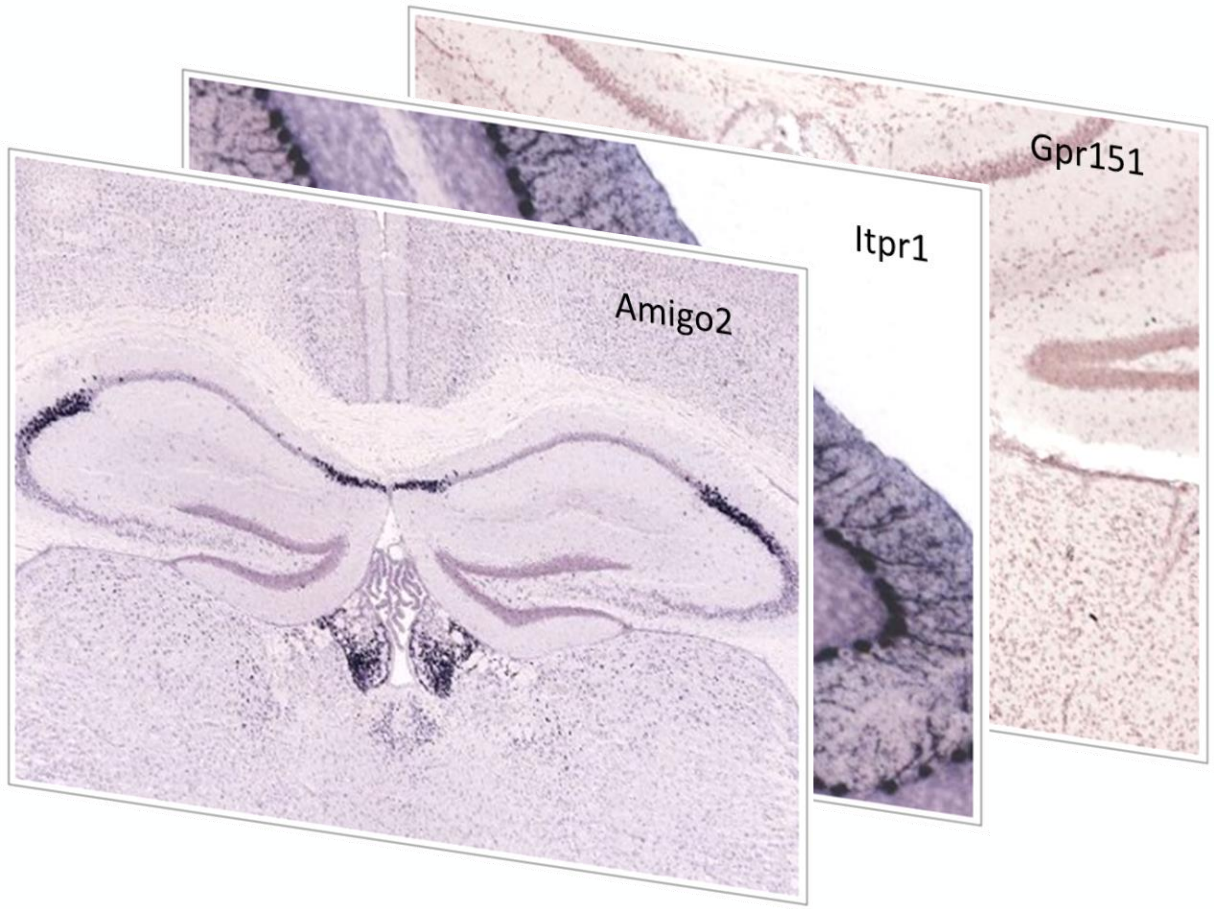
Microarray & Bulk RNA-seq



Adapted from Shalek and Regev (2016)

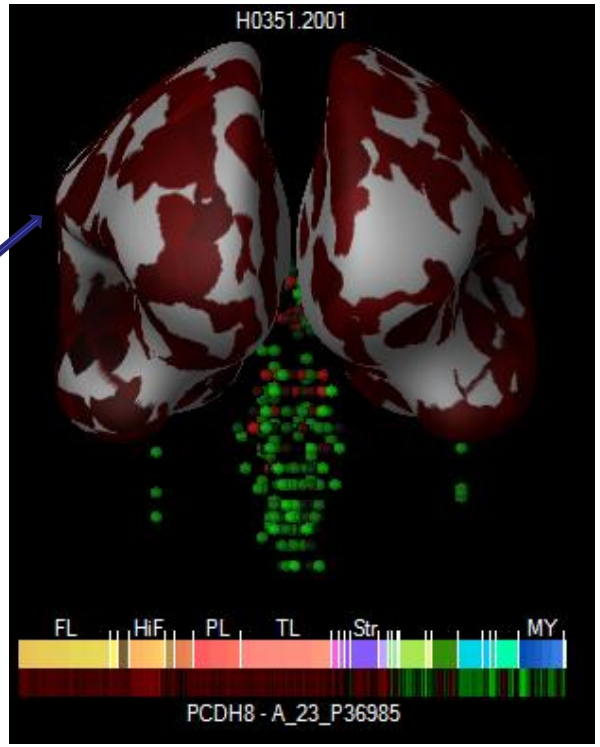
genome-wide, brain-wide atlases in adult mouse and human/

Mouse



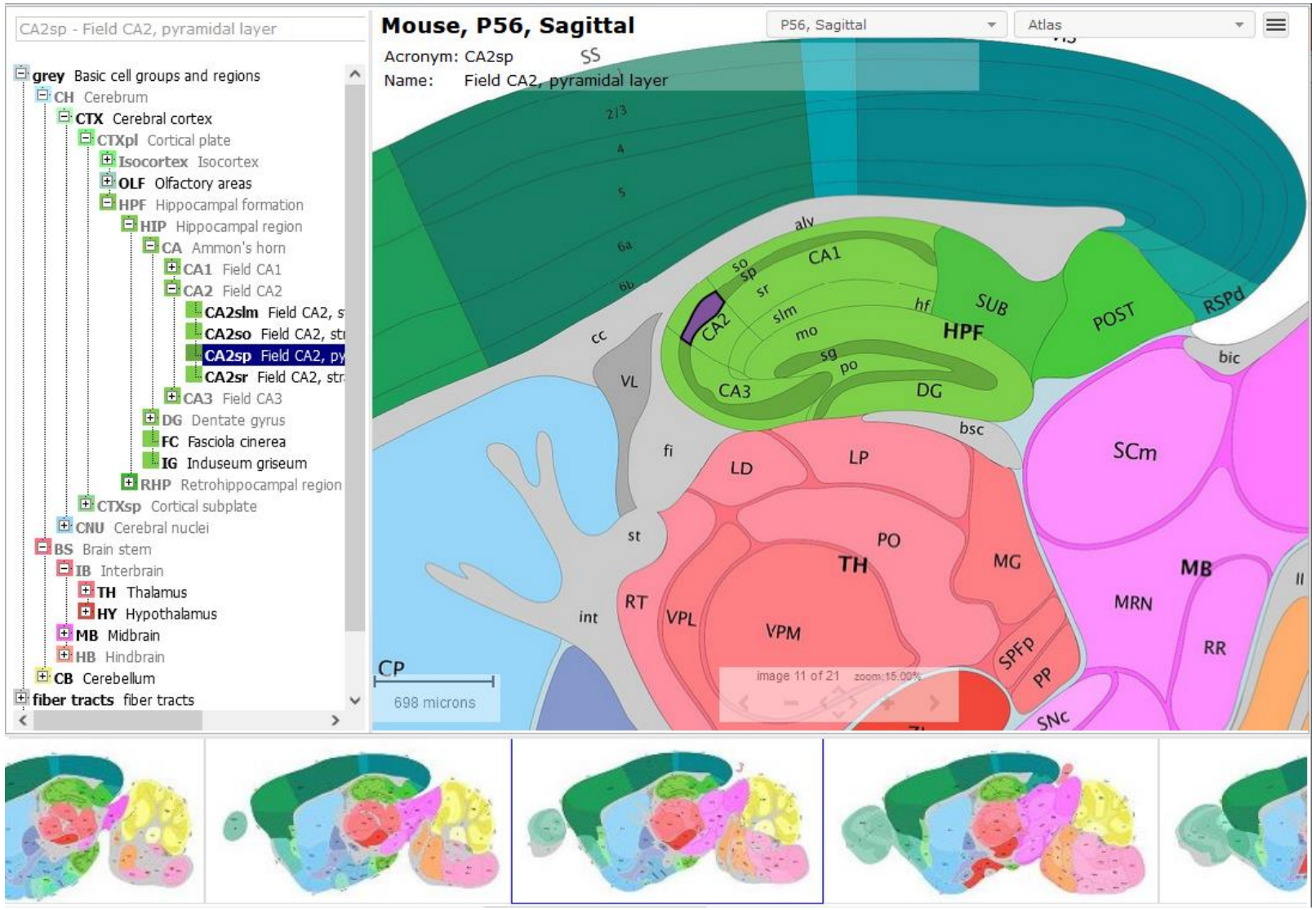
Human

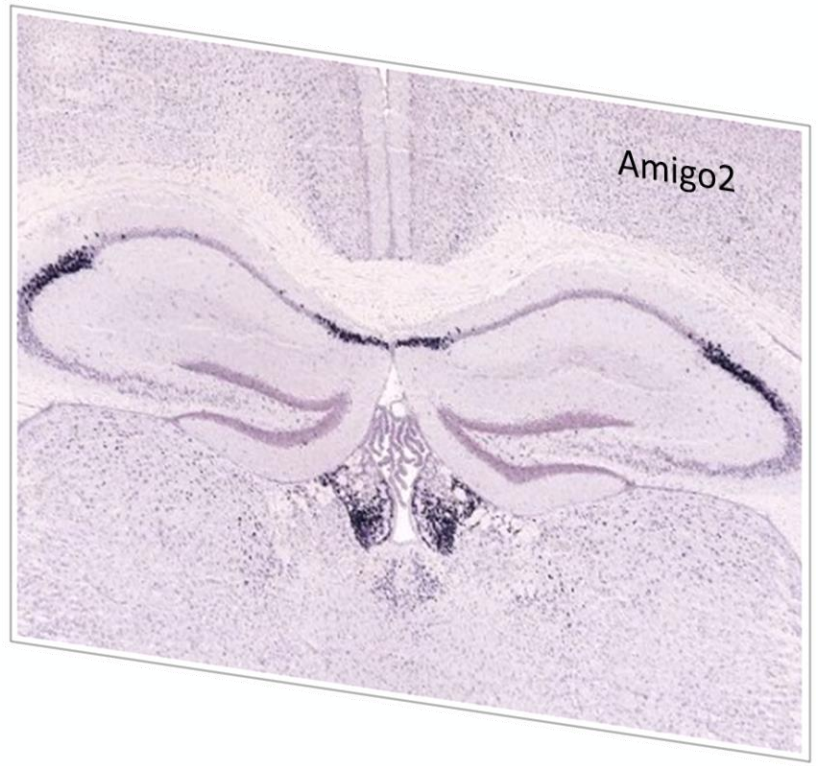
Gene Symbol	Probe Name
ACTA1	A_23_P1102
ACTA1	CUST_15163_PI416261804
ACTA2	A_23_P150053
ACTA2	CUST_15154_PI416261804
ACTB	A_23_P135769
ACTB	A_23_P31323
ACTB	A_24_P226554
ACTB	A_32_P118686
ACTB	A_32_P137939
ACTBL2	A_24_P6903
ACTBL2	CUST_4789_PI416261804
ACTC1	A_23_P205894
ACTC1	A_24_P121535
ACTG1	A_23_P44037



(Also atlases of development and non-human primate, not shown)

Allen Brain Reference Atlases/ maps of brain region locations and their names/

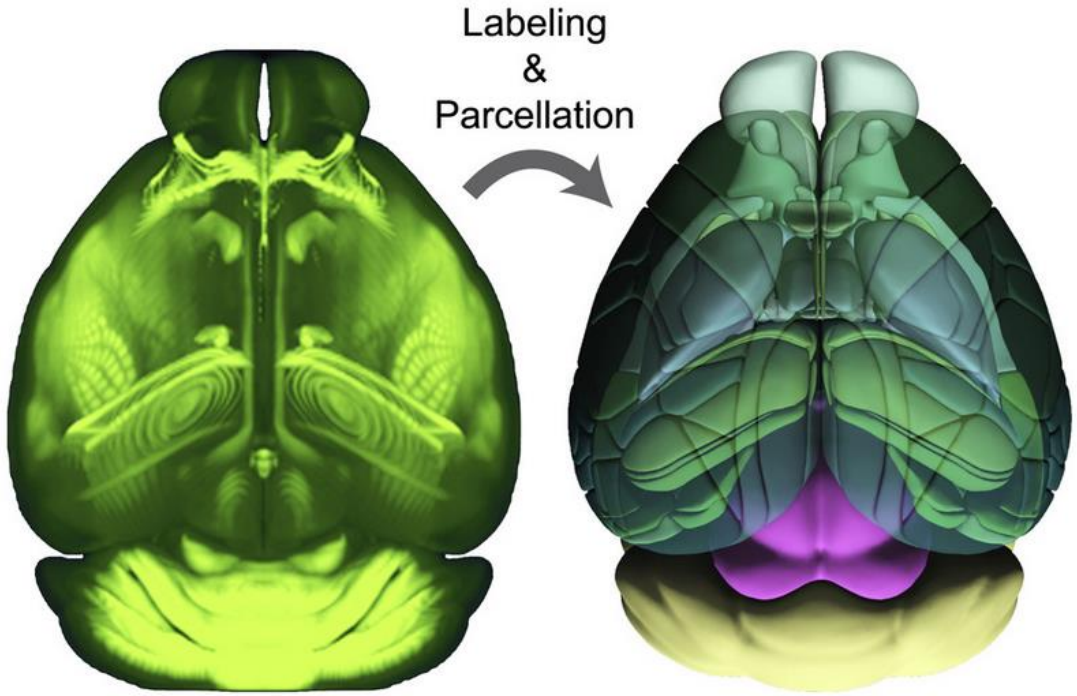


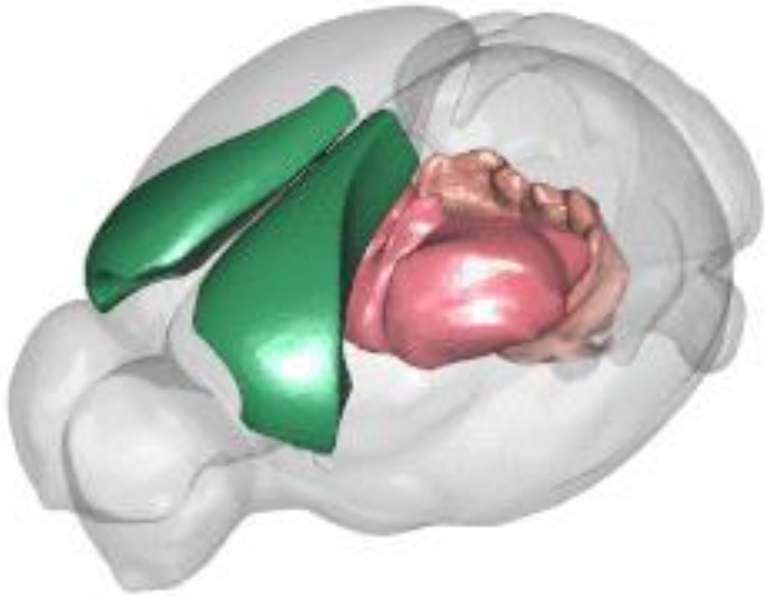


- Hierarchy of brain regions
- Mouse and human
- Adult and development
- 2D and 3D versions

Ng et al 2009, Nature Neuroscience

common coordinate framework ('CCF')/ 3D space for aligning <u>anything</u> in the brain (imagine the coordinates of a globe)/





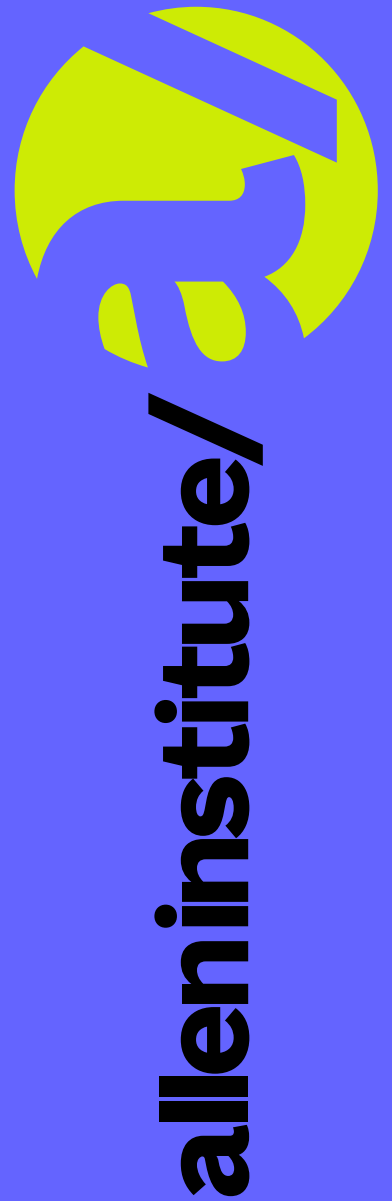
- **Can put gene expression, cell type, and anatomical info into this 3D space for comparison across studies**

(For human/NHP, CCF is in MRI space.)

Wang, Ding, Li, et al., 2020, Cell

cell typing at the allen institute/

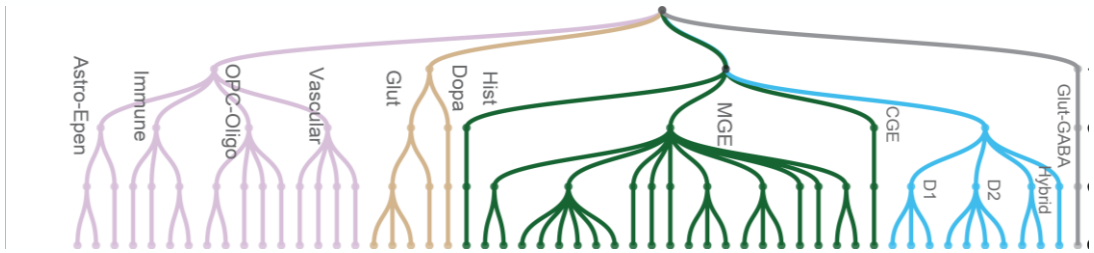
Jeremy Miller



key methods for cell type atlasing/

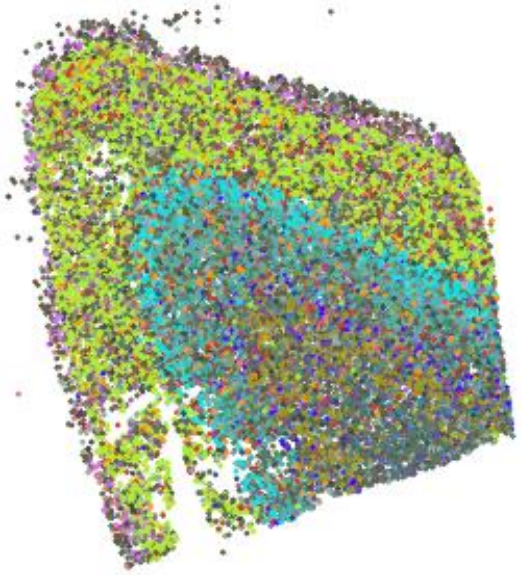
Single cell -omics

- All genes per experiment
- Up to millions of cells per study
- Dissection-level resolution
- Used for defining cell types



Spatial transcriptomics

- <1000 genes per data set
- Cellular resolution
- A few regions (human) or brain-wide (mouse)
- Can assign types to all cells



Cell type atlases are typically **defined** using a combination of both methods and **refined** using other types of data.

key methods for cell type atlasing/

Single cell -omics

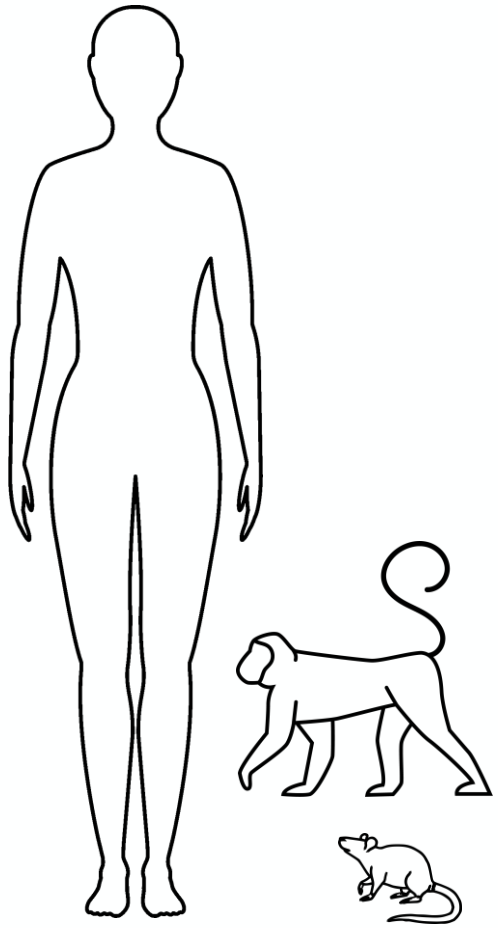


Spatial transcriptomics



Adapted from Shalek and Regev (2016)

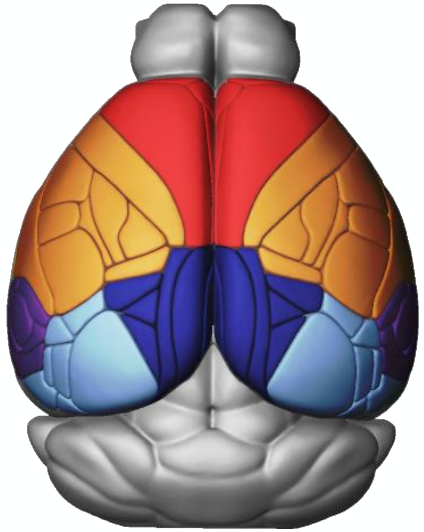
goal/integrated “uber atlas” for cells, spanning:



SPECIES

- Transcriptomics
- Epigenomics
- Spatial distribution
- Targeting tools
- Proteomics
- Morphology
- Connectivity
- Physiology
- Function
- Disease

MODALITIES

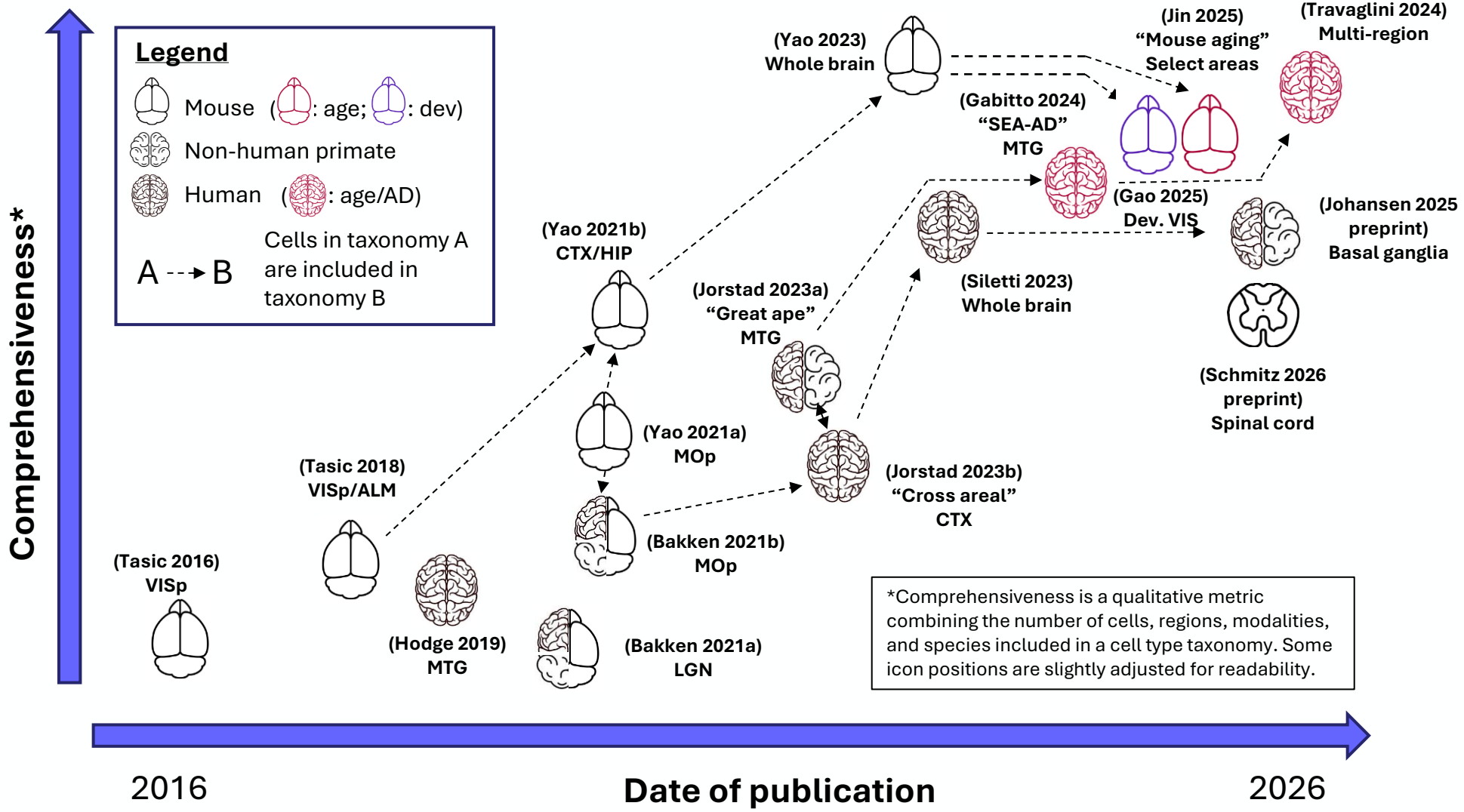


SPACE



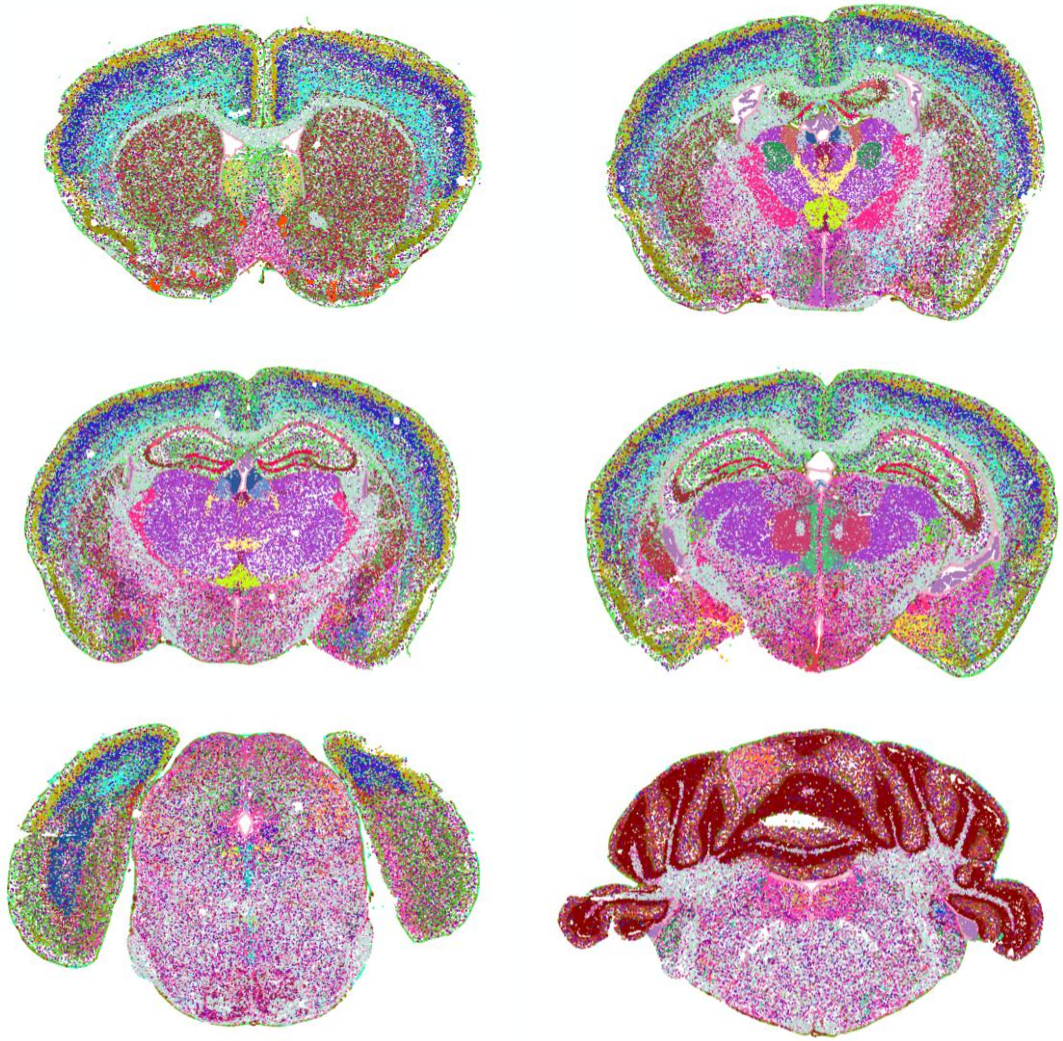
TIME

more than a decade of cell typing efforts at the Allen institute span brain regions and species/

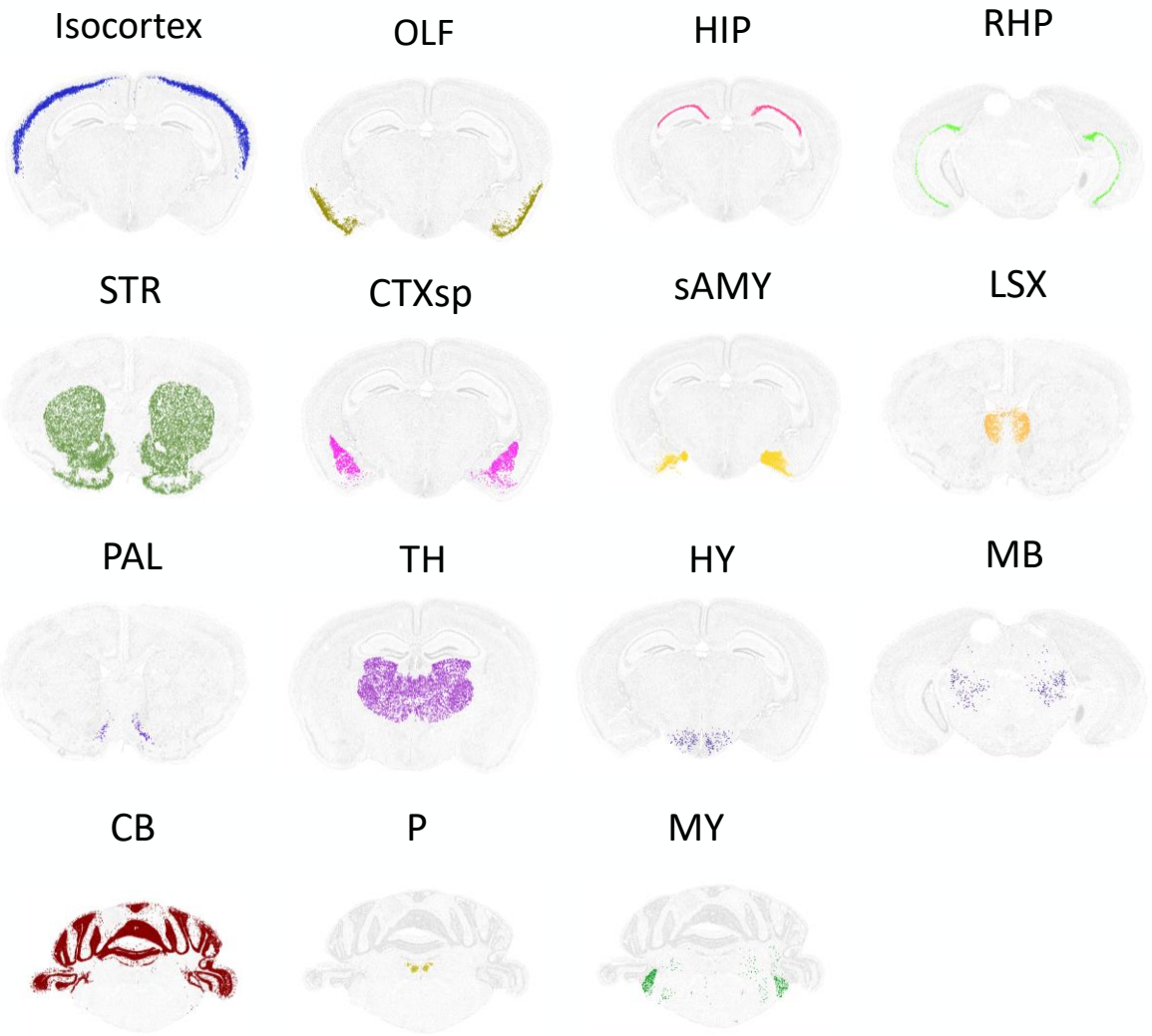


whole mouse brain (cell type) atlas/ created by combining single cell and spatial transcriptomics brain-wide/

(Cells color-coded by cell type)

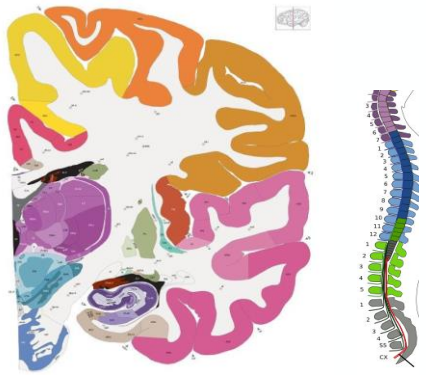


(Cells color-coded by brain region)



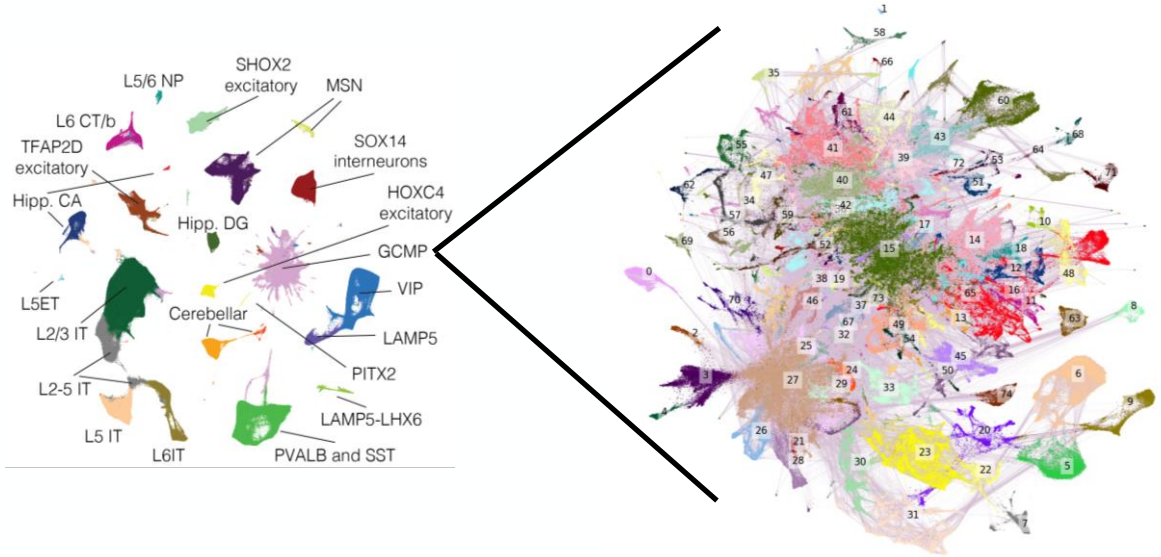
Yao et al., 2023, Nature

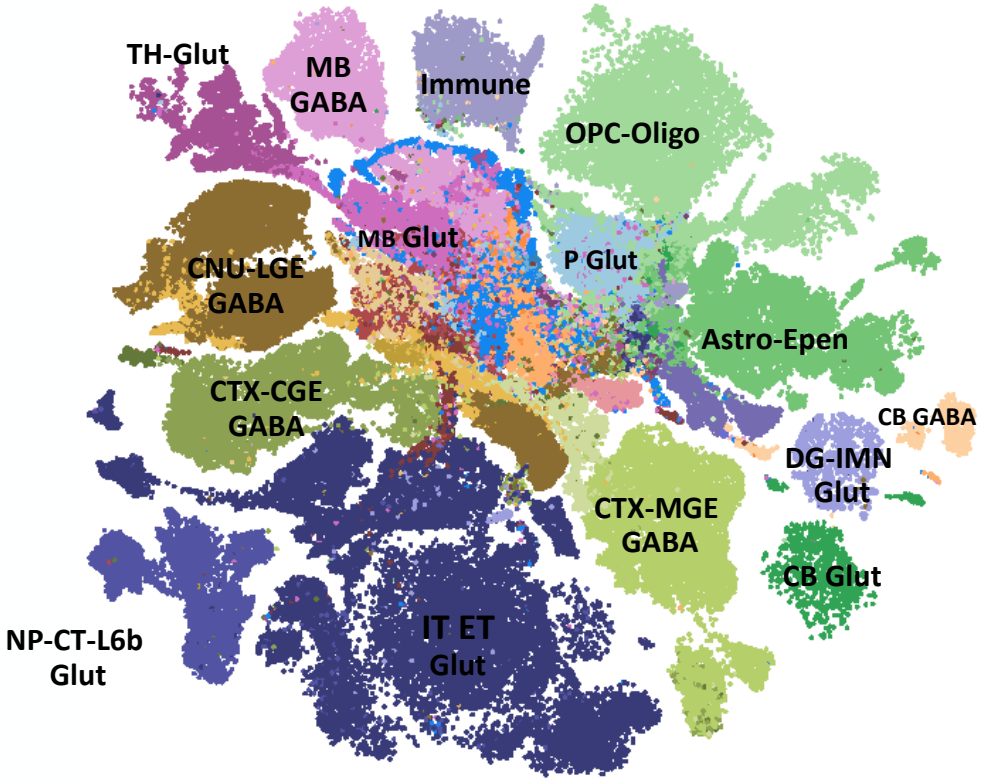
efforts to define cell types across the whole human brain are ongoing/



Published:

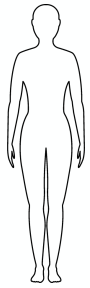
- ~3,000 cell types in the whole brain based on ~3,000,000 cells
- Minimal spatial transcriptomics data





In process:

- Add more scRNA-seq data (now >5,000 types)
- Spatial transcriptomics



Siletti et al., 2023, Science

(expected late 2027)

studies of healthy adult brain cells act as anchor for defining cell types in development, aging, and disease/

Create cell type atlases in **healthy** developing and adult mouse, human, and non-human primate brain

BICCN

BICAN

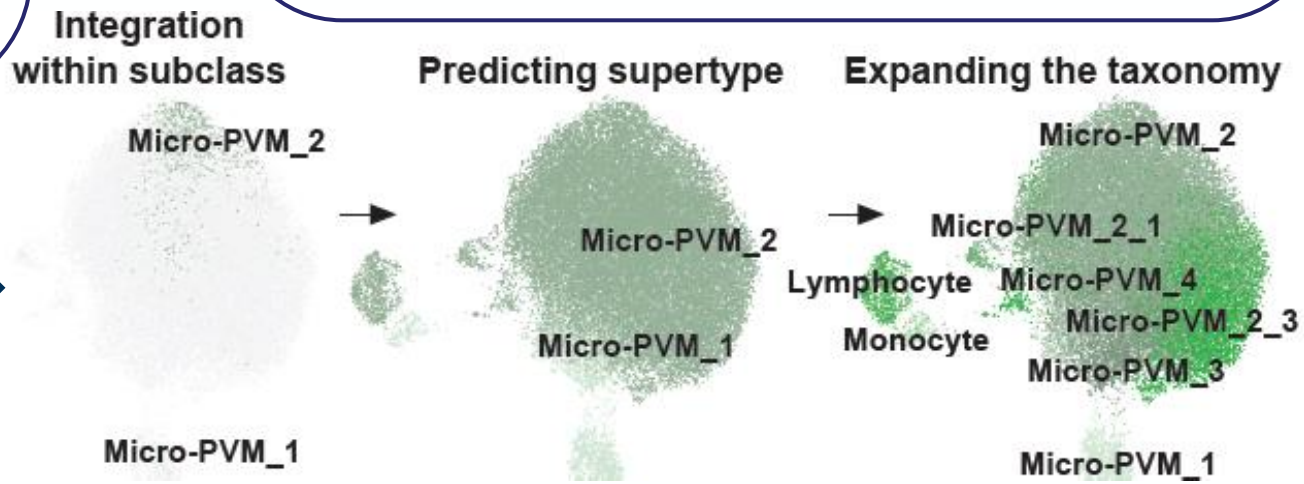
The BRAIN Initiative®

HMBA

Extend such atlases to the aged human brain to understand early molecular and cellular changes in Alzheimer's disease.

SEA-AD

Example human → microglia



Disease

studies of healthy adult brain cells act as anchor for defining cell types in development, aging, and disease/

Create cell type atlases in **healthy** developing and adult mouse, human, and non-human primate brain

BICCN

BICAN

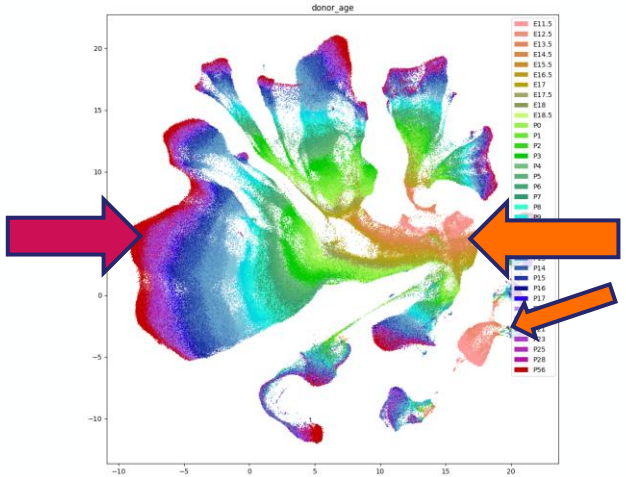
The BRAIN Initiative®

HMBA

Similar strategies used for cell type in development and aging across species.



TIME



Cells from adult mouse brains

Cells from developing mice include some "progenitor" types not found in adult.

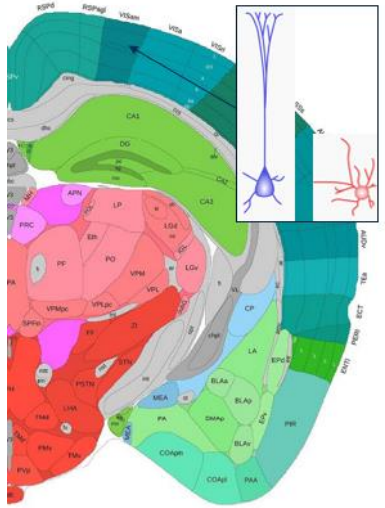
TIME

data from other modalities can be anchored to gene expression and cell location/

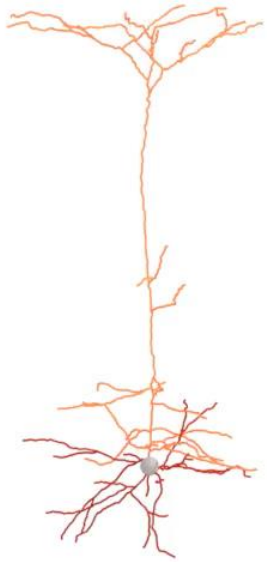
Gene expression
(transcriptomics)



Cell location
(anatomy)



Cell shape
(morphology)



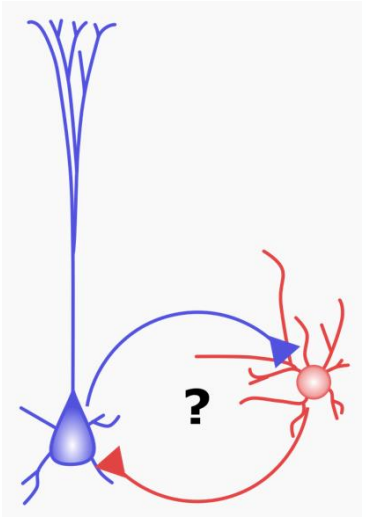
Firing properties
(electrophysiology)



Chromatic accessibility
(epigenetics / targeting)



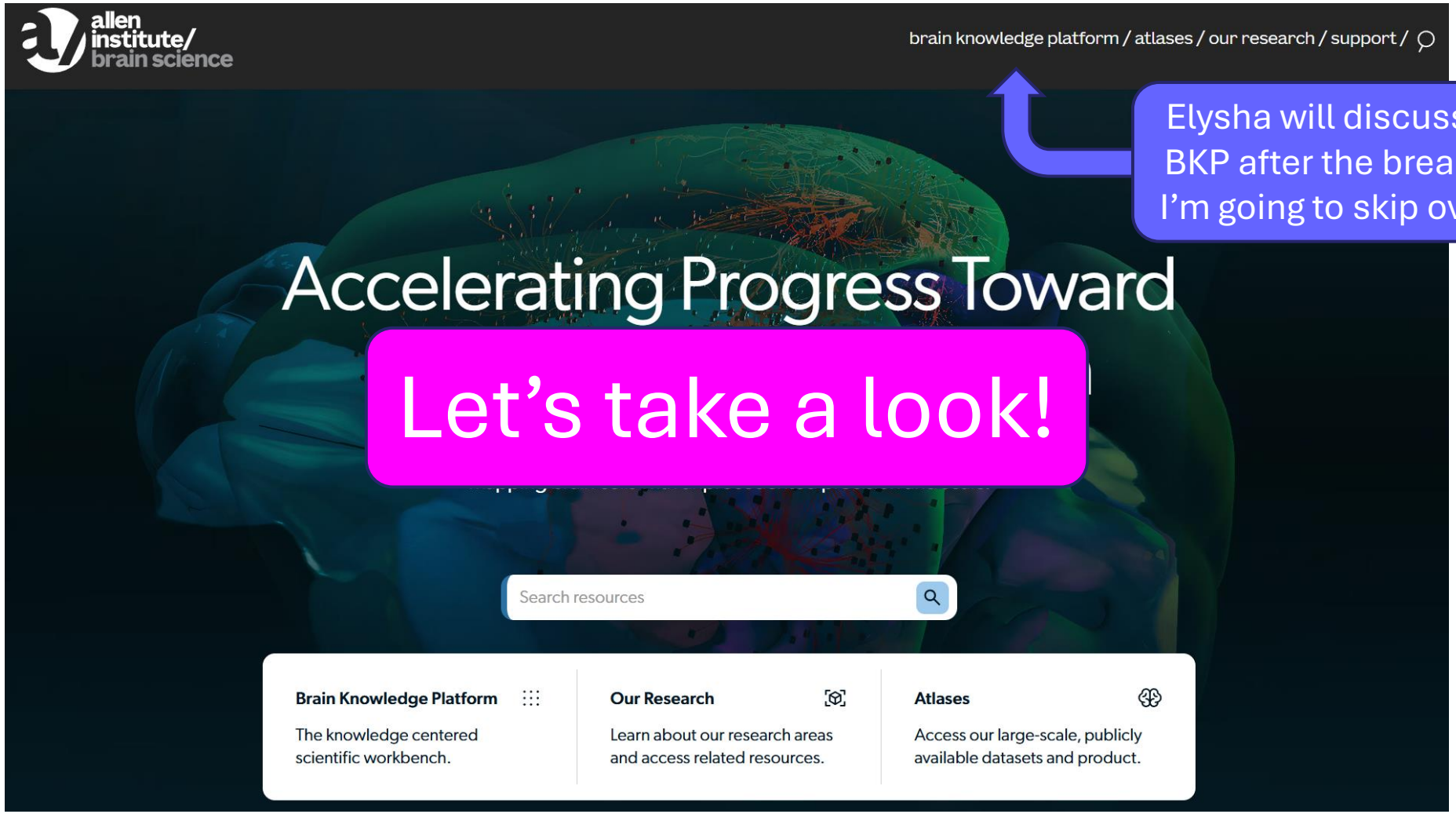
Cell communication
(connectivity)



Cell function
(in response to behavior)

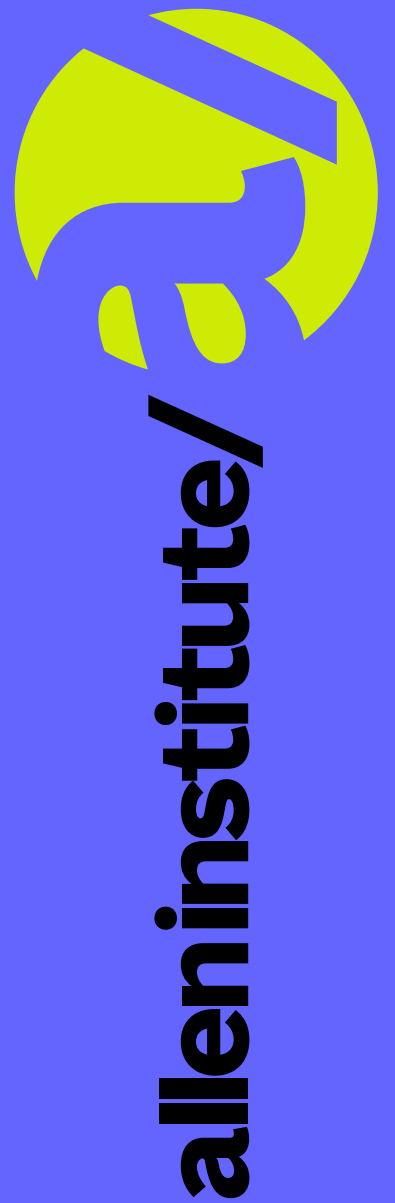
MODALITIES

all of this can be explored for free online/



brain-map.org

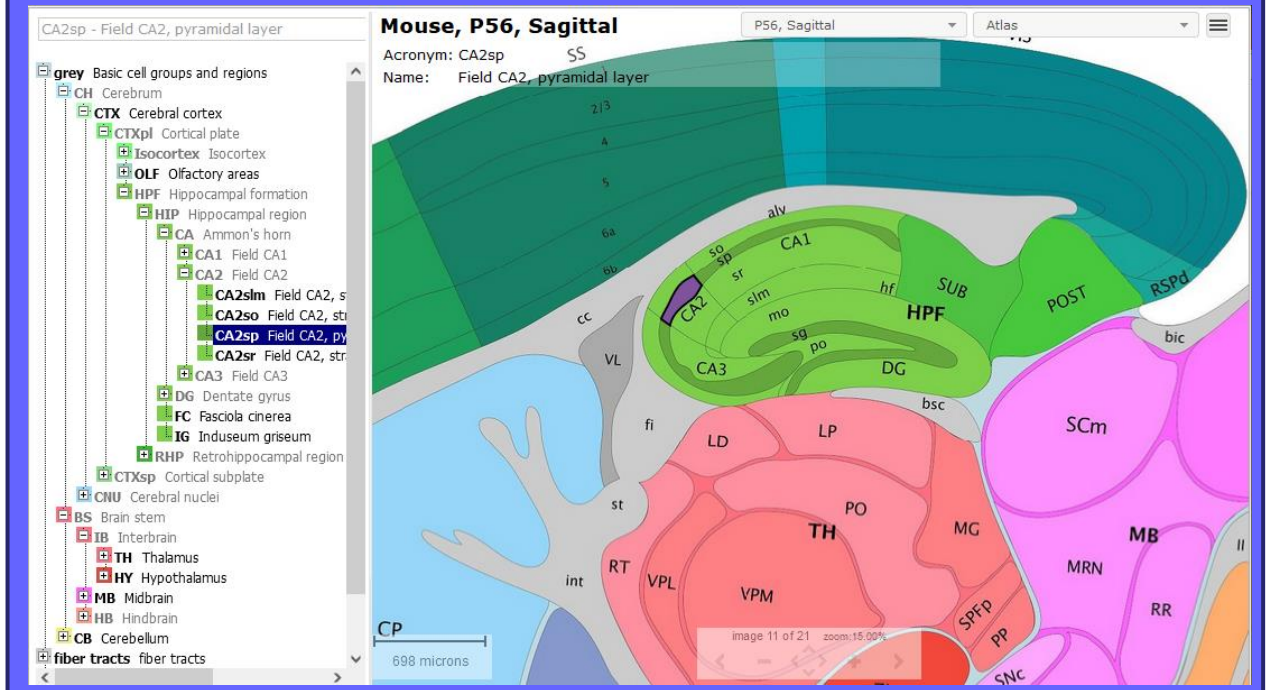
cell type use cases/



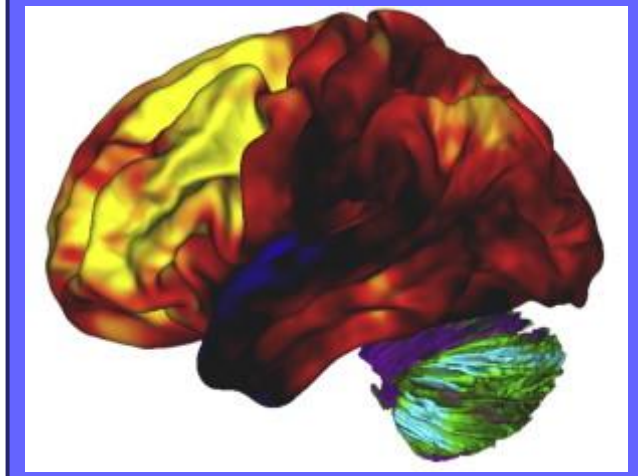
use cases/localization

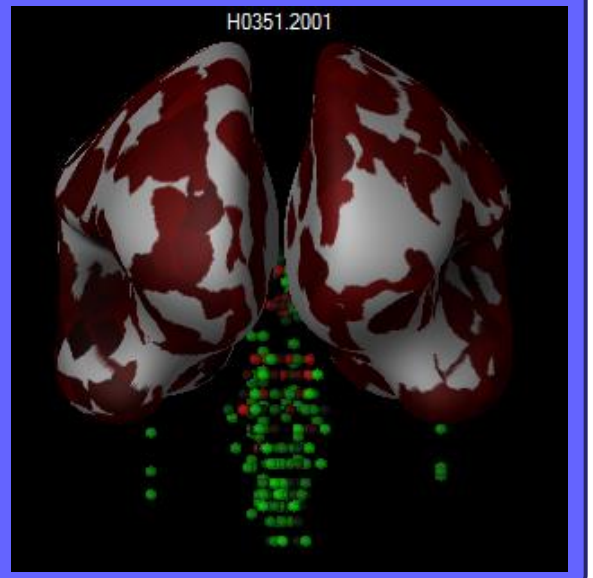
- What is this brain region called and where in the brain is it located?
- How does it connect to other brain regions in mouse? In human?

Allen Brain Reference Atlas (mouse/human/NHP)

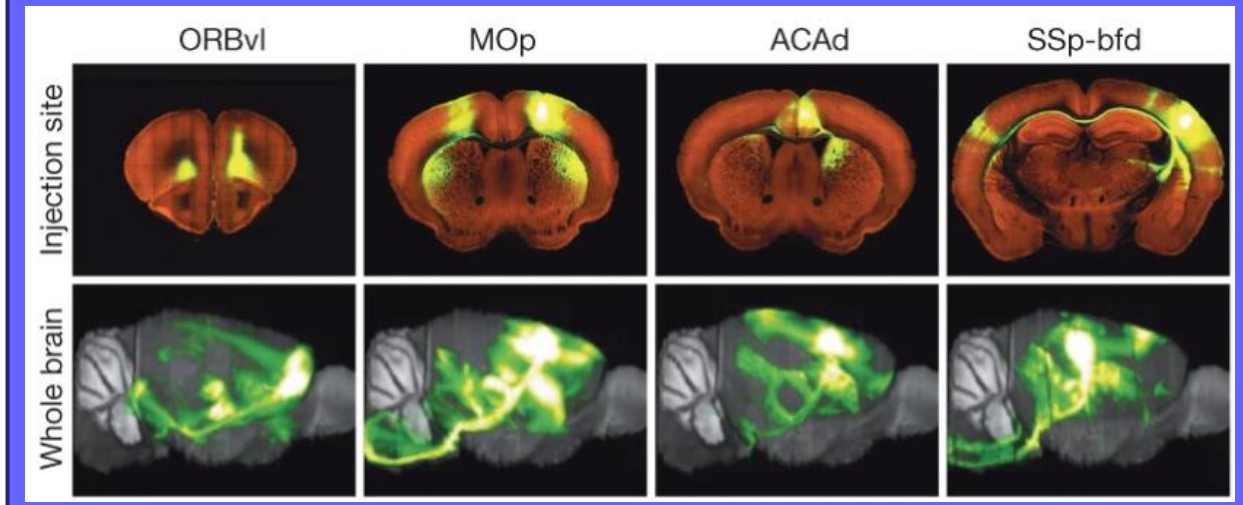


(Indirectly by linking structural MRI and gene expression from Allen Human Brain Atlas with external fMRI)





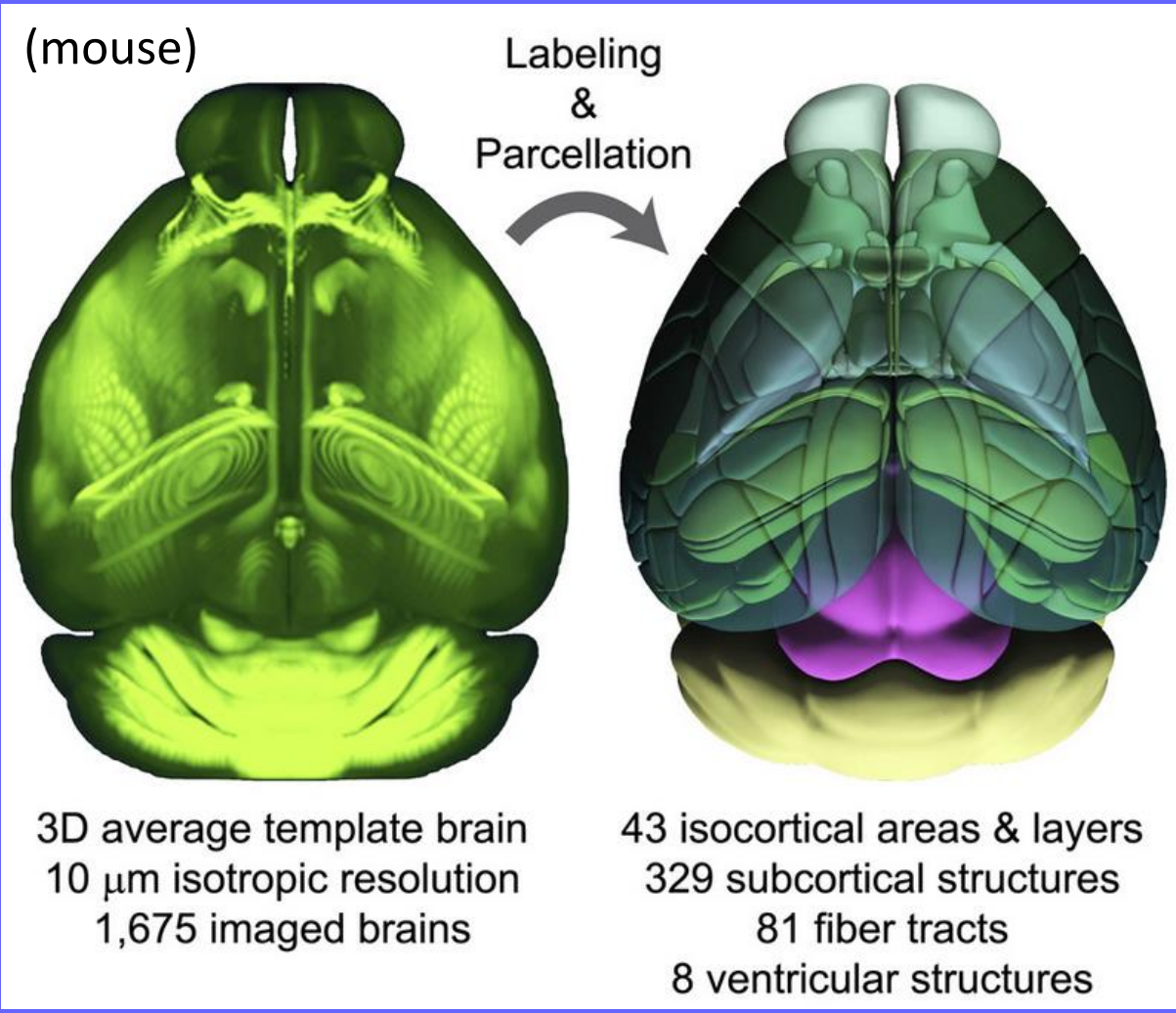
Allen Connectivity Atlas (mouse)



use cases/localization

- What is this brain region called and where in the brain is it located?
- How does it connect to other brain regions in mouse? In human?
- How can we align brain regions (and any other data modalities) to the same set of coordinates?

Allen Common Coordinate Frameworks (CCF)

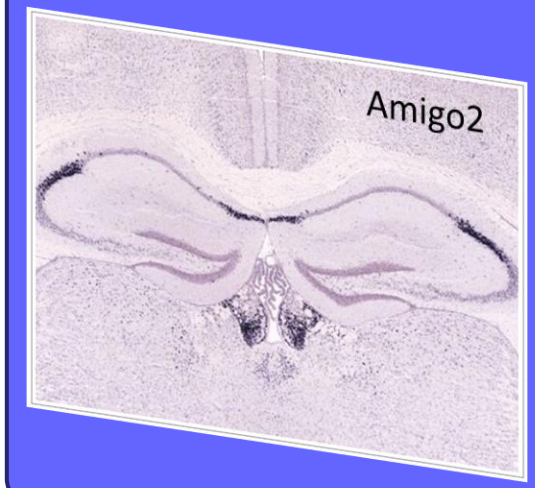


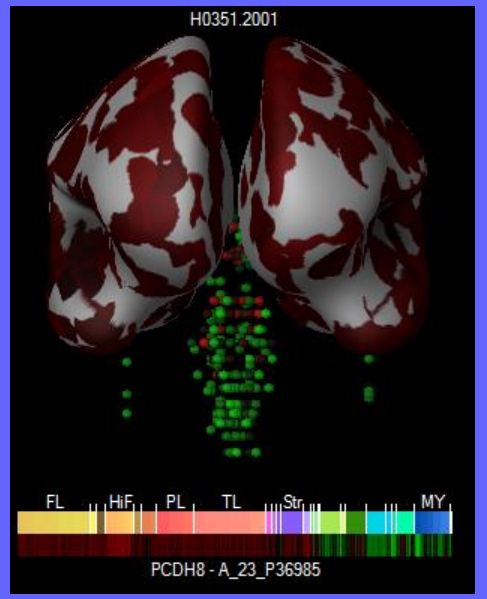
(CCFs in human and NHP based on MRI data exist or are in progress via HMBA.)

use cases/gene expression

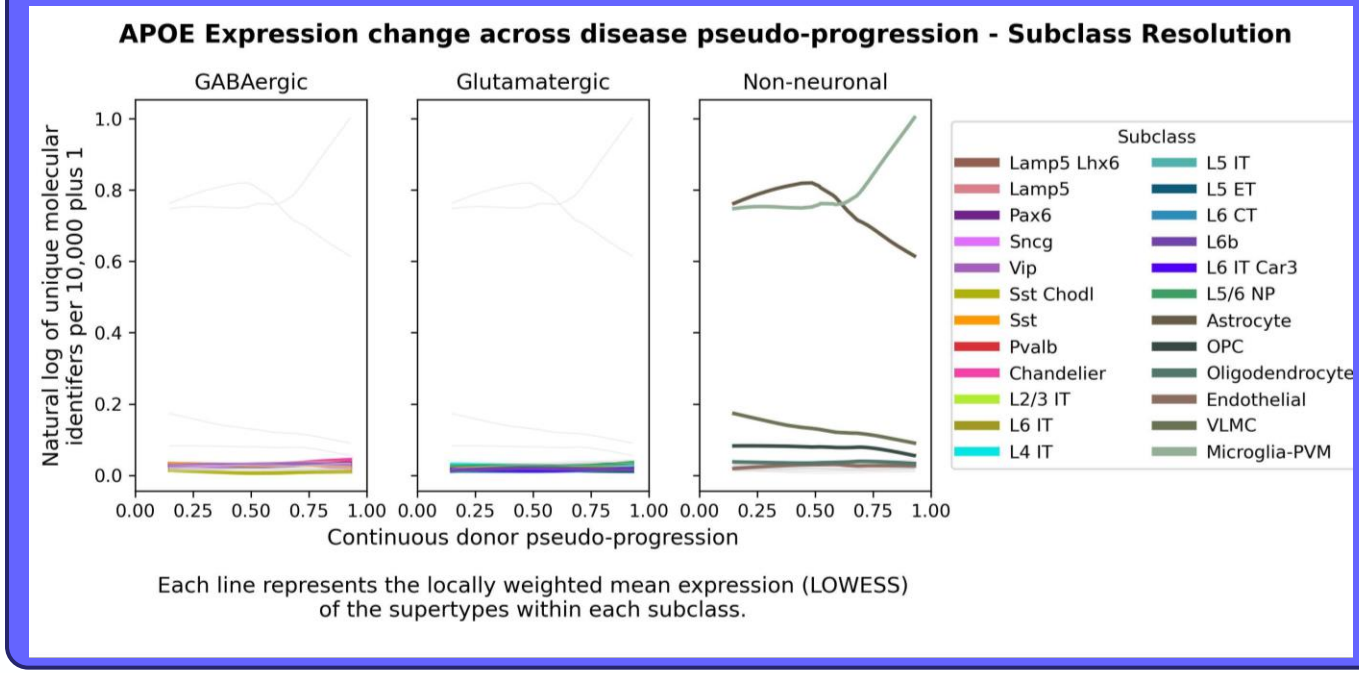
- Where in the brain is my gene expressed?
- And in which cell types?
- How does this change in AD?

Historic atlases

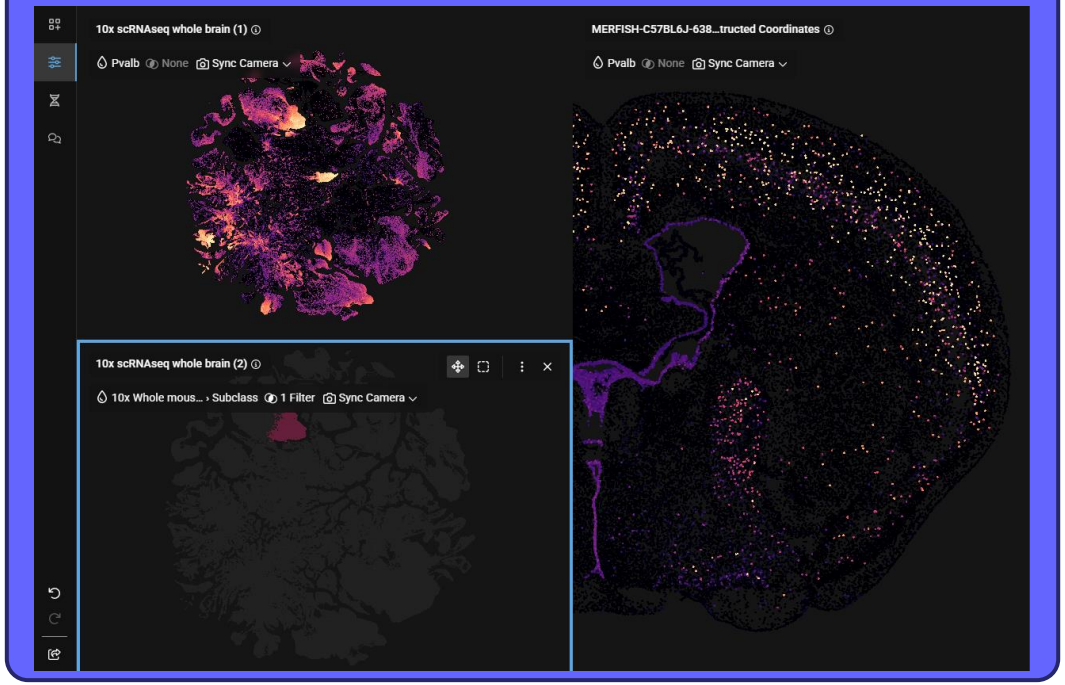




SEA-AD gene trajectory viewer



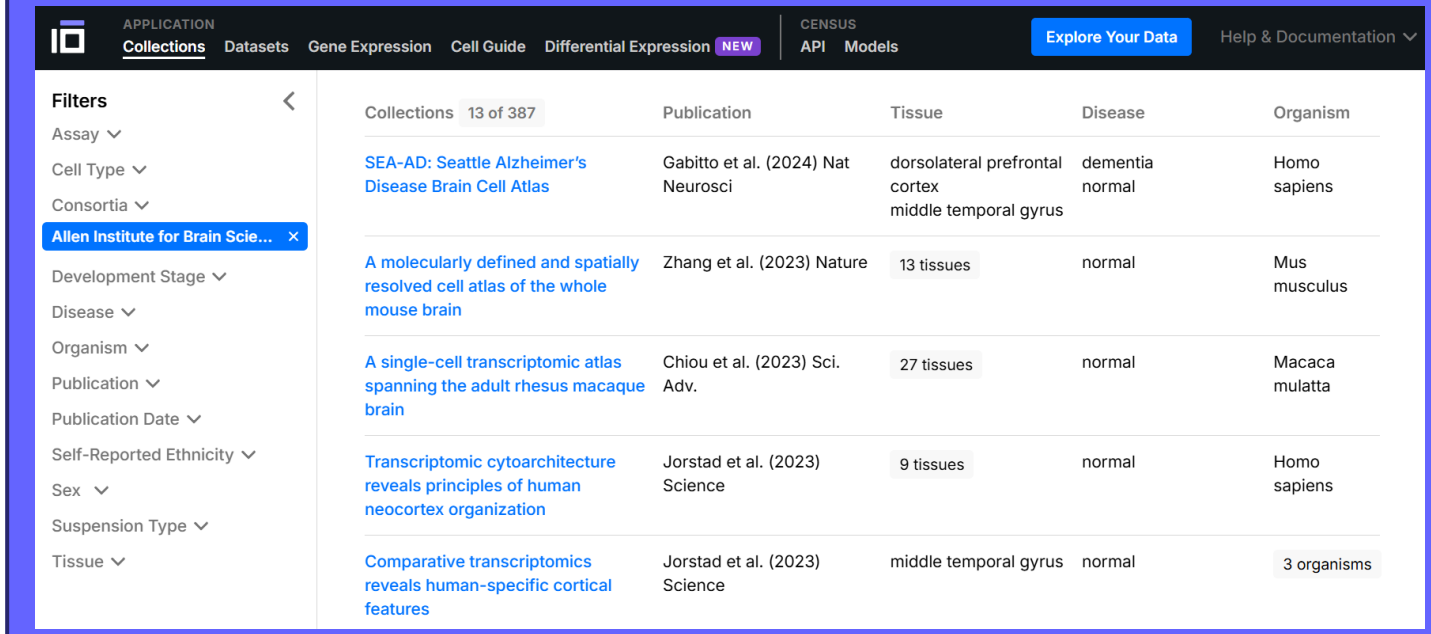
ABC Atlas (much more on this later!)



use cases/gene expression

- Where in the brain is my gene expressed?
- And in which cell types?
- How does this change in AD?
- Which other genes are patterned similarly?

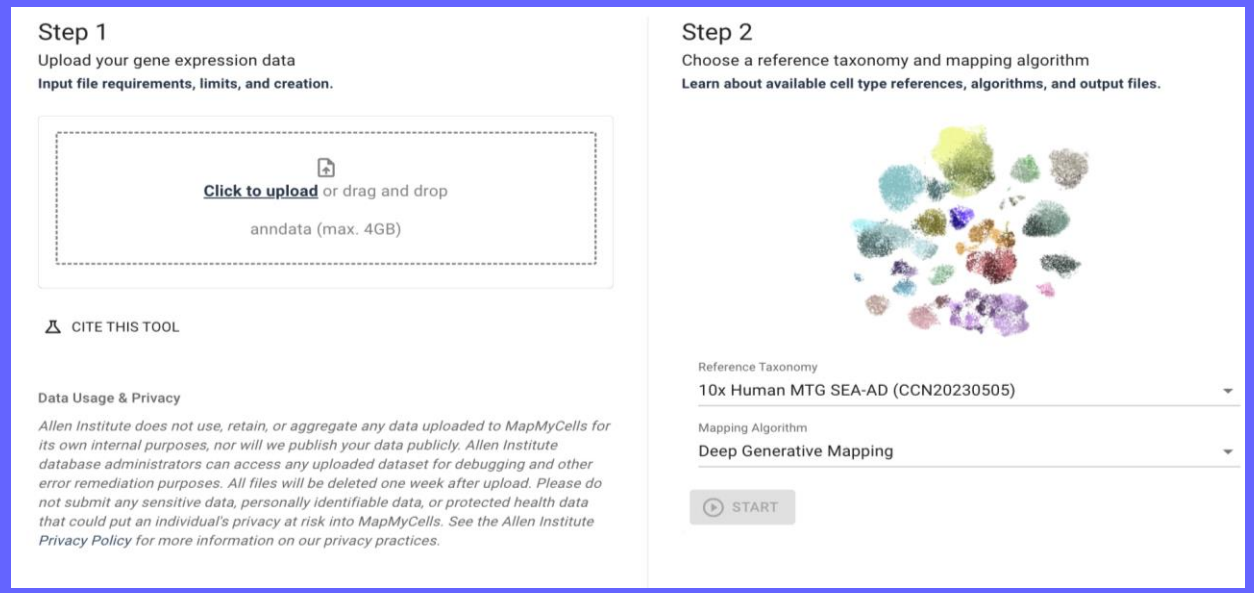
External tools like CELLxGENE (or code)



use cases/cell types

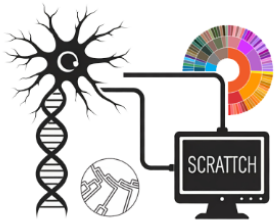
- How do I localize cells in the brain?
- ~~How do I define my own cell types?~~
- How do I assign Allen Institute cell type labels to my own cells?

MapMyCells (more on this later too!)



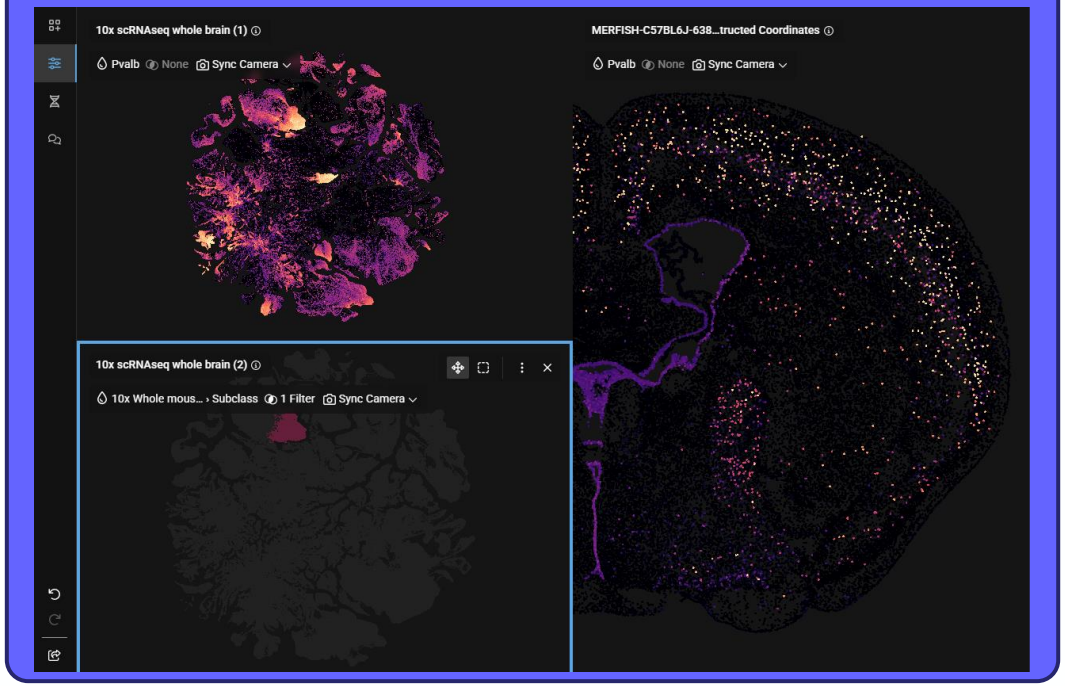
scrattch

Single-**c**ell **R**NA-seq **a**nalysis for **t**ranscriptomic **t**ype **ch**aracterization (or "scrattch") is a suite of R and python scripts from the Allen Institute for Brain Science. The core R libraries are linked in the umbrella package called `scrattch`, which is modeled after the `tidyverse` package. You can use `scrattch` to automatically install or update some of the underlying packages and can run the remaining packages in docker environments. This page describes all core and adjacent scrattch content.



- **Clustering**
- **Taxonomy building**
- **Visualization**
- **Mapping**

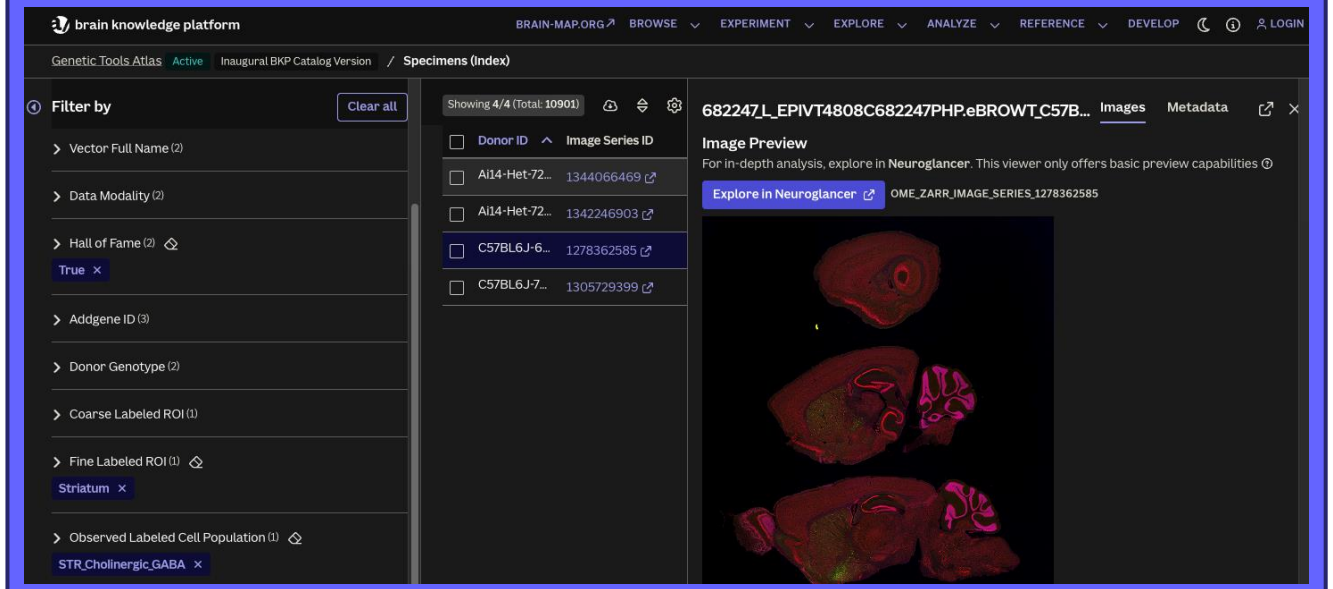
ABC Atlas (much more on this later!)



use cases/cell types

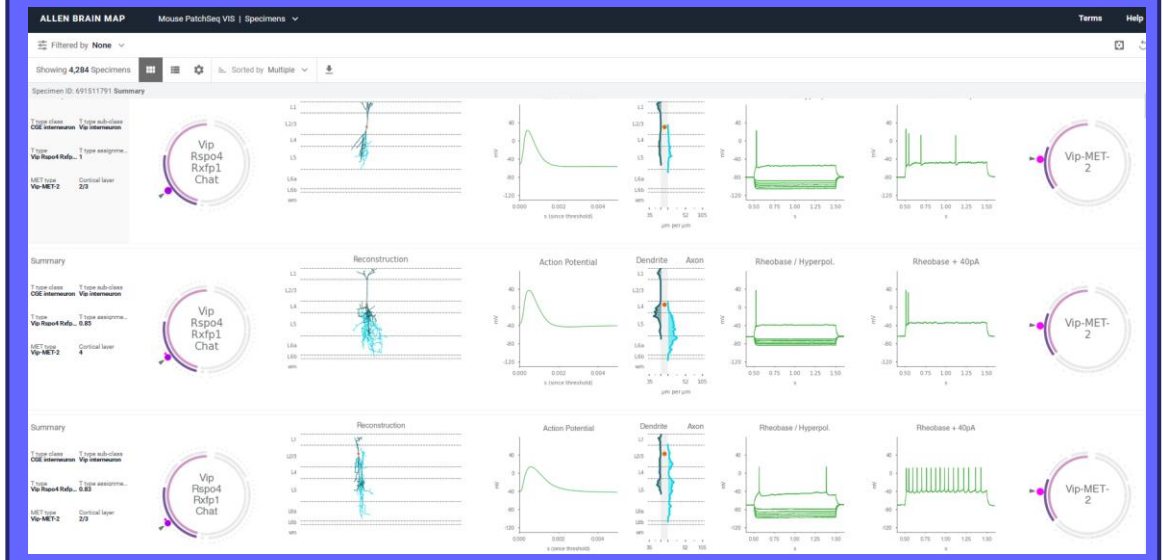
- How do I localize cells in the brain?
- How do I define my own cell types?
- How do I assign Allen Institute cell type labels to my own cells?
- Which genetic tools target specific cell types?
- What are their morphoelectric features?

Genetic tools atlas (more on this later too!)

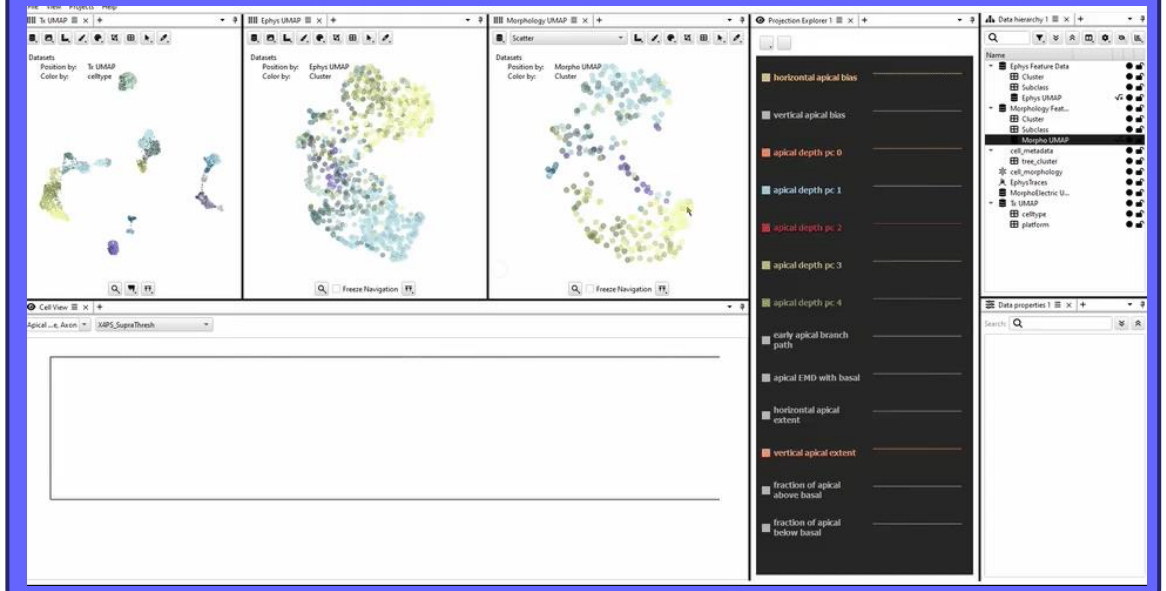


Cell Type Knowledge Explorer (not shown)

Patch-seq Viewer



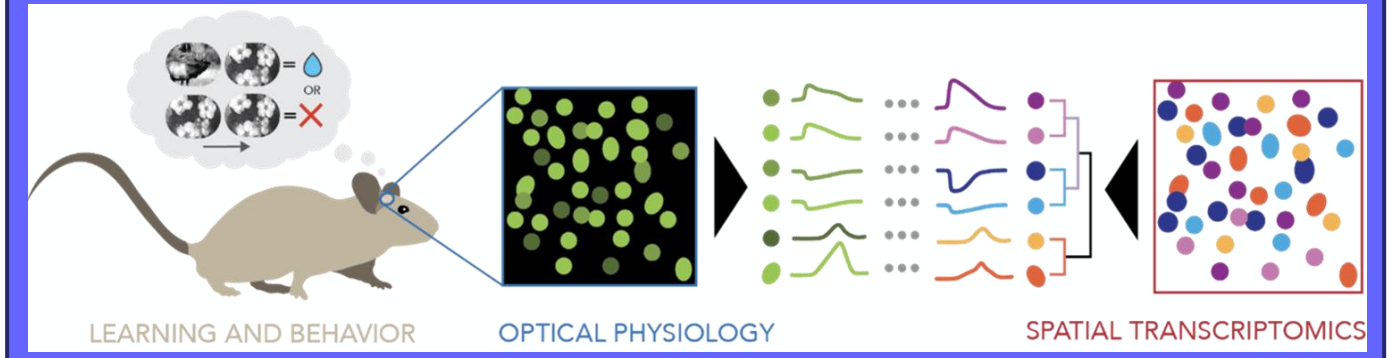
Cytosplore Viewer



use cases/cell types

- What are good cell type markers?
- How do different cell types respond to different visual behaviors?

allen institute/neural dynamics

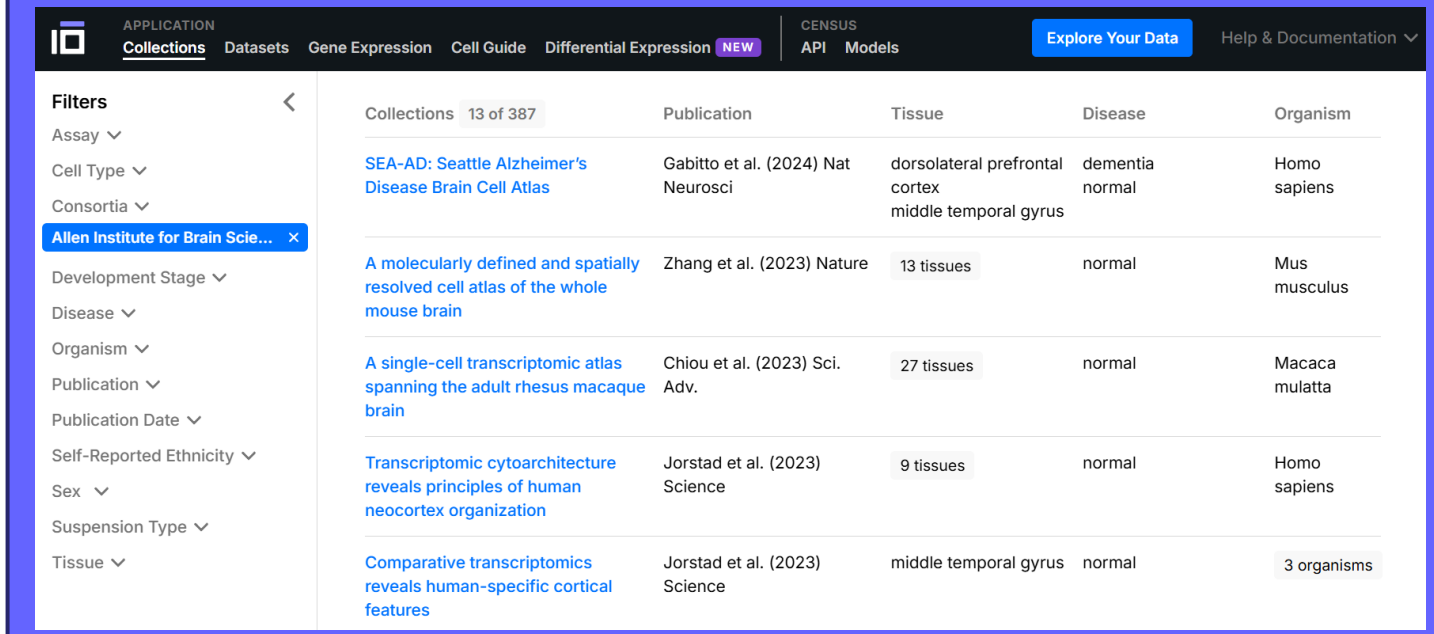


Mouse does something

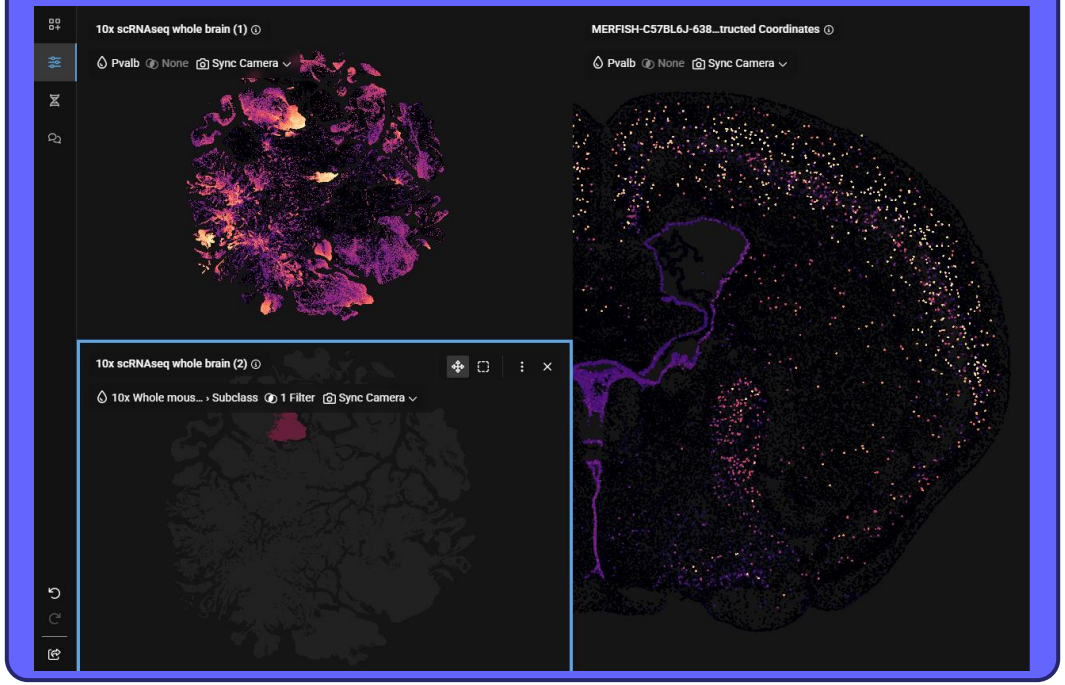
Measure cells while mouse does something

After the experiments, see what kind of cells were measured

External tools like CELLxGENE (or code)



ABC Atlas (much more on this later!)

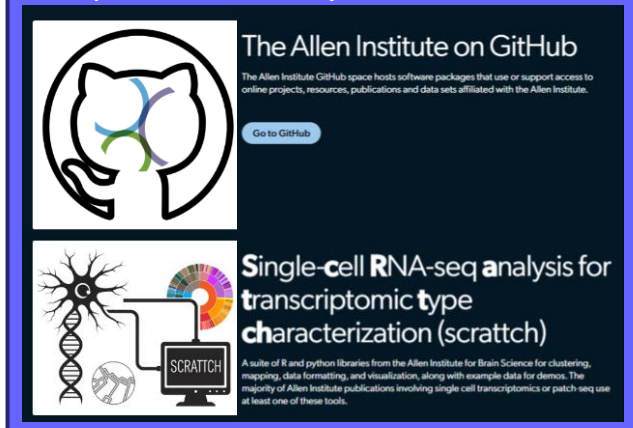


use cases/data access

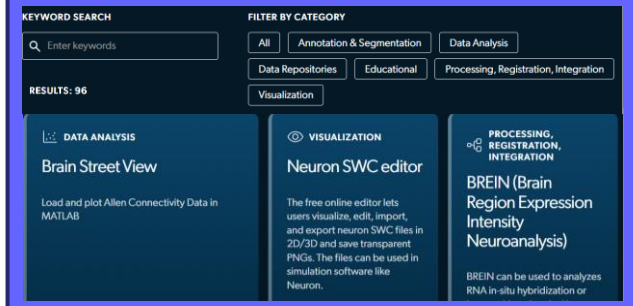
- How do I access the data myself?
- What resources exist for doing analyses off-line using python or R?

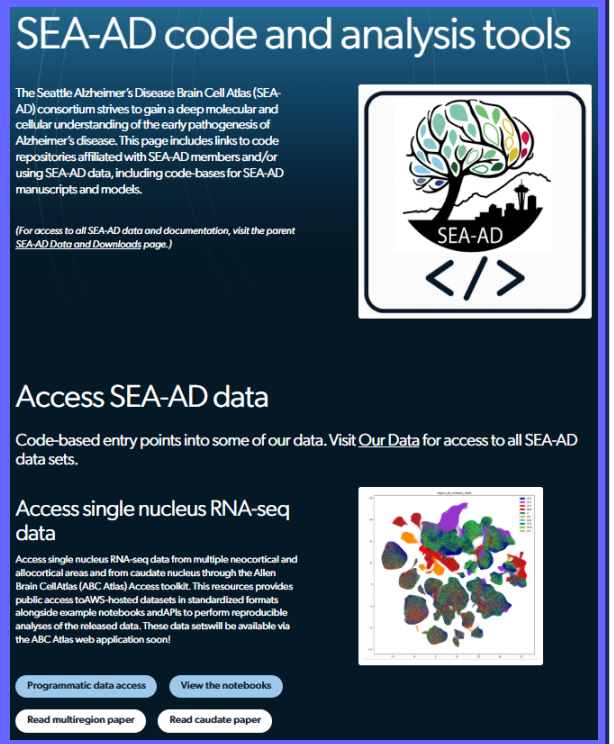
Code landing pages

Computational Analysis Toolbox



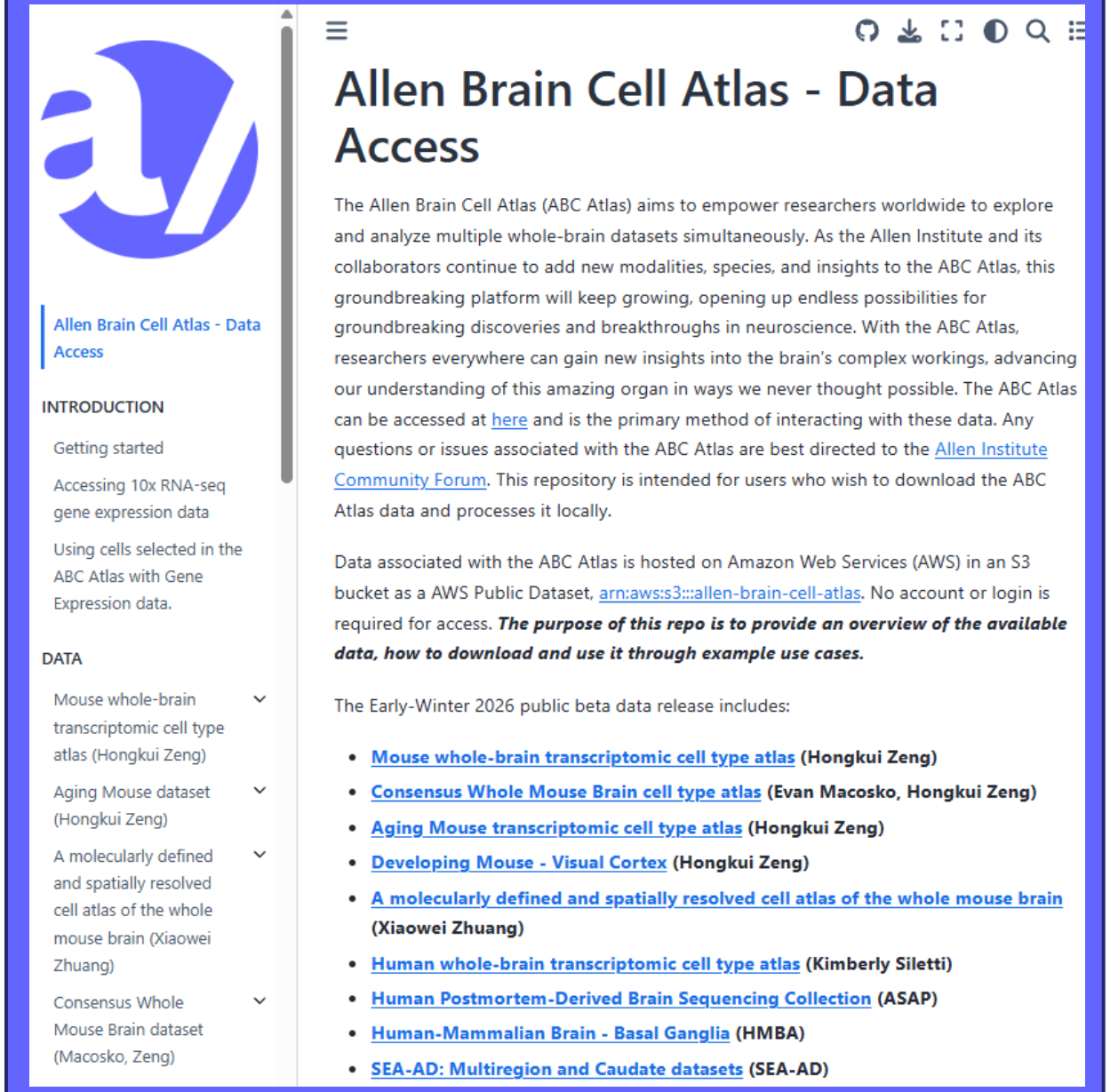
Community Tools





ABC Atlas Access

(and resource-specific download pages not shown)



use cases/???

- Please ask us about any additional use cases you may have during one of the interactive sessions!

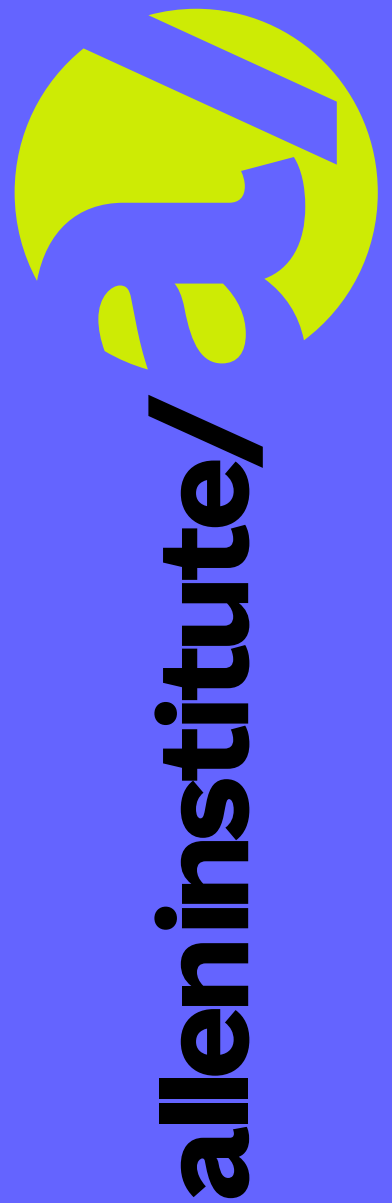


any questions so far/

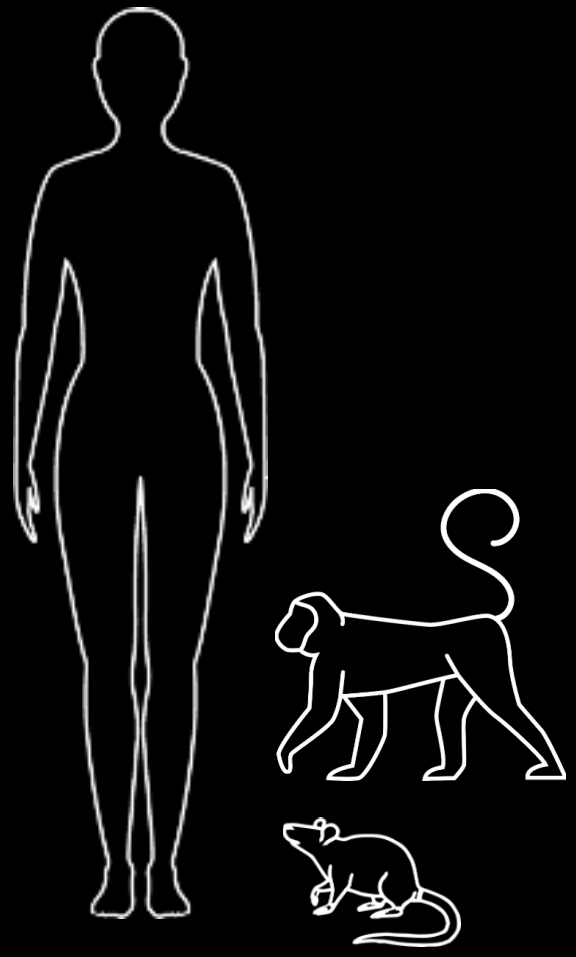


overview of the brain knowledge platform/

Elysha Fiabane



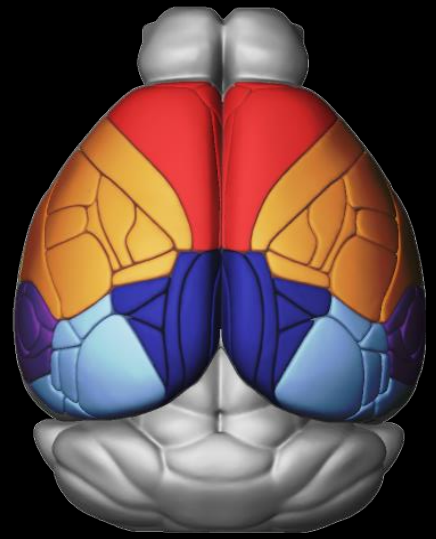
integrated “uber atlas” for cells, spanning:



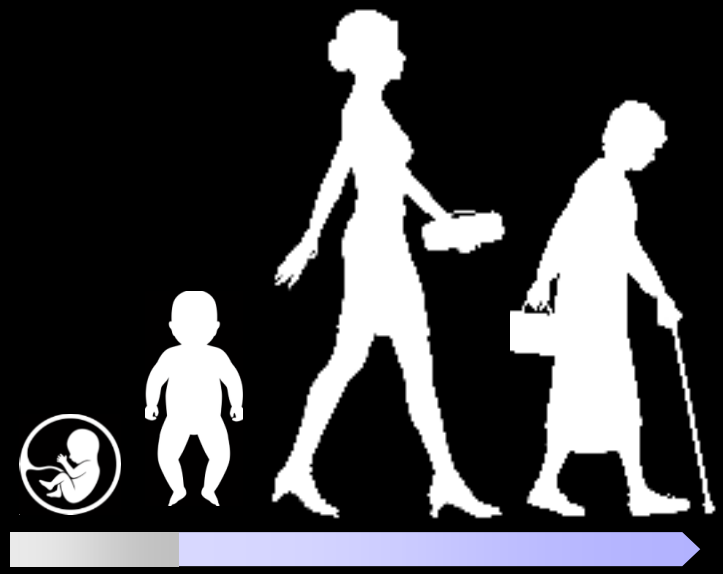
SPECIES

- Transcriptomics
- Epigenomics
- Spatial distribution
- Targeting tools
- Proteomics
- Morphology
- Connectivity
- Physiology
- Function
- Disease

MODALITIES



SPACE



TIME

Brain Knowledge Platform

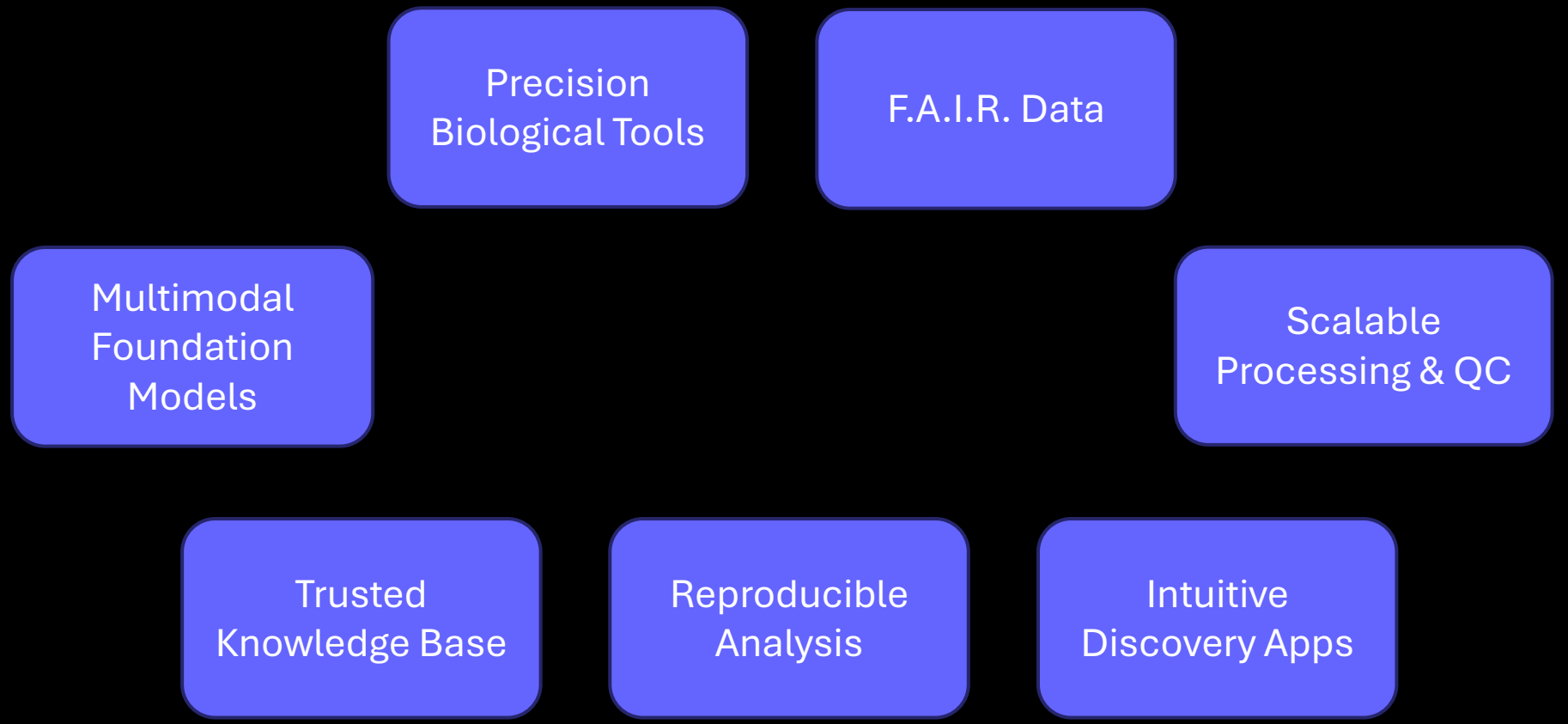
An open, AI-powered neuroscience tool developed by the Allen Institute

It harmonizes data from over 34 million brain cells across 22 species into a single, searchable map, providing scientists with a universal language to study both healthy and diseased mammalian brains.

The platform combines extensive data catalogs with advanced analytical software and genetic tools to help researchers accelerate discoveries in neurological conditions like Alzheimer's and Parkinson's diseases.



technology components of the BKP



AI integrated Dynamic Infrastructure

Optimized data flow and computation

Fast feedback loops

Shared semantics

Provenance driven

Resource governance

Programmatic access

AI-assisted coordination

Support data sharing and reuse

https://brain-map.org/bkp

Brain Knowledge Platform

browse

Find data in the Data Catalog

explore

Visualize and compare data in ABC Atlas

analyze

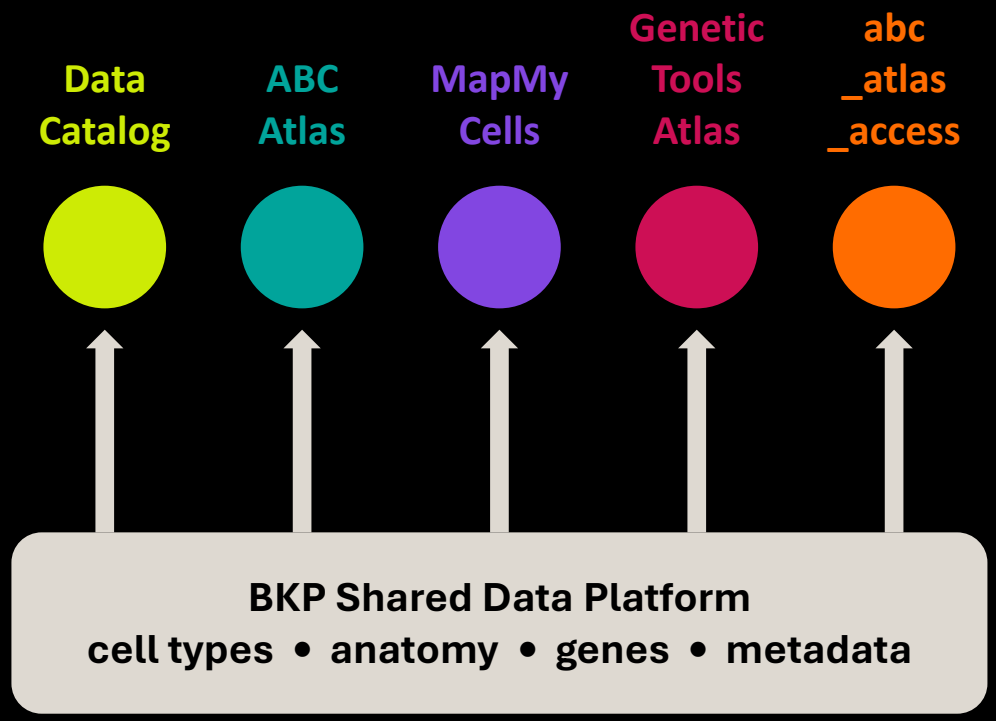
Label and align your data with MapMyCells

experiment

Design experiments with Genetic Tools Atlas

develop

Download and work with data programmatically with abc_atlas_access



BKP/browse

https://brain-map.org/bkp



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Compare your data to Allen Institute's with MapMyCells

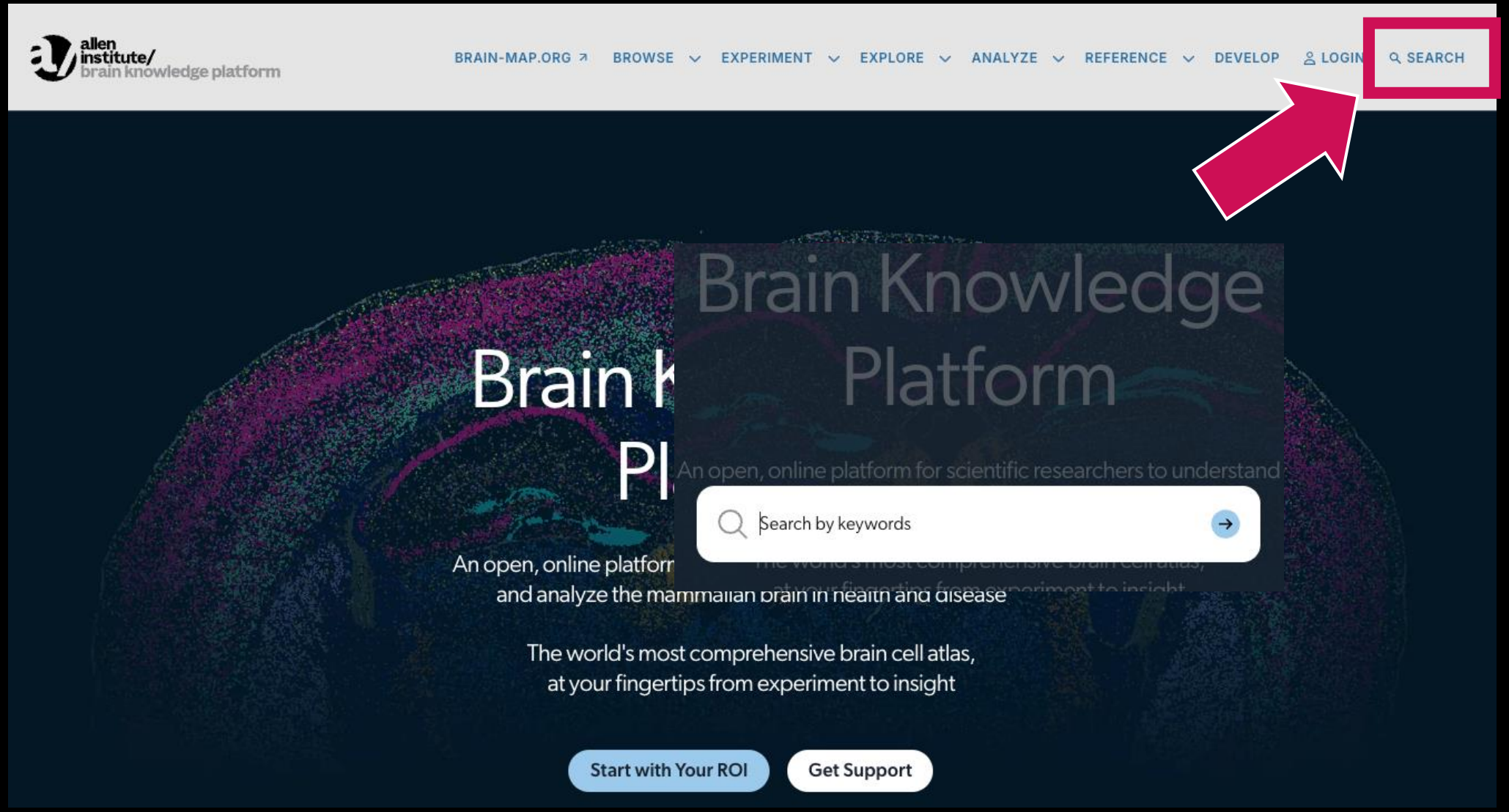
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Design experiments with Genetic Tools Atlas

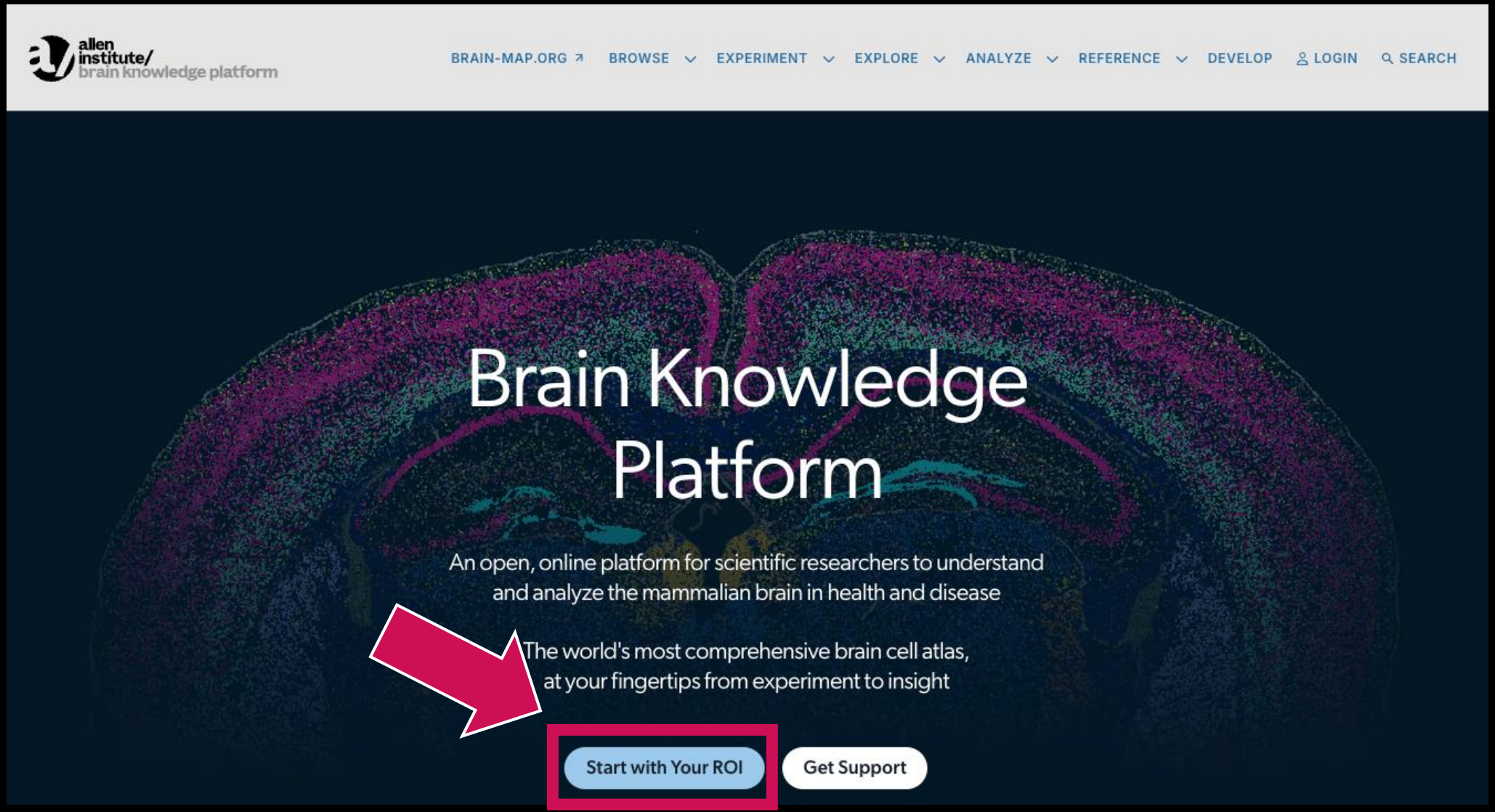
develop

Download and work with data programmatically with abc_atlas_access

BKP/browse


allen institute/ brain knowledge platform
BRAIN-MAP.ORG
BROWSE
EXPERIMENT
EXPLORE
ANALYZE
REFERENCE
DEVELOP
LOGIN
SEARCH
Brain Knowledge Platform
An open, online platform for scientific researchers to understand
Search by keywords
and analyze the mammalian brain in health and disease
The world's most comprehensive brain cell atlas,
at your fingertips from experiment to insight
Start with Your ROI
Get Support

BKP/browse

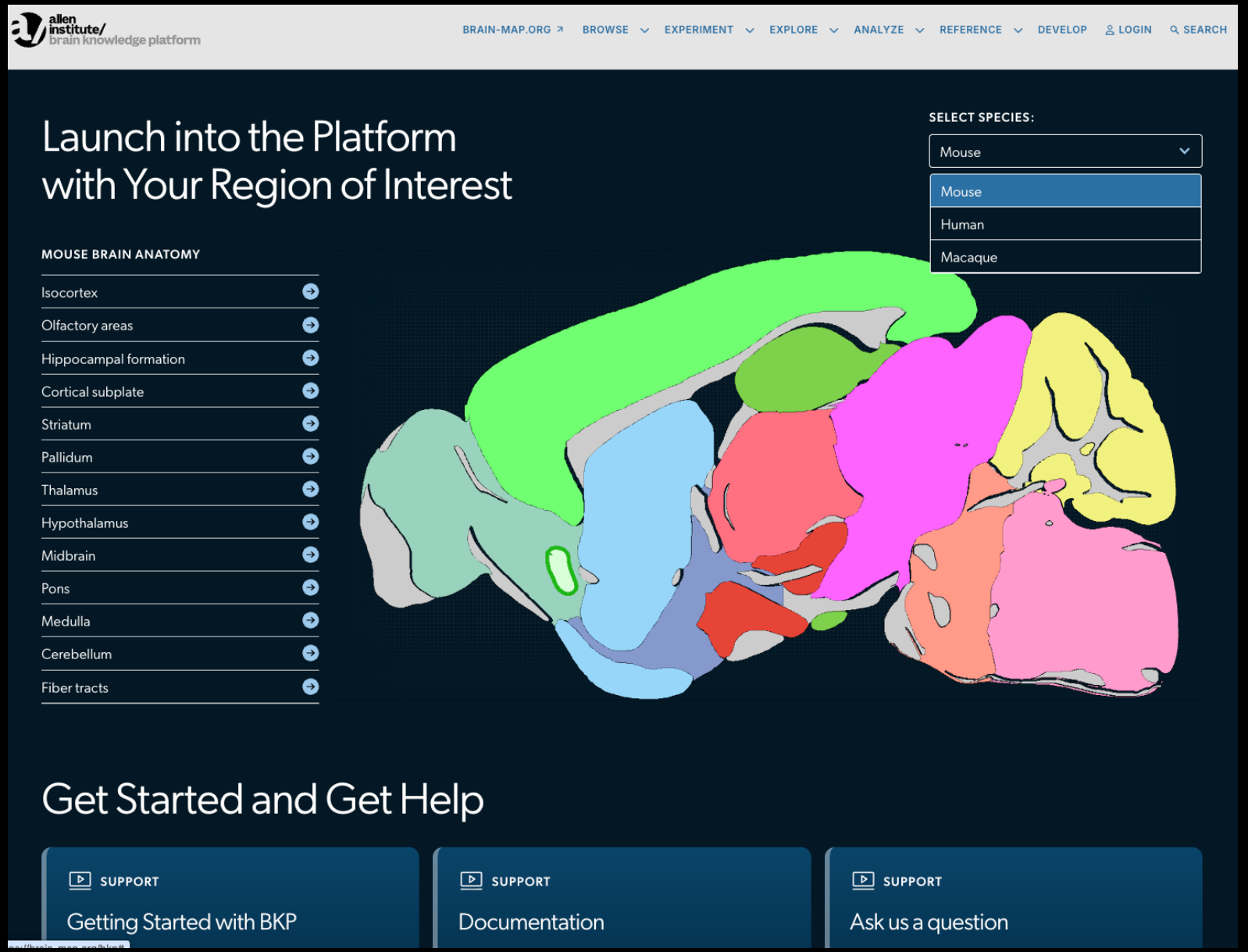

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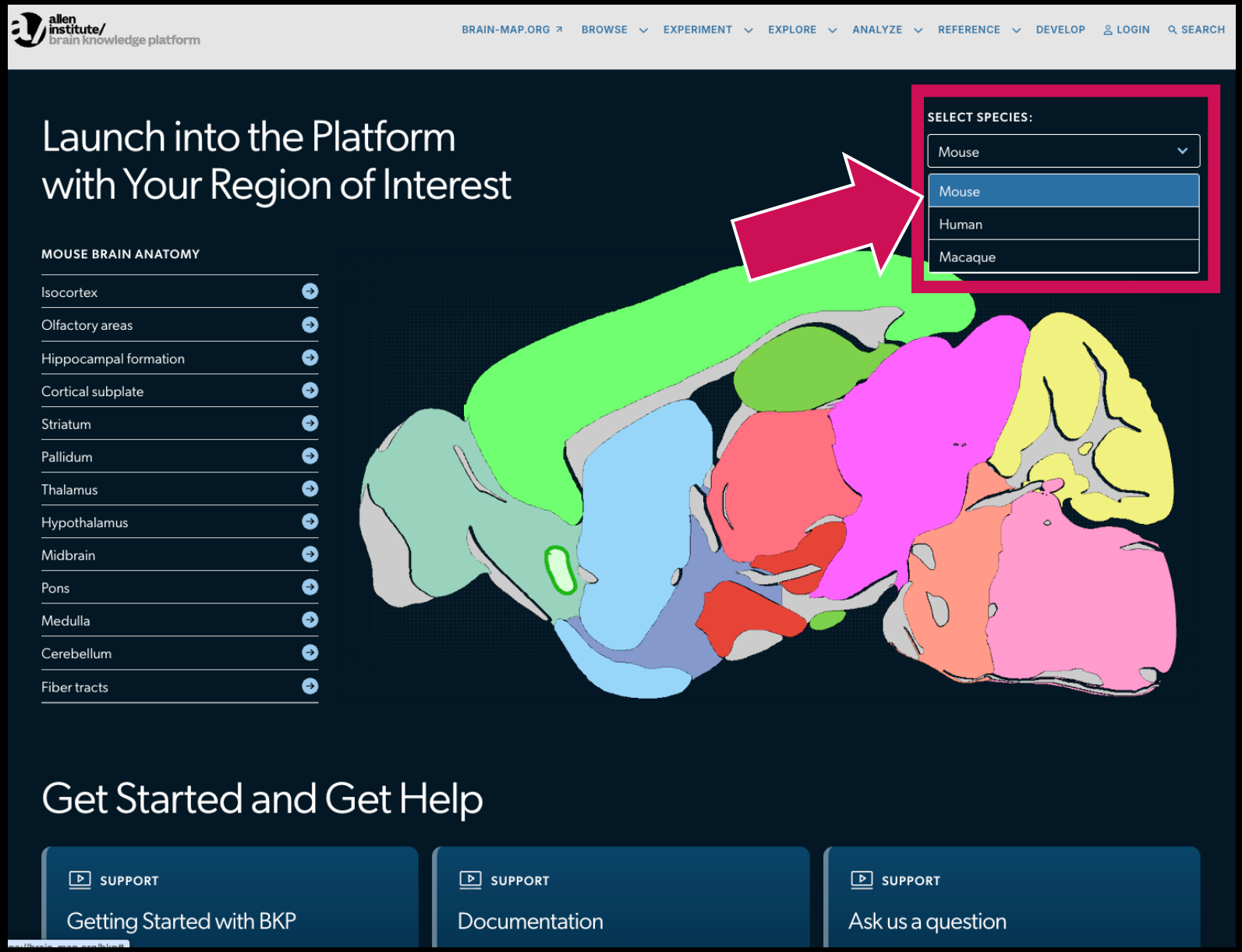
The Brain Knowledge Platform's Data Catalog offers access to a wide range of projects & data collections.

Find relevant-to-you datasets to accelerate your research.

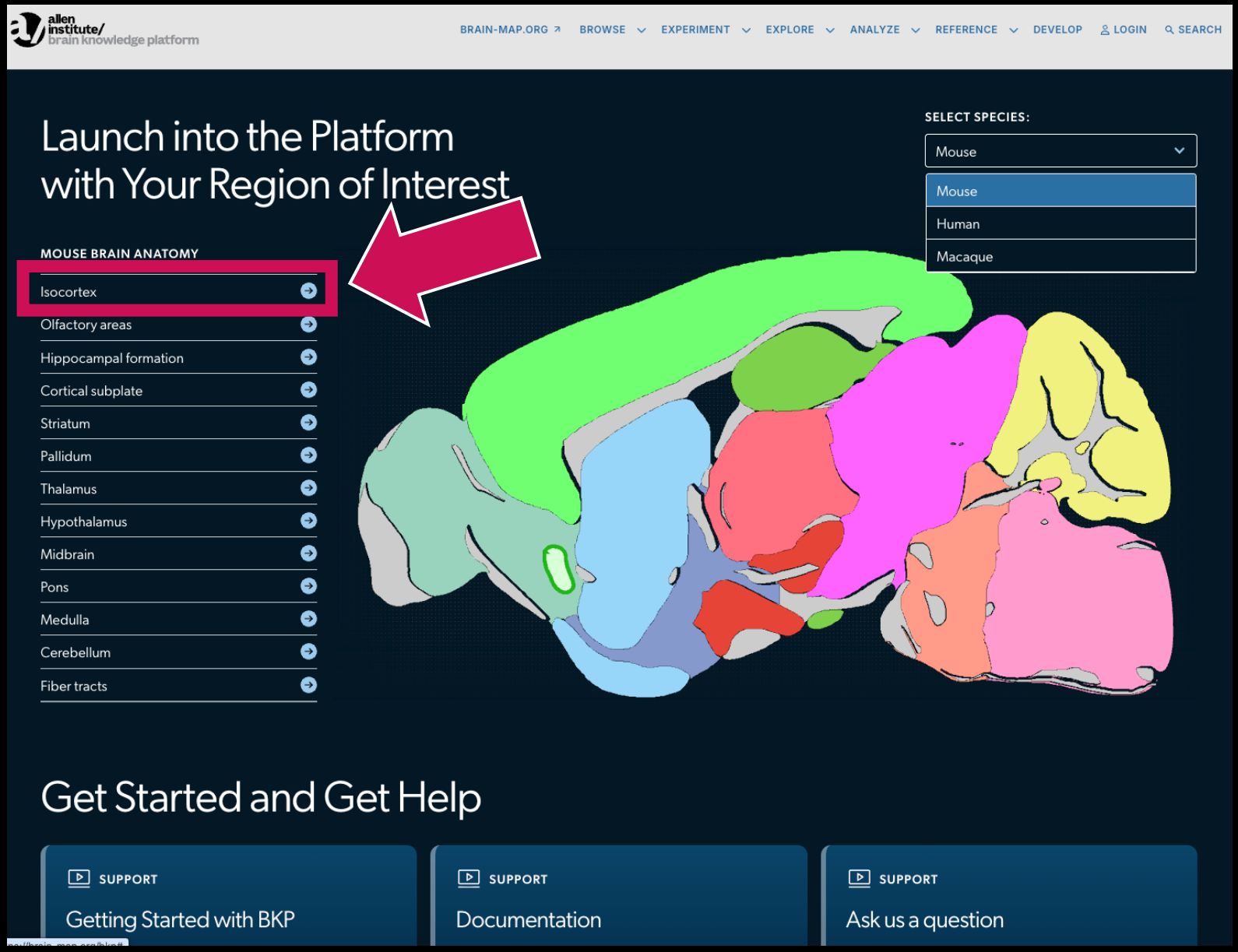
Filter and search by species, modality, technique, author, specimen type, and more.



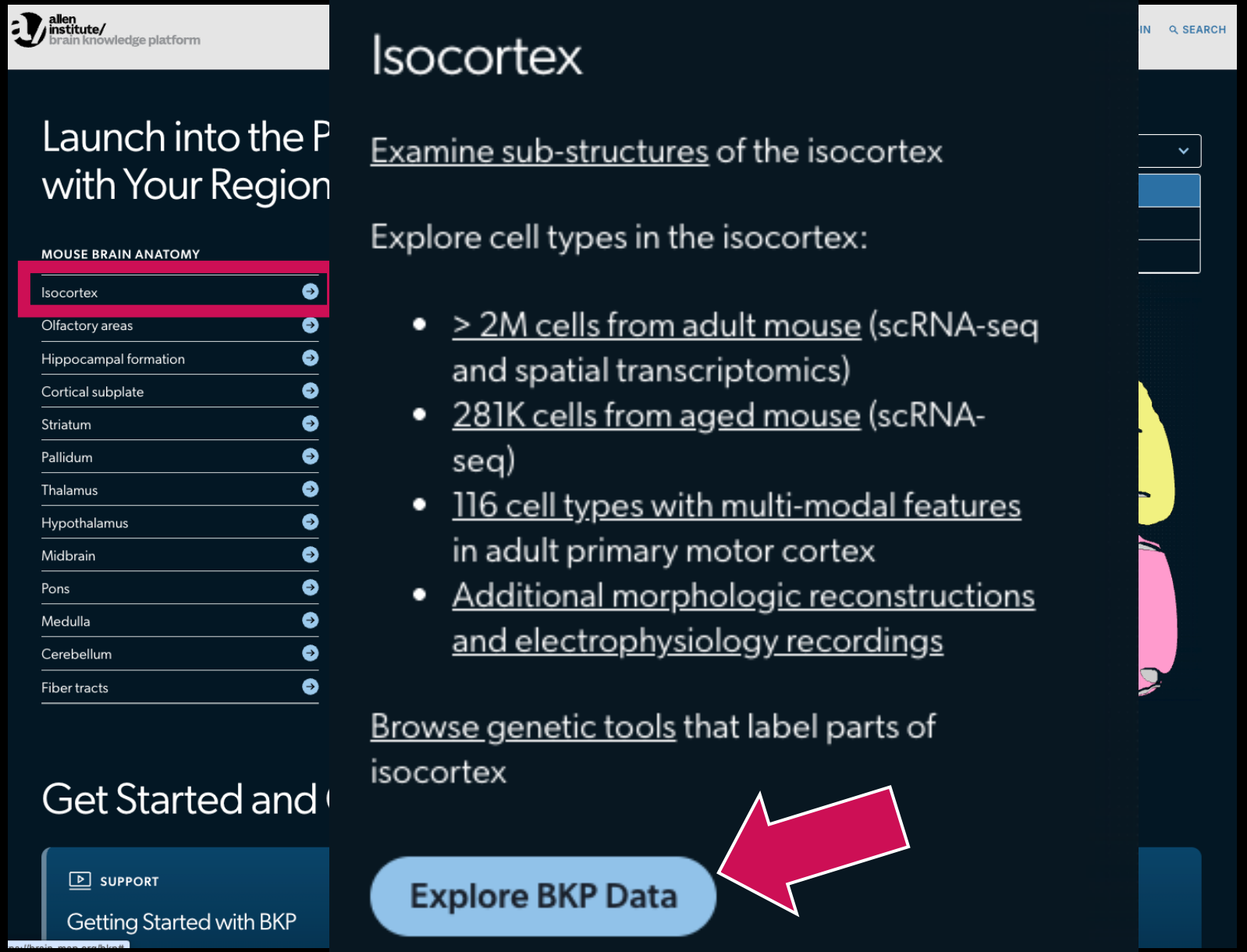
BKP/browse

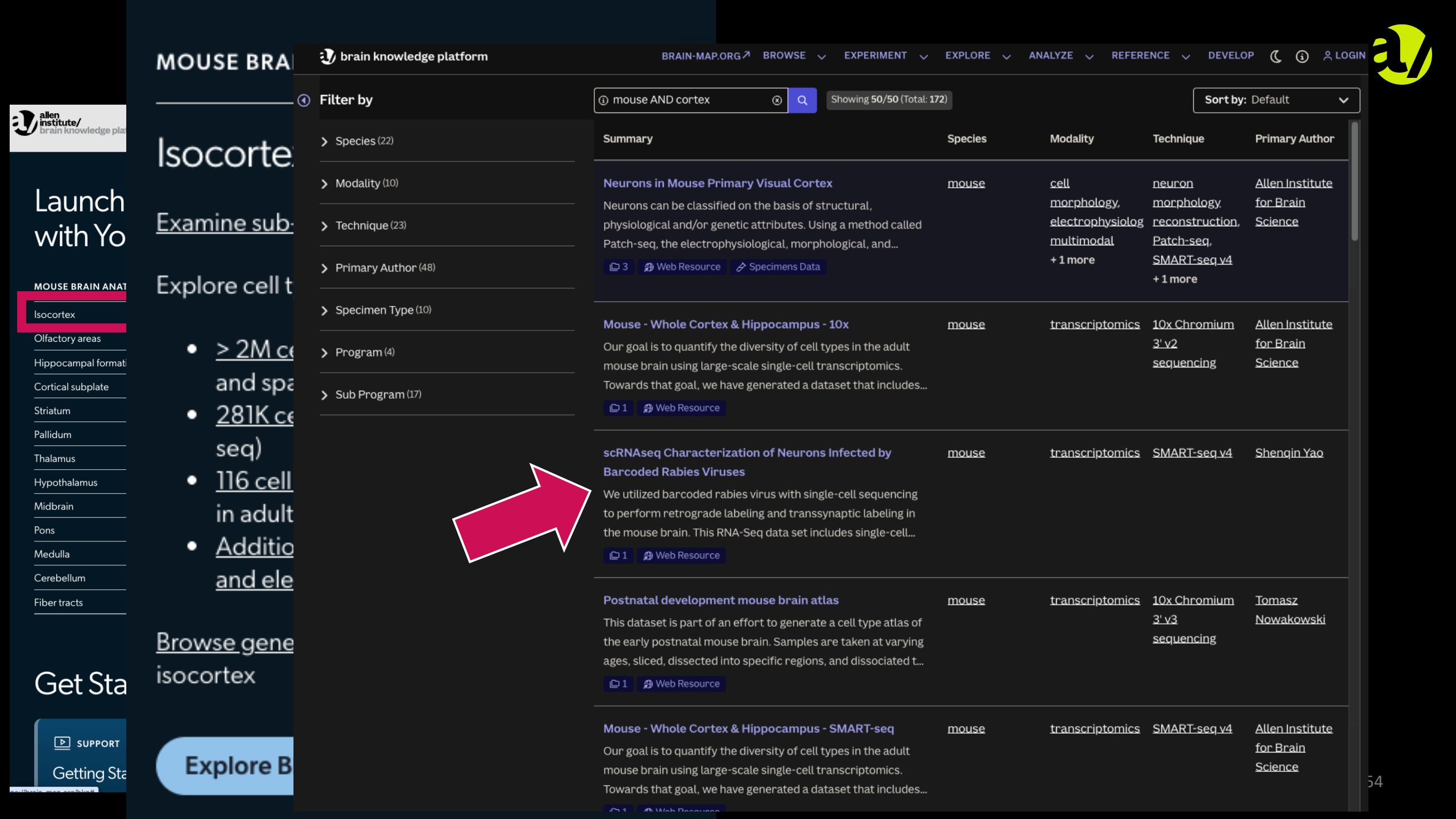

allen institute/ brain knowledge platform
BRAIN-MAP.ORG
BROWSE
EXPERIMENT
EXPLORE
ANALYZE
REFERENCE
DEVELOP
LOGIN
SEARCH
Launch into the Platform with Your Region of Interest
SELECT SPECIES:
Mouse
Mouse
Human
Macaque
MOUSE BRAIN ANATOMY
Isocortex
Olfactory areas
Hippocampal formation
Cortical subplate
Striatum
Pallidum
Thalamus
Hypothalamus
Midbrain
Pons
Medulla
Cerebellum
Fiber tracts
Get Started and Get Help
SUPPORT
Getting Started with BKP
SUPPORT
Documentation
SUPPORT
Ask us a question

BKP/browse


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Get Started and
SUPPORT
Getting Started with BKP
MOUSE BRAIN ANATOMY
Isocortex
Examine sub-structures of the isocortex
Explore cell types in the isocortex:
> 2M cells from adult mouse (scRNA-seq and spatial transcriptomics)
281K cells from aged mouse (scRNA-seq)
116 cell types with multi-modal features in adult primary motor cortex
Additional morphologic reconstructions and electrophysiology recordings
Browse genetic tools that label parts of isocortex
Explore BKP Data
SEARCH


MOUSE BRA
brain knowledge platform
BRAIN-MAP.ORG
BROWSE
EXPERIMENT
EXPLORE
ANALYZE
REFERENCE
DEVELOP
LOGIN
Filter by
mouse AND cortex
Showing 50/50 (Total: 172)
Sort by: Default
Species (22)
Modality (10)
Technique (23)
Primary Author (48)
Specimen Type (10)
Program (4)
Sub Program (17)
Summary
Species
Modality
Technique
Primary Author
Neurons in Mouse Primary Visual Cortex
Neurons can be classified on the basis of structural, physiological and/or genetic attributes. Using a method called Patch-seq, the electrophysiological, morphological, and...
3
Web Resource
Specimens Data
mouse
cell morphology, electrophysiolog multimodal
+ 1 more
neuron morphology reconstruction, Patch-seq, SMART-seq v4
+ 1 more
Allen Institute for Brain Science
Mouse - Whole Cortex & Hippocampus - 10x
Our goal is to quantify the diversity of cell types in the adult mouse brain using large-scale single-cell transcriptomics. Towards that goal, we have generated a dataset that includes...
1
Web Resource
mouse
transcriptomics
10x Chromium 3' v2 sequencing
Allen Institute for Brain Science
scRNAseq Characterization of Neurons Infected by Barcoded Rabies Viruses
We utilized barcoded rabies virus with single-cell sequencing to perform retrograde labeling and transsynaptic labeling in the mouse brain. This RNA-Seq data set includes single-cell...
1
Web Resource
mouse
transcriptomics
SMART-seq v4
Shenqin Yao
Postnatal development mouse brain atlas
This dataset is part of an effort to generate a cell type atlas of the early postnatal mouse brain. Samples are taken at varying ages, sliced, dissected into specific regions, and dissociated t...
1
Web Resource
mouse
transcriptomics
10x Chromium 3' v3 sequencing
Tomasz Nowakowski
Mouse - Whole Cortex & Hippocampus - SMART-seq
Our goal is to quantify the diversity of cell types in the adult mouse brain using large-scale single-cell transcriptomics. Towards that goal, we have generated a dataset that includes...
mouse
transcriptomics
SMART-seq v4
Allen Institute for Brain Science
allen institute/ brain knowledge pla
Launch with Yo
MOUSE BRAIN ANAT
Isocortex
Olfactory areas
Hippocampal formati
Cortical subplate
Striatum
Pallidum
Thalamus
Hypothalamus
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SUPPORT
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Isocorte
Examine sub
Explore cell t
> 2M ce
and spa
281K ce
seq)
116 cell
in adult
Additio
and ele
Browse gene
isocortex
Explore B

BKP/explore

https://brain-map.org/bkp



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https://brain-map.org/bkp/explore

what is the ABC Atlas/

A comprehensive data visualization platform

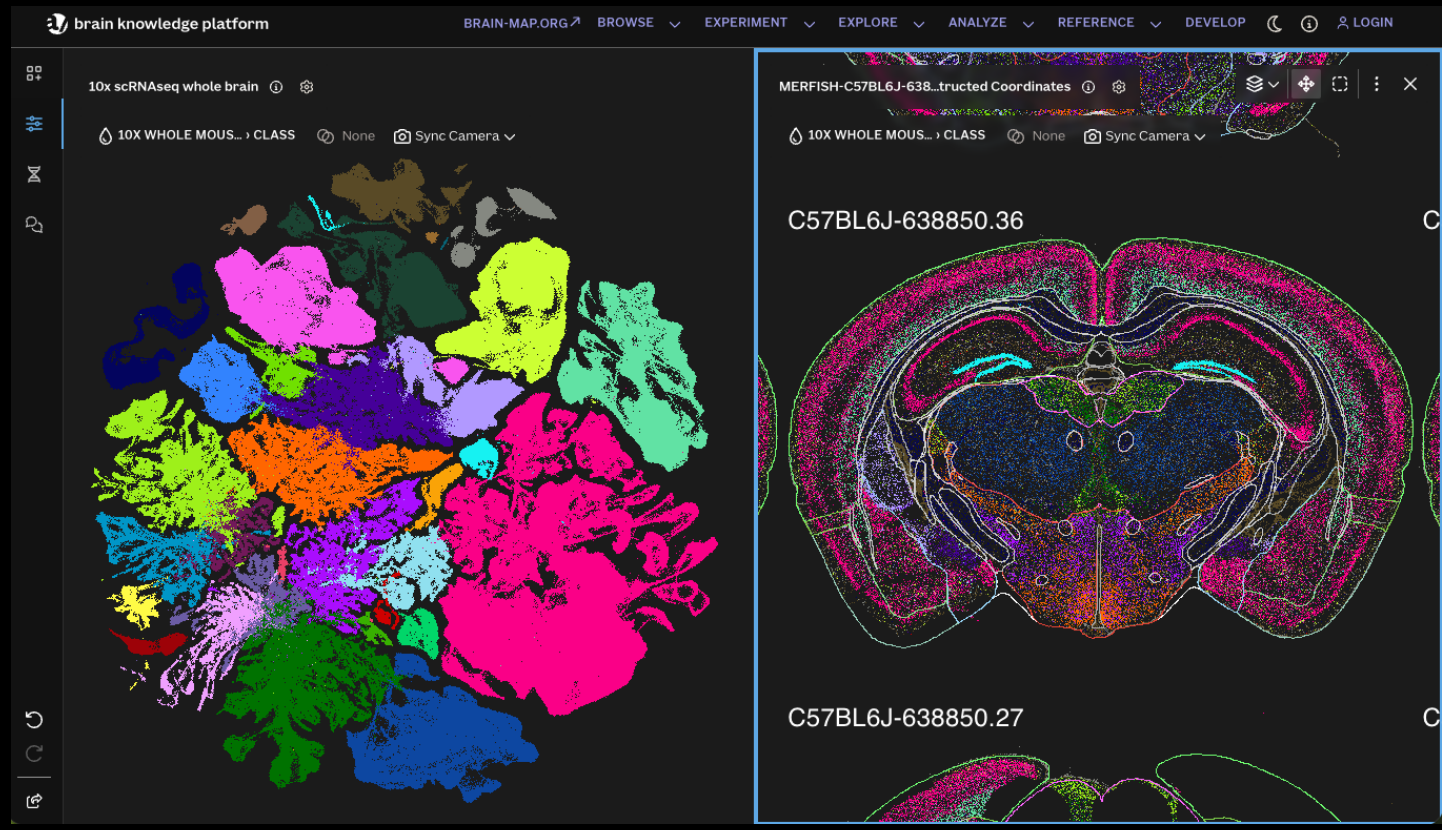
28 high-quality data sets

20+ million cells

Performance at scale Supports viewing hundreds of millions of points simultaneously and many thousands of cell type clusters

Integrated Visualize connected data across species, modality, development, and condition, linked by shared metadata

Accessible Free, web-based access serving ~3,500 monthly users



what's in the ABC Atlas/

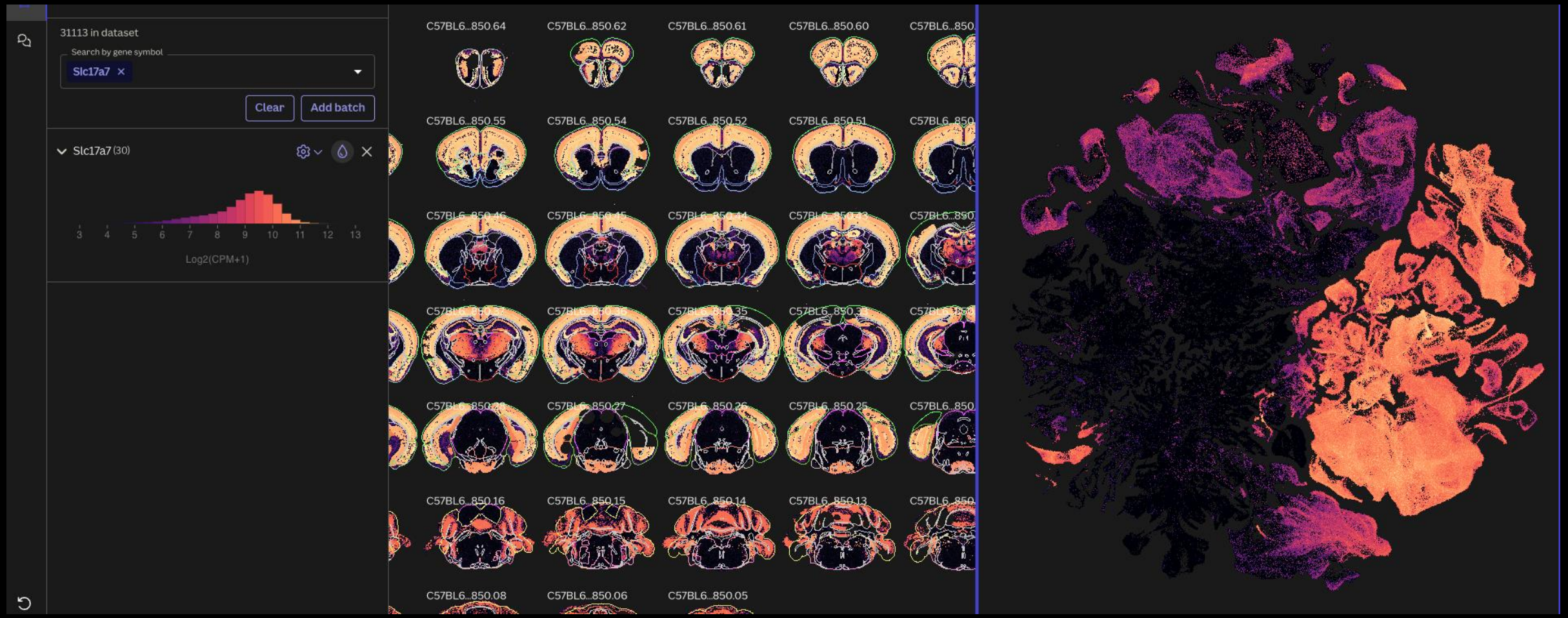
mouse

- **Whole Mouse Brain** (scRNAseq and MERFISH data)
  - Yao et al., 2023, Zhang et al., 2023 – Nature
- **Consensus Whole Mouse Brain** (scRNAseq)
  - Yao et al., 2023, Langlieb et al., 2023 – Nature
- **Aging Mouse** (scRNAseq)
  - Jin et al., 2025 – Nature.
- **Developing Mouse Visual Cortex** (scRNAseq)
  - Gao et al., 2025 – Nature

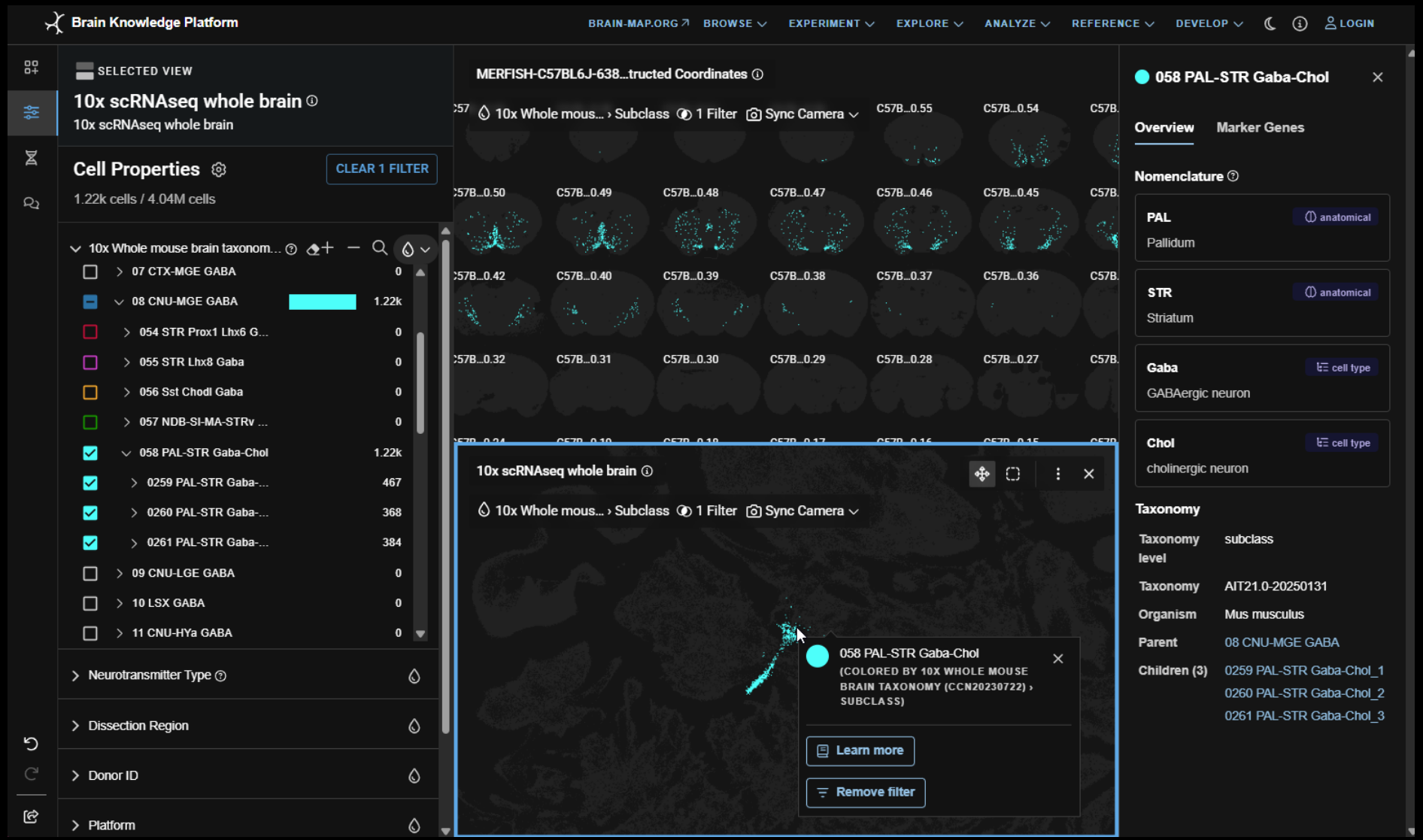
human & primate

- **Whole Human Brain** (snRNAseq)
  - Siletti et al., 2023 – Science
- **Human Alzheimer's Disease** – MTG and DLPFC (SEA-AD) (snRNAseq and MERFISH data)
  - Gabitto & Travaglini et al., 2024 – Nature Neuroscience
- **Human Parkinson's Disease** (ASAP Human Postmortem BrainSeq) (snRNAseq)
  - (No publication—collaboration with MJFF/ASAP)
- **Cross-species HMBA Basal Ganglia** (snRNAseq and MERFISH data)
  - Johansen & Fu et al, 2026; Hewitt & Turner et al, 2026 - bioRxiv

what can you do?/view gene expression



what can you do?/explore cell types

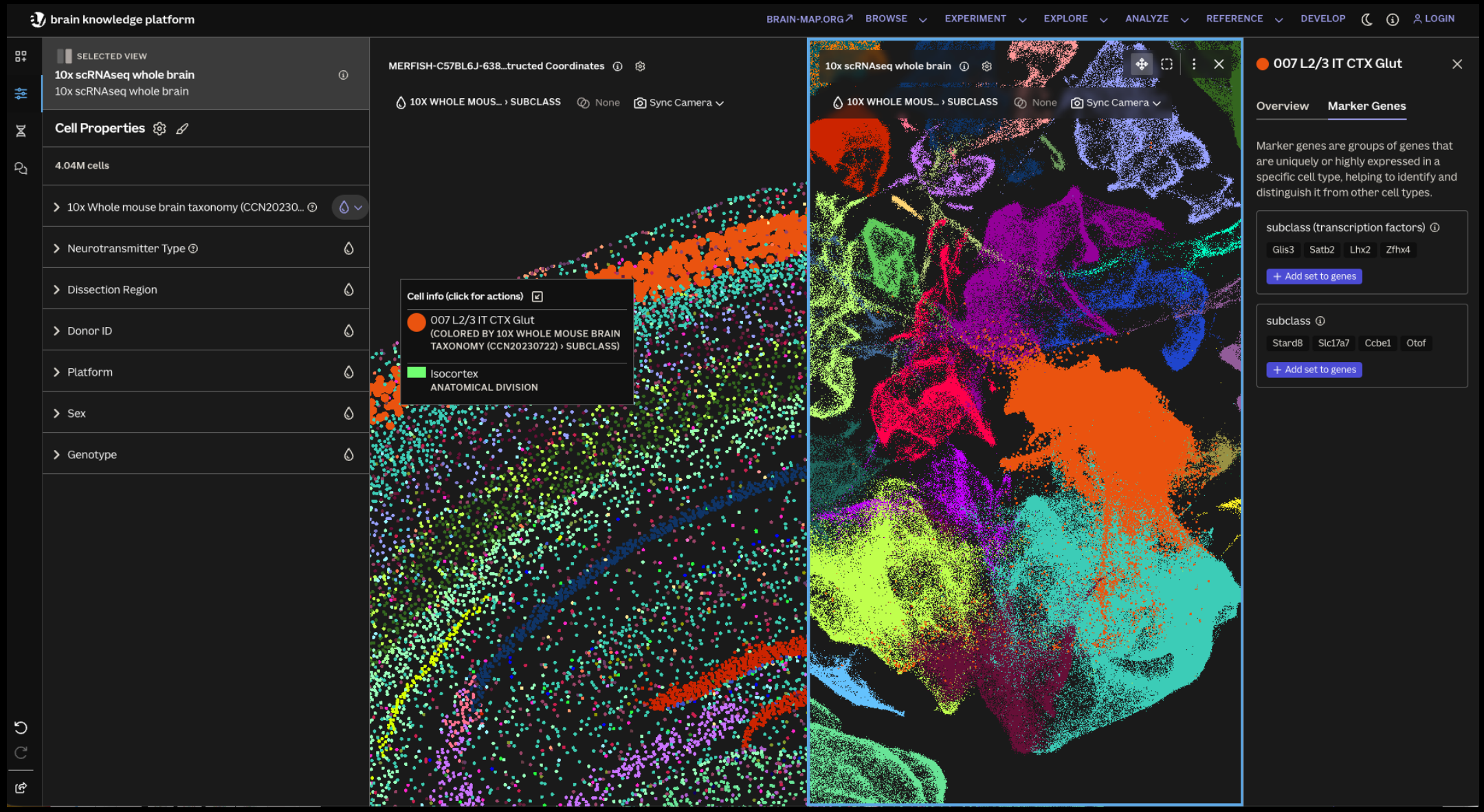


What do the acronyms mean?

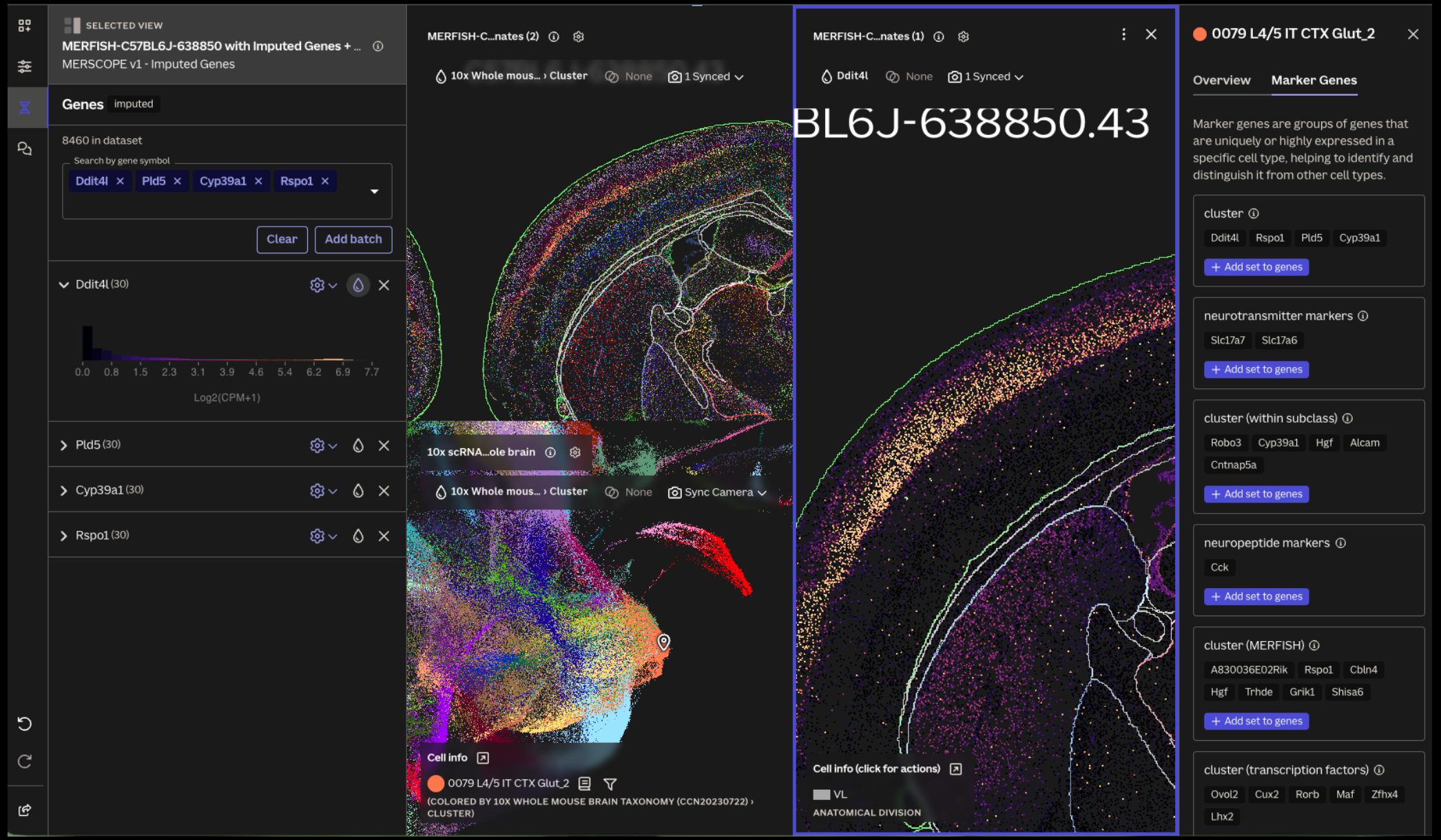
How is this cell type defined?

What else is known about this cell type?

what can you do?/explore cell types



what can you do?/explore cell types & genes



what can you do?/compare data

Parkinson's data (MJFF/ASAP) was mapped to whole human brain cell types (Siletti et al) and well-defined Alzheimer's cell types (SEA-AD) using MapMyCells, unlocking new comparative visualization and analysis



View in ABC Atlas

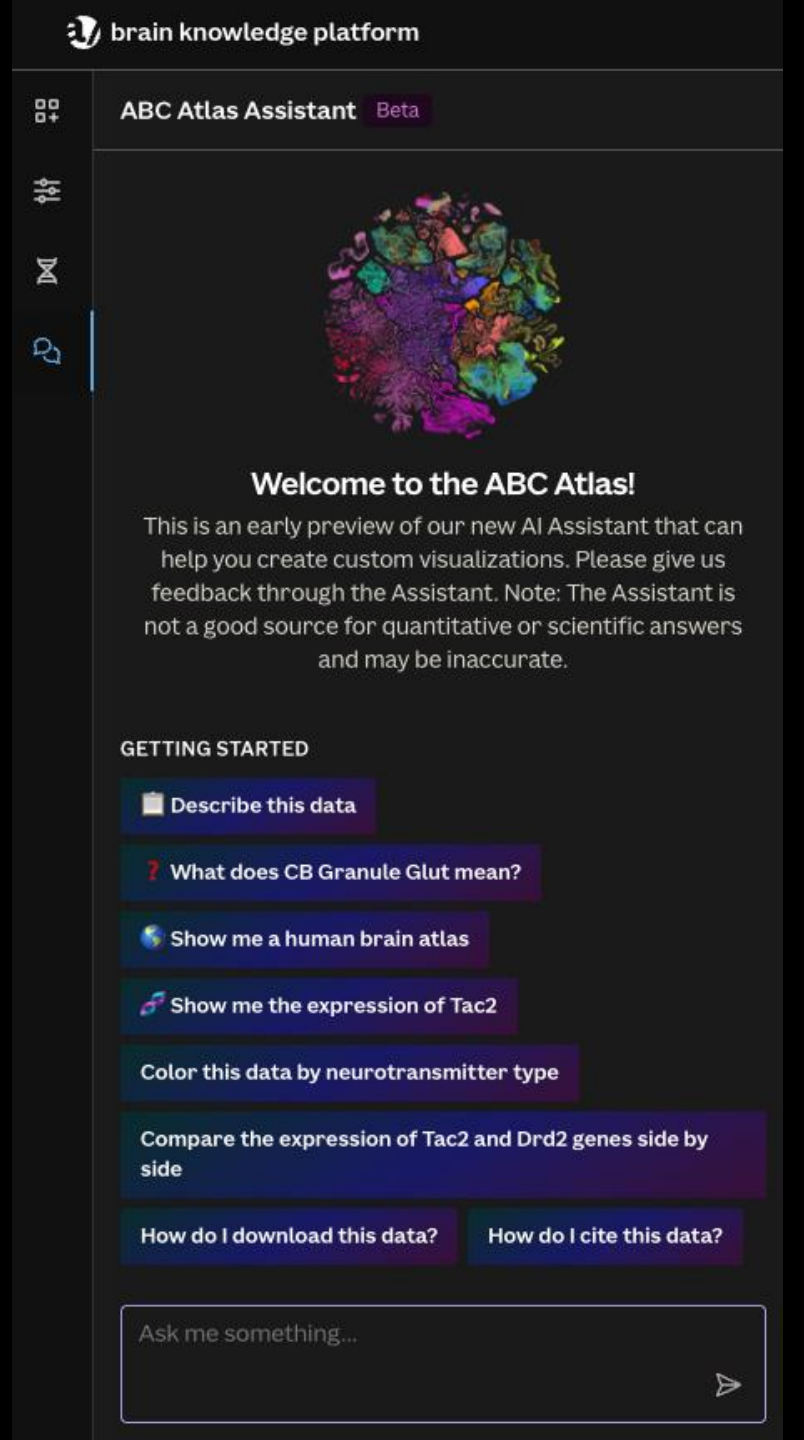
ask the ABC Atlas Assistant!

The AI Assistant uses **natural language input** to help users create custom visualizations faster

- **simplifies** complex tasks for new users
- **accelerates** advanced controls for experienced users

The Assistant can:

- Describe, suggest, and open data sets
- Adjust colors and filters across multiple data sets based on user instruction
- Offer “how to” instructions



BKP/analyze

https://brain-map.org/bkp



browse

Find data in the Data Catalog

explore

Visualize and compare data in ABC Atlas

analyze

Compare your data to Allen Institute's with MapMyCells

experiment

Design experiments with Genetic Tools Atlas

develop

Download and work with data programmatically with abc_atlas_access

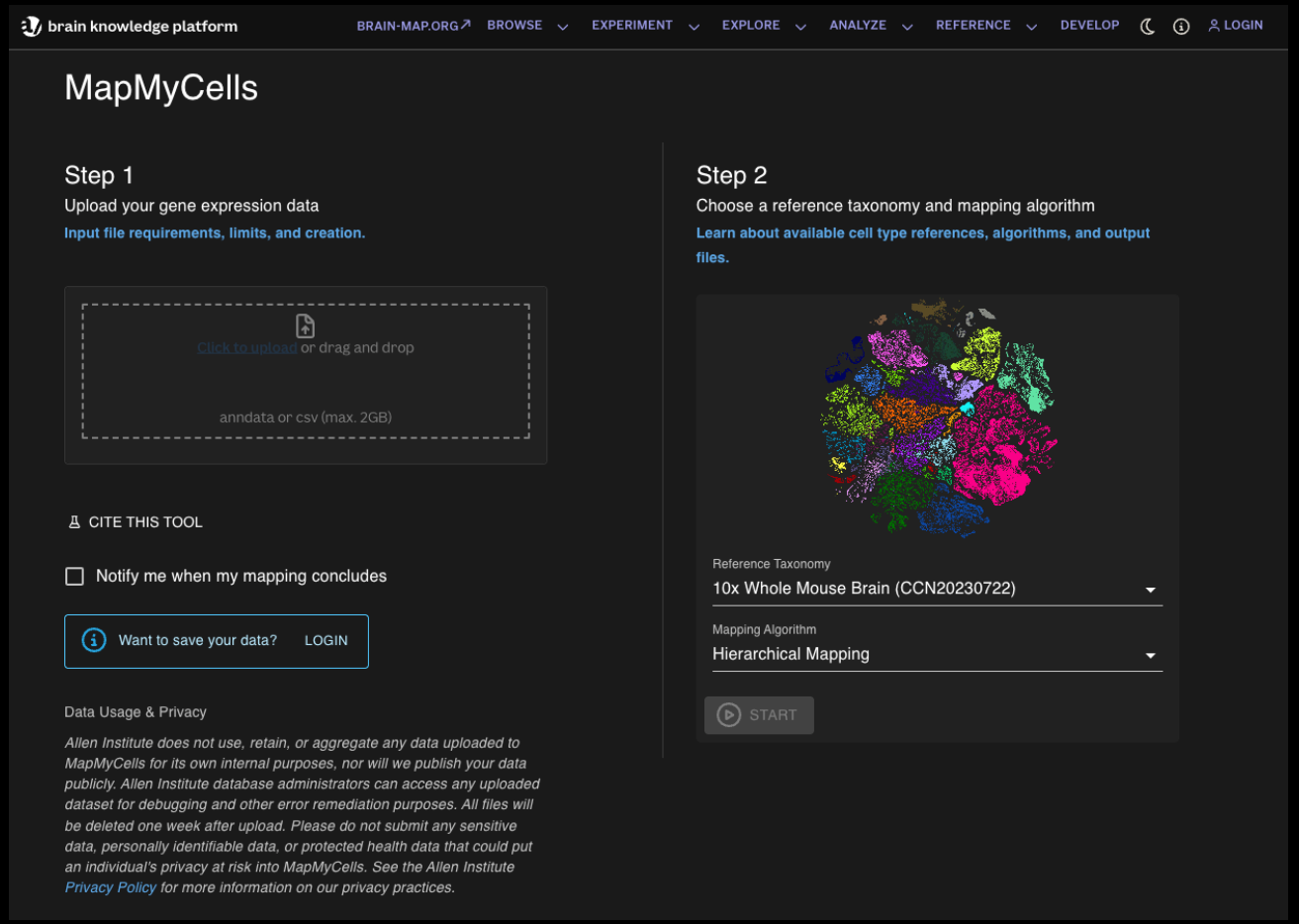
https://brain-map.org/bkp/analyze

MapMyCells: Map your RNAseq data onto an Allen reference taxonomy

6 Cell Type Taxonomies
3 Mapping Algorithms

Input: **Drag and drop** .h5ad file, csv, cellxgene matrix

Output: .csv file,
cell IDs with Allen Institute taxonomy labels & probabilities
.JSON file, metadata



Also available for offline programmatic access:
https://github.com/AllenInstitute/cell_type_mapper

Preprint:

BKP/experiment

https://brain-map.org/bkp



browse

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explore

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analyze

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experiment

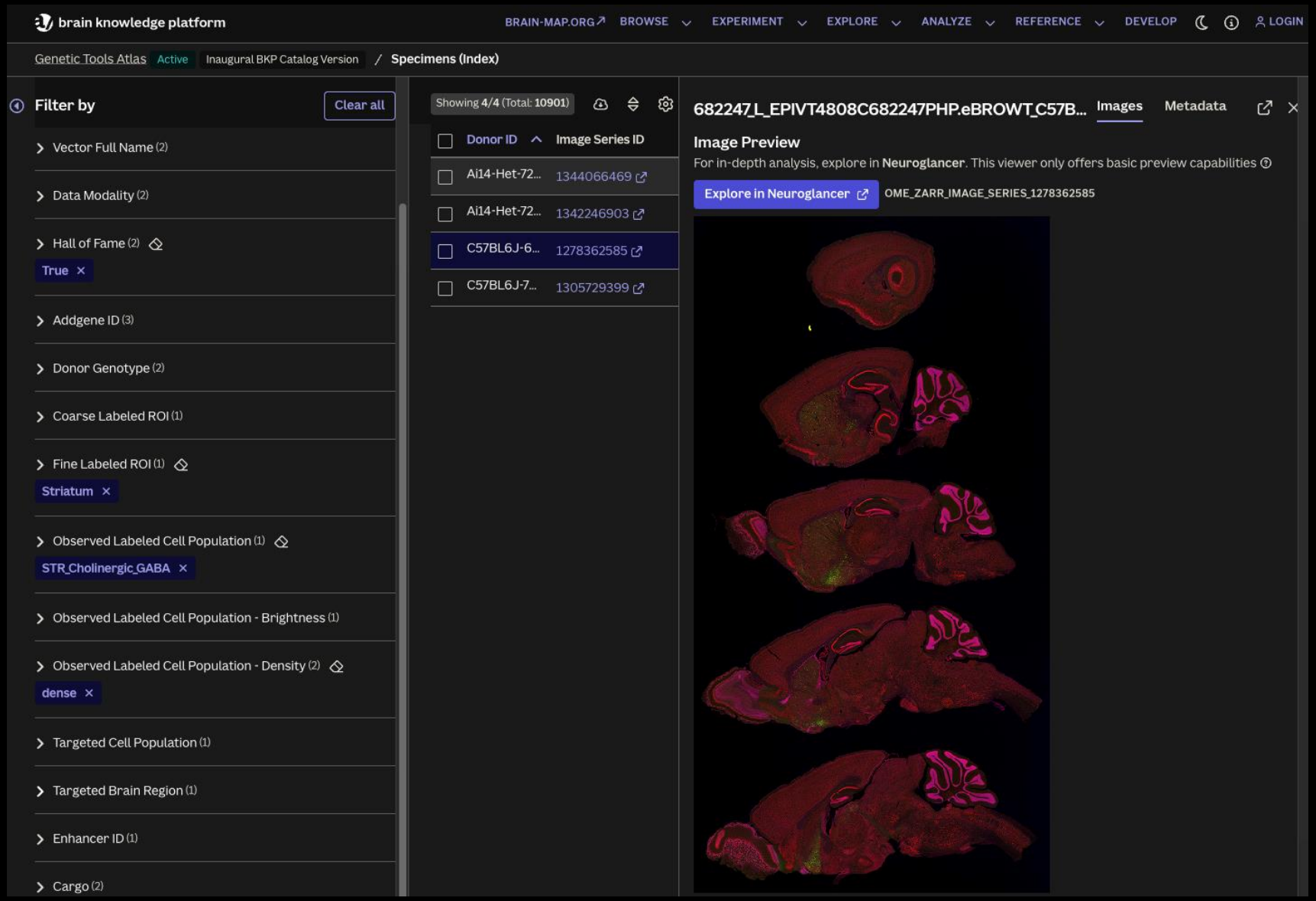
Design experiments with Genetic Tools Atlas

develop

Download and work with data programmatically with abc_atlas_access

Genetic Tools Atlas

https://brain-map.org/bkp/experiment



- The GTA offers a **large genetic toolkit** for selective gene expression in brain cell types of interest.

- Searchable web tool representing information and data on enhancer-adeno-associated viruses (enhancer AAVs) and mouse transgenes **developed and tested at the Allen Institute**.

BKP/develop

https://brain-map.org/bkp



browse

Find data in the Data Catalog

explore

Visualize and compare data in ABC Atlas

analyze

Compare your data to Allen Institute's with MapMyCells

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develop

Download and work with data programmatically with abc_atlas_access

abc_atlas_access/

https://brain-map.org/bkp/develop



https://alleninstitute.github.io/abc_atlas_access/intro.html

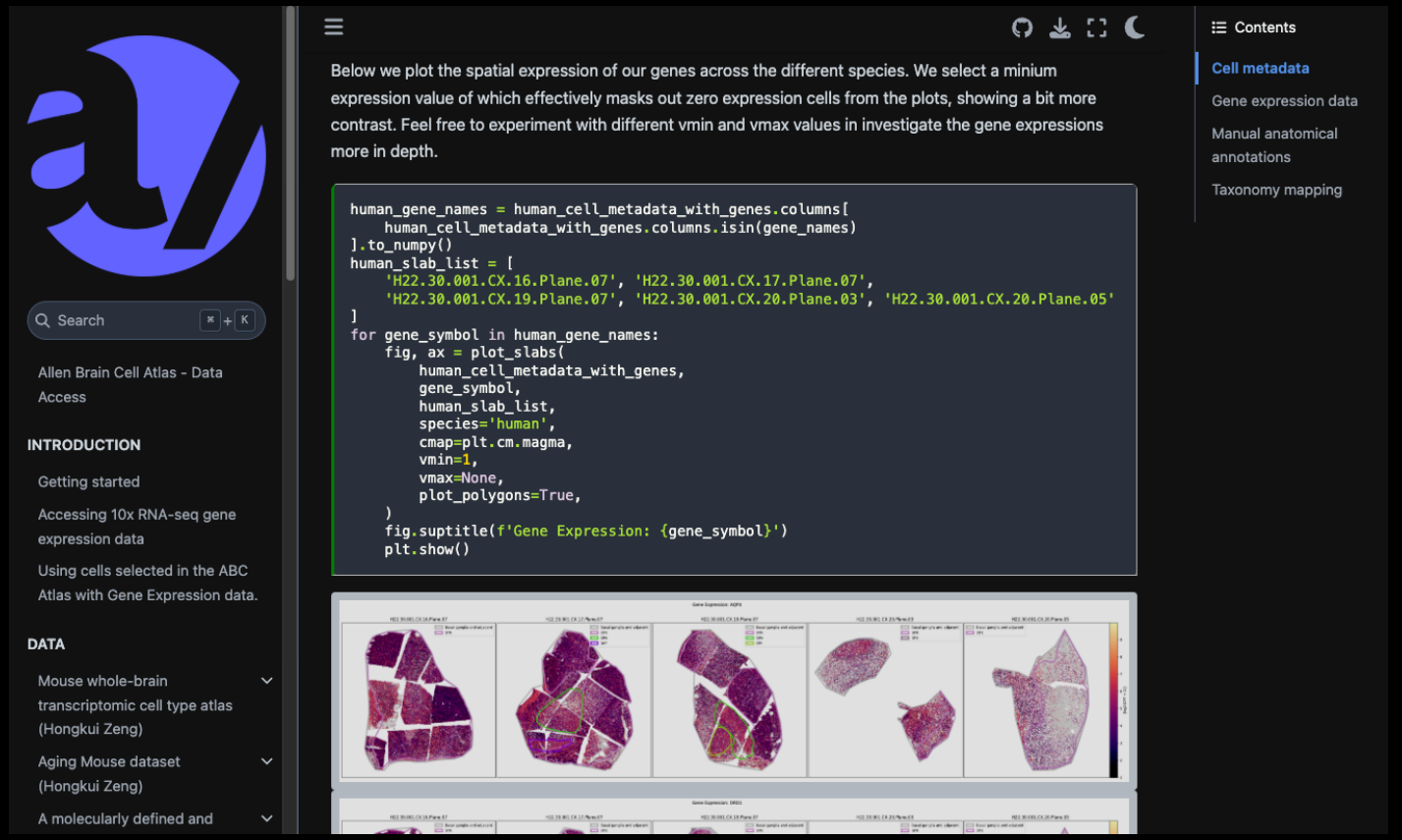
abc_atlas_acess

How do I create custom differential gene expression analysis?

How do I create custom visualizations using Python?

The Allen Institute GitHub repo provides an overview of the available data and how to download and use it through example use cases.

- Download full and partitioned data sets
- Access data definitions and schemas
- Tutorial Python notebooks with example queries



Data associated with the ABC Atlas is hosted in an S3 bucket as an AWS Public Dataset: arn:aws:s3:::allen-brain-cell-atlas. No account or login is required for access.

community use of BKP tools and data/

Title: Evolution of Cajal-Retzius Cells in Vertebrates from an Ancient Class of Tp73+ Neurons

Authors: Elias Gumnit[1], Giacomo Gattoni[1], Jamie Woych[1], Astrid Deryckere[1#], Prabesh Bhattarai[2,3], J Andrew Gillis[4], Caghan Kizil[2,3], Maria Antonietta Tosches[1*]

Next, we investigated whether the molecular and spatial heterogeneity of salamander *Tp73*+ neurons has counterparts in mammals. We isolated mouse *Tp73*+ neurons from the Allen Brain Cell (ABC) atlas (*34*) and compared them to the salamander *Tp73*+ neuron dataset. The

A unified analysis of atlas single-cell data

Hao Chen,[1,2] Nam D. Nguyen,[1] Matthew Ruffalo,[1] and Ziv Bar-Joseph[1,3]

To further test the ability of GIANT on integrating data from different modalities, we collected single-cell mouse brain data from the Allen Brain Cell Atlas (Yao et al. 2023), which contains 2,341,350 cells from scRNA-seq and 3,938,808 cells from MERFISH spatial transcriptomics (Chen et al. 2015). Both modali-

Morphine regulates astrocyte transcriptional dynamics in the ventral tegmental area by stimulation of glucocorticoid signaling

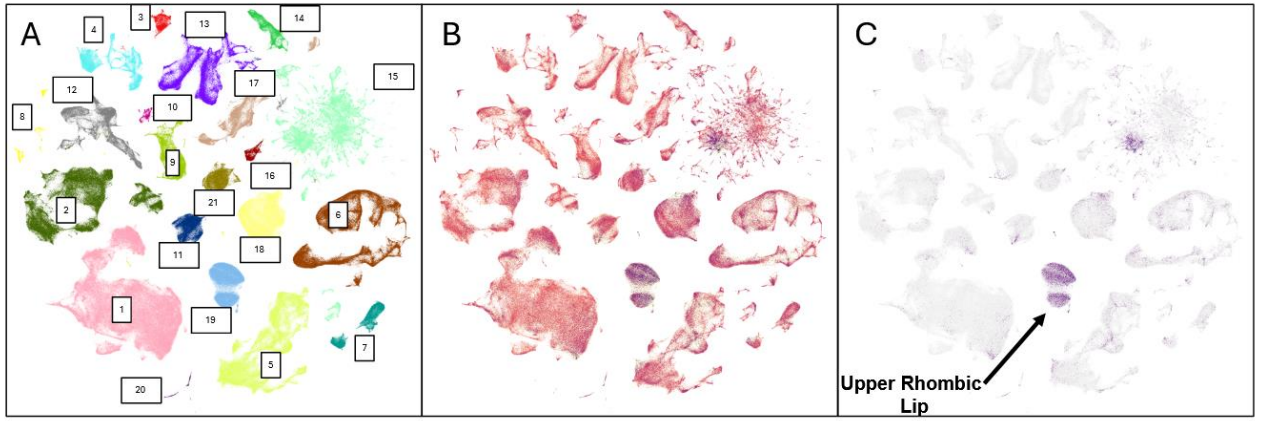
Jennifer J. Tuscher[1,3]*, Angela Cleere[1,3], Robert A. Phillips III[1,4], Catherine E. Newman[1], Guy Twa[1], Nathaniel J. Robinson[1], Lara Ianov[1], Robert E. Sorge[2], & Jeremy J. Day[1]*

clusters using lisi v1.0 in R[83]. To refine resolution of VTA neuronal subtypes and assign cells to classes, we followed the Allen Institute's Whole Mouse Brain taxonomy using the MapMyCells web interface[84] (RRID:SCR_024672). After removing additional low-

Review

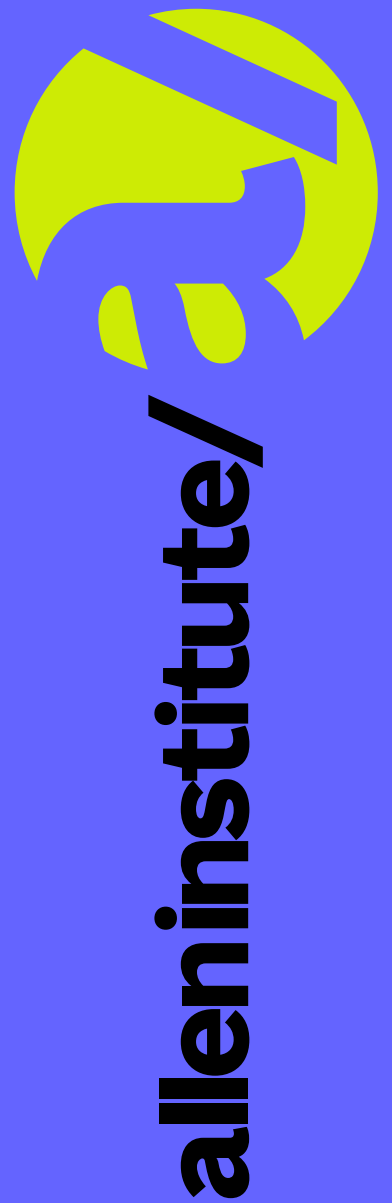
Using the Allen Brain Cell Atlas of the Human Brain to Gain Insights into C-Terminal-Binding Protein 1 (CtBP1)'s Potential Function

Suhjin Lee [1] and Uthayashanker R. Ezekiel [2,*]



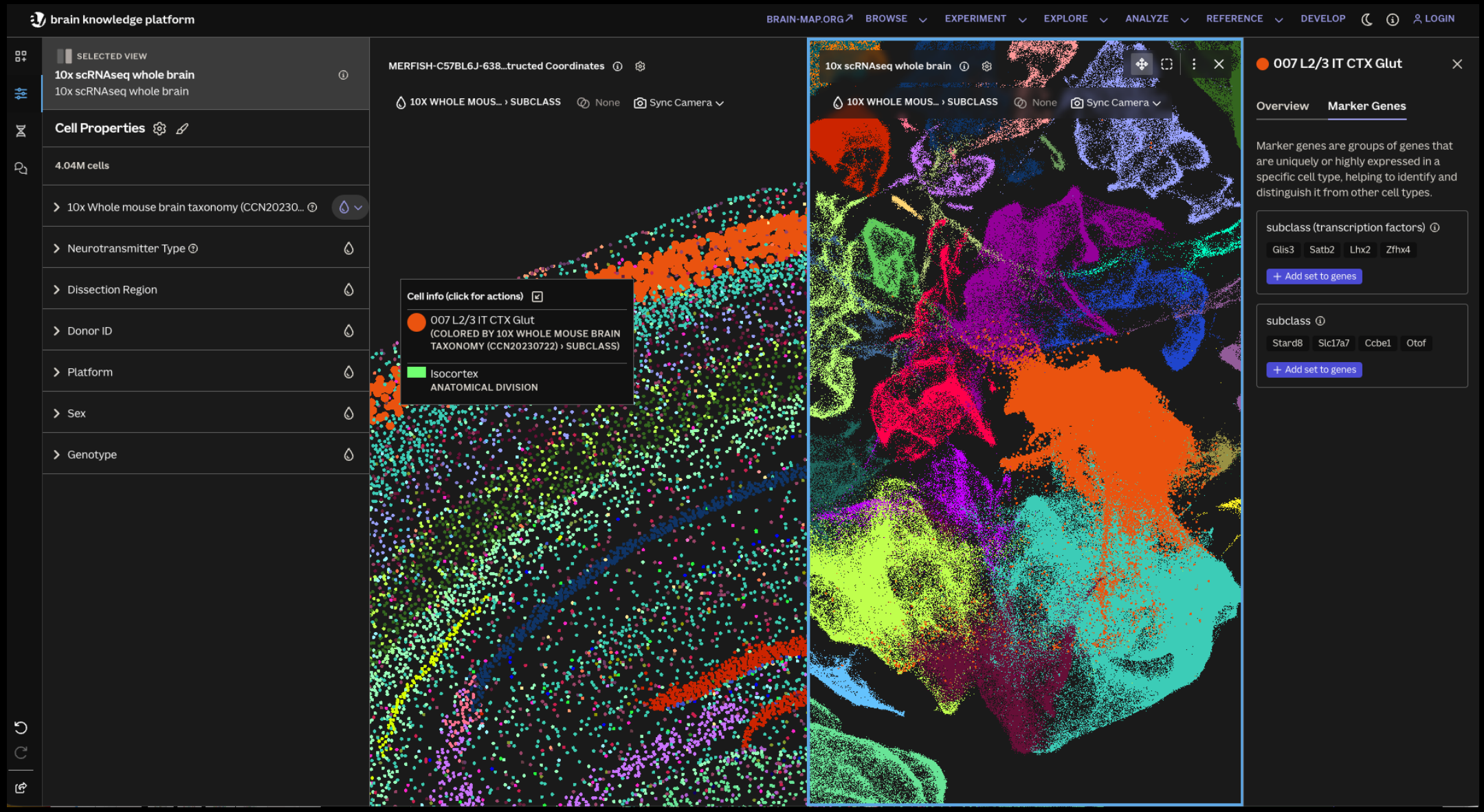
allen brain cell atlas demo/

Elysha Fiabane

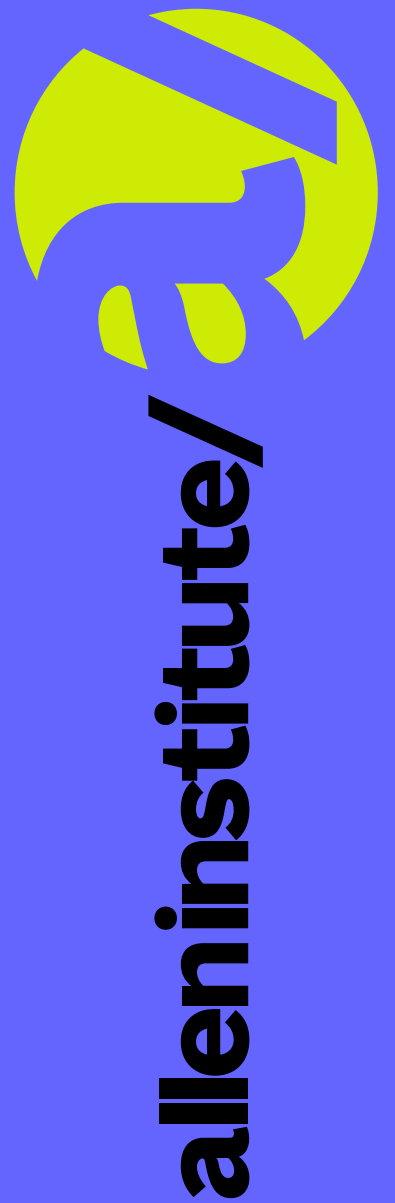


ABC Atlas demo/

https://brain-map.org/bkp/explore/abc-atlas



user exploration/



instructions/

- Go to **brain-map.org**
- Explore what interests you
- If you run into issues or have any questions, let us know
  - *(If we are busy, ask a neighbor or try something else, and find us later — there will be more opportunities to explore and seek help this afternoon!)*
- While you are exploring, think about what works well for you and what doesn't – we are always looking for ways to improve our site

(Session ends at 12:00)

lunch/we'll resume at 1:00

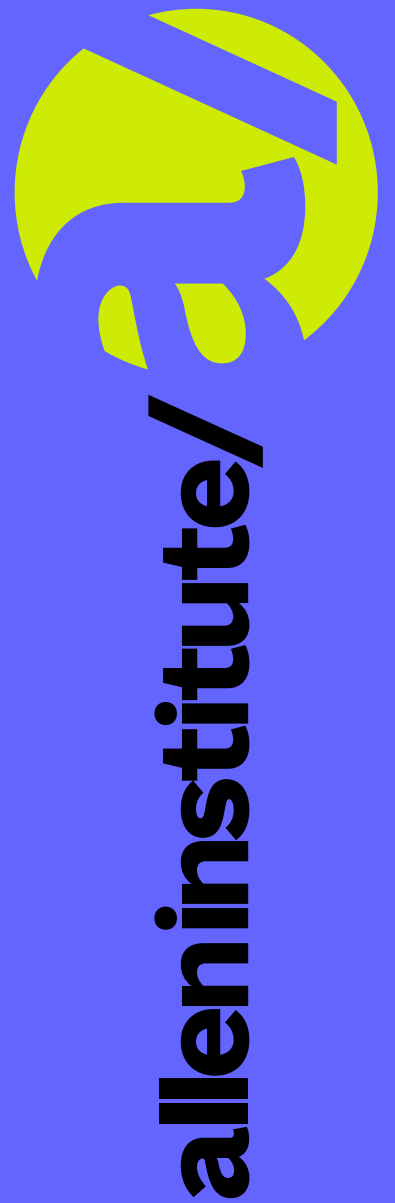


agenda/afternoon

- **12:00-1:00 | Lunch**
- **1:00-3:00 |** Breakout sessions
  - Knowledgebases and data repositories across the BRAIN initiative – Satra & Dorota
  - Computational access to Allen Institute data and knowledge – Elysha & Jeremy
- **3:00-3:15 |** Break
- **3:15-4:00 |** Coming soon / feedback / next steps

  - - - - -
- **4:00-5:00 |** User exploration of Allen Institute tools
- **5:00-6:00 |** Networking reception

breakout sessions/



computational access to allen institute content/



Chris Morrison



Scott Daniel

The python workflows to be presented today were created by **Chris Morrison** (ABC Atlas Access) and **Scott Daniel** (MapMyCells). *Scott will be available remotely to address any questions we can't.*

alleninstitute/

agenda/offline access to Allen Institute data

- **1:00-1:10 |** Prepping for computational analysis
  - *Finding the data and setting up your environments*

- **1:10-1:30 |** Overview of ABC Atlas Access data sets and data structure
- **1:30-1:45 |** Walk-though of whole mouse brain notebook
- **1:45-2:15 |** ABC Atlas Access Q&A / user exploration

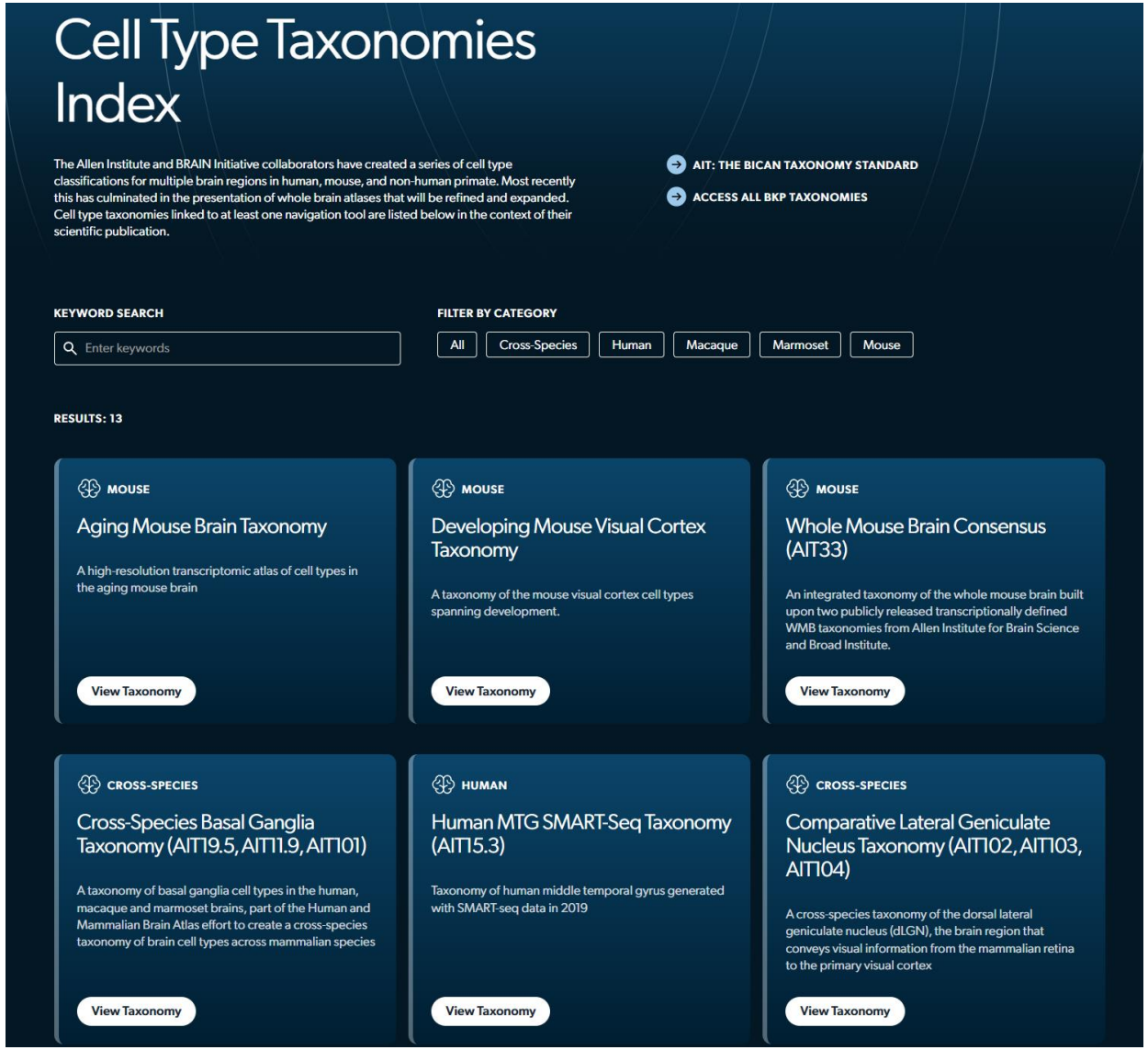
- **2:15-2:45 |** (Transcriptomic) user data in the context of Allen Brain Map
  - *Building your own cell type taxonomy*
  - *Mapping your data to cell type taxonomies on Allen Brain Map*
- **2:45-3:00 |** User data Q&A

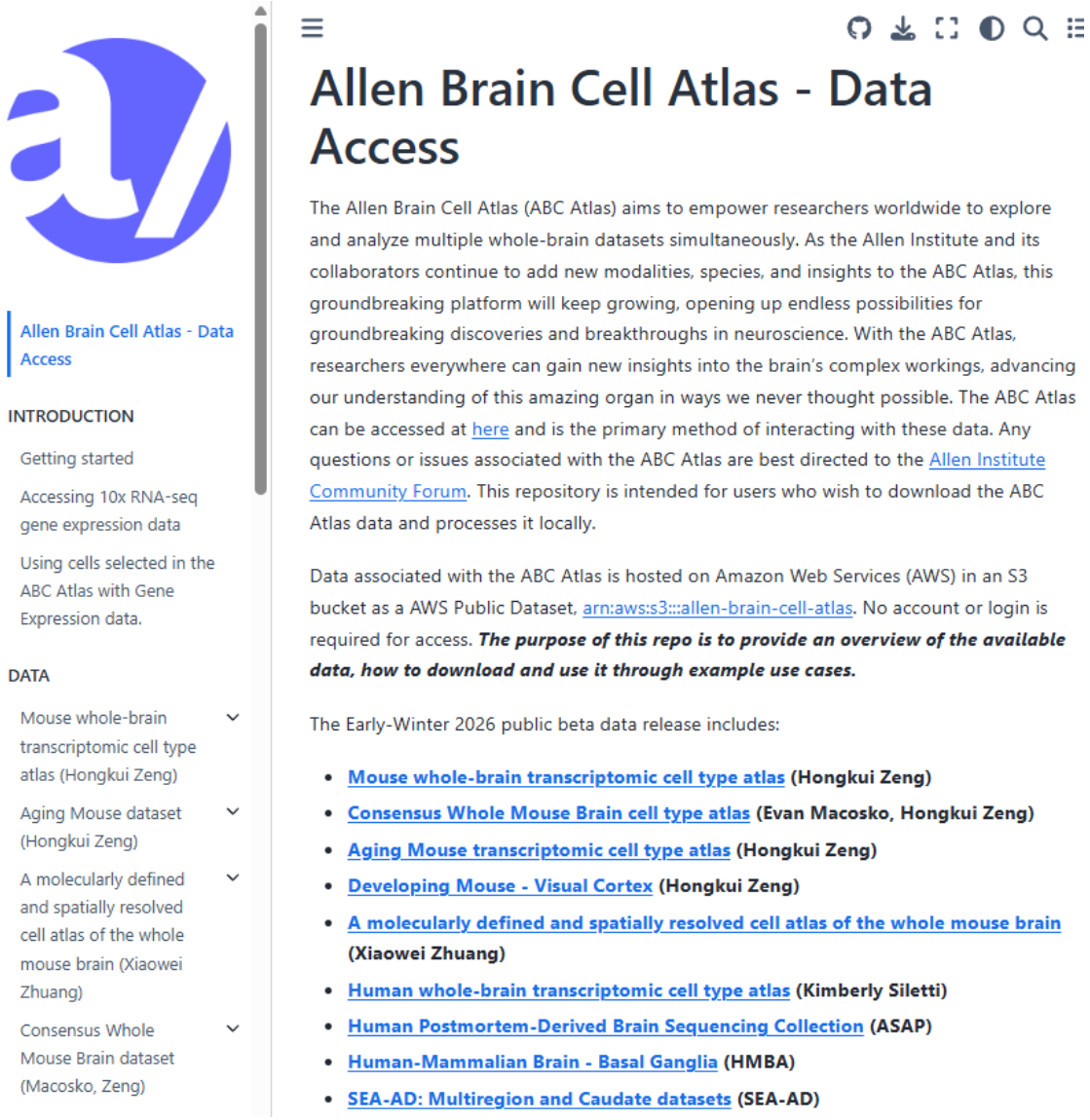
prepping for analysis/



alleninstitute/

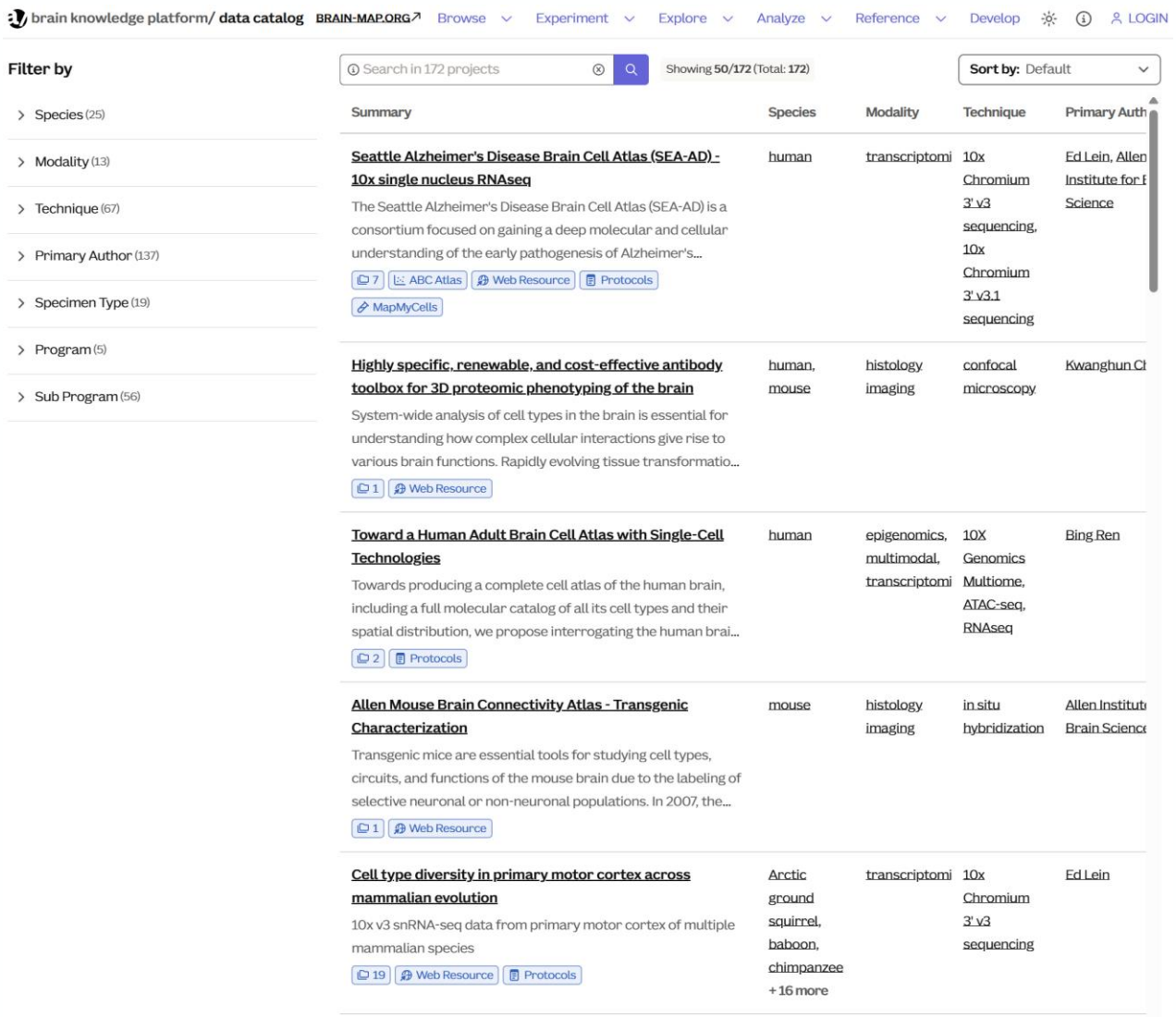
find allen brain map data/single cell + spatial omics



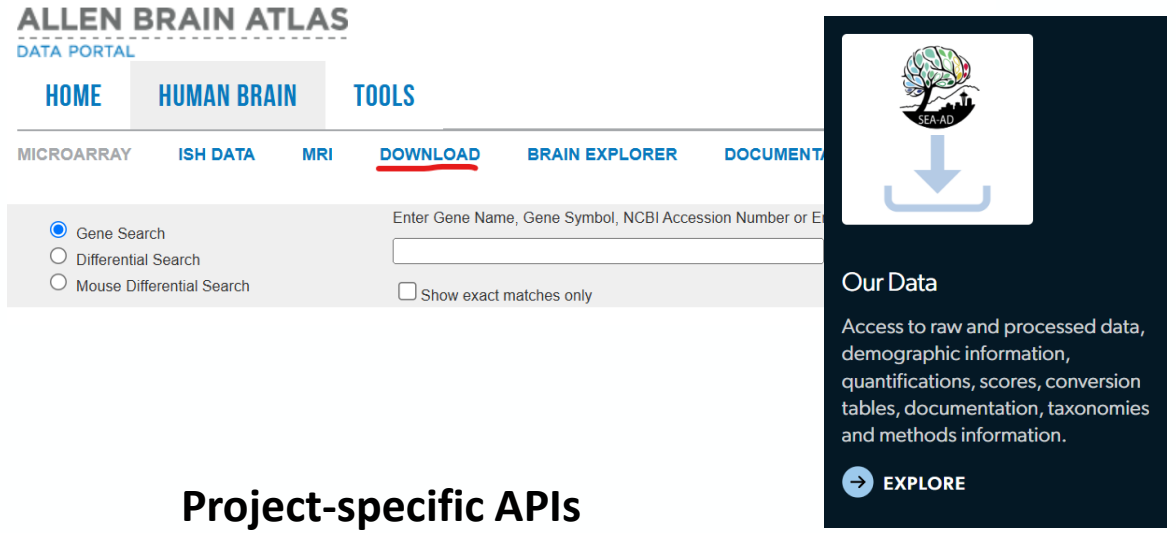


find allen brain map data/other data

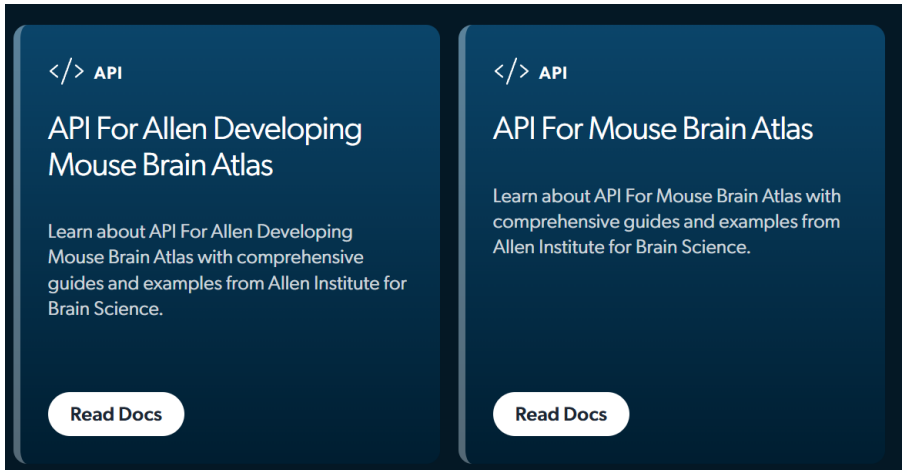
BKP data catalog



Project-specific download pages

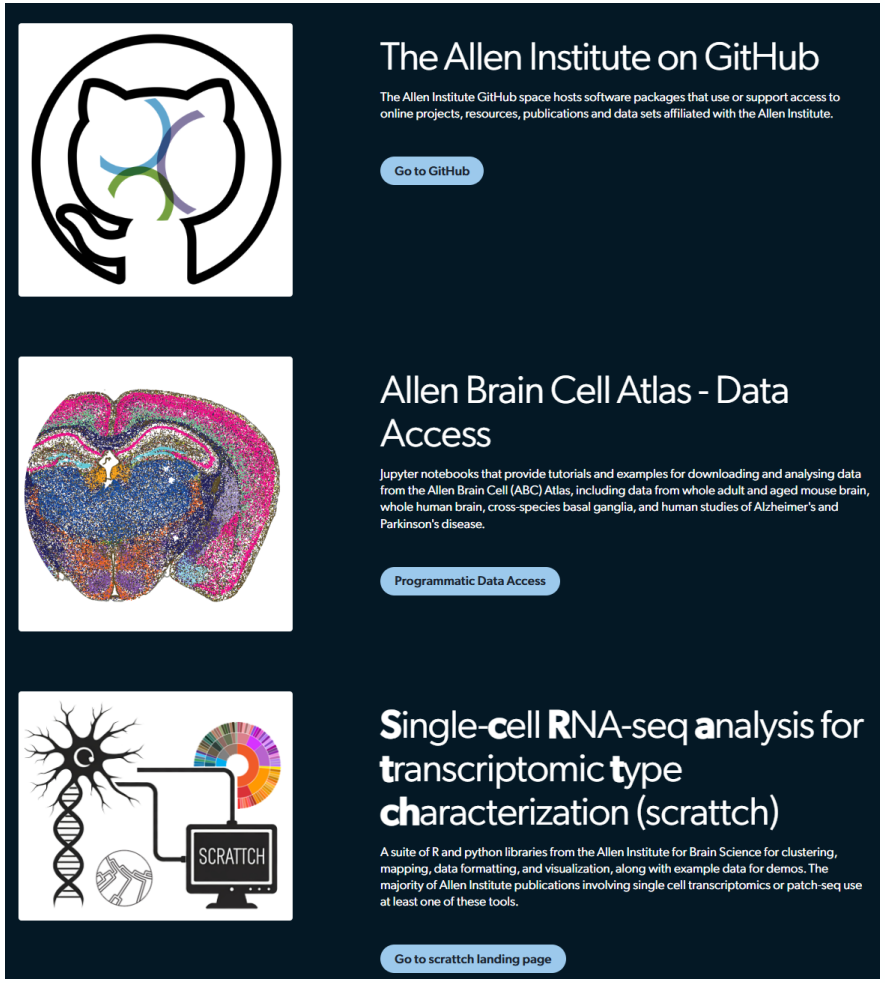


Project-specific APIs

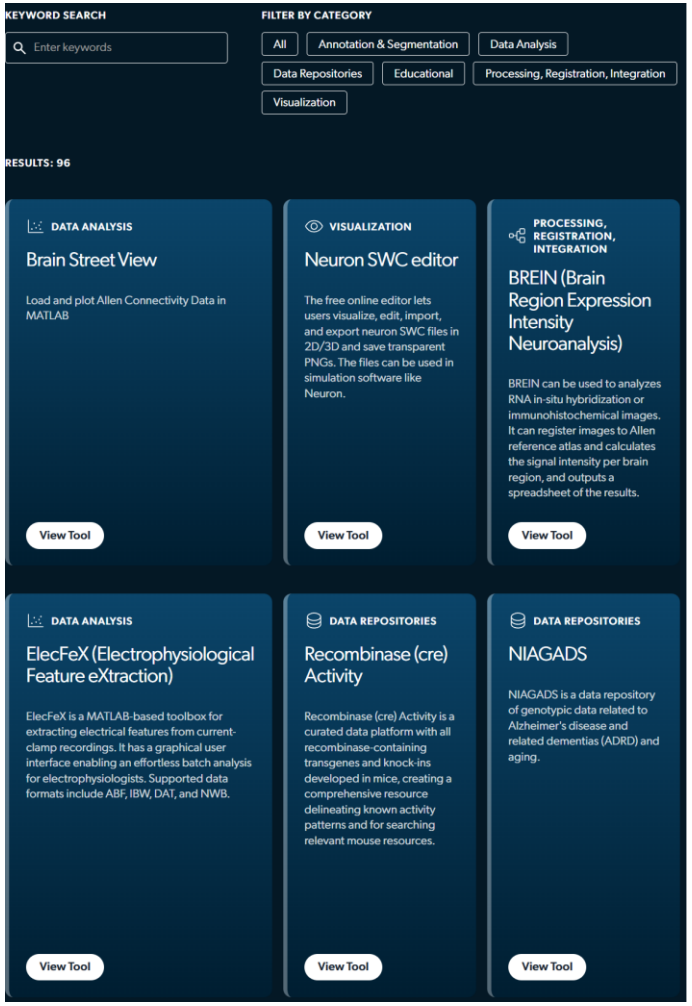


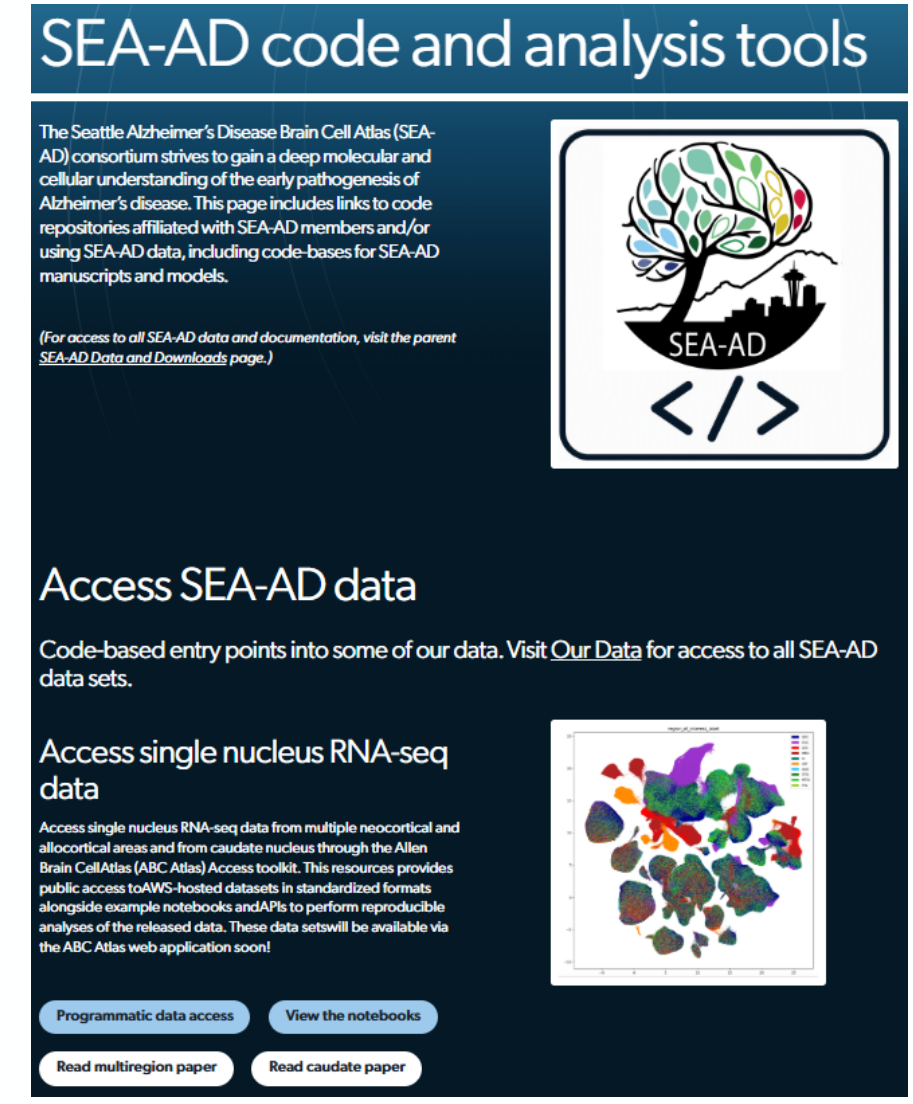
run allen brain map code on your own/

Computational Analysis Toolbox



Community Tools





setting up your python environment/

- This is beyond the scope of this workshop…

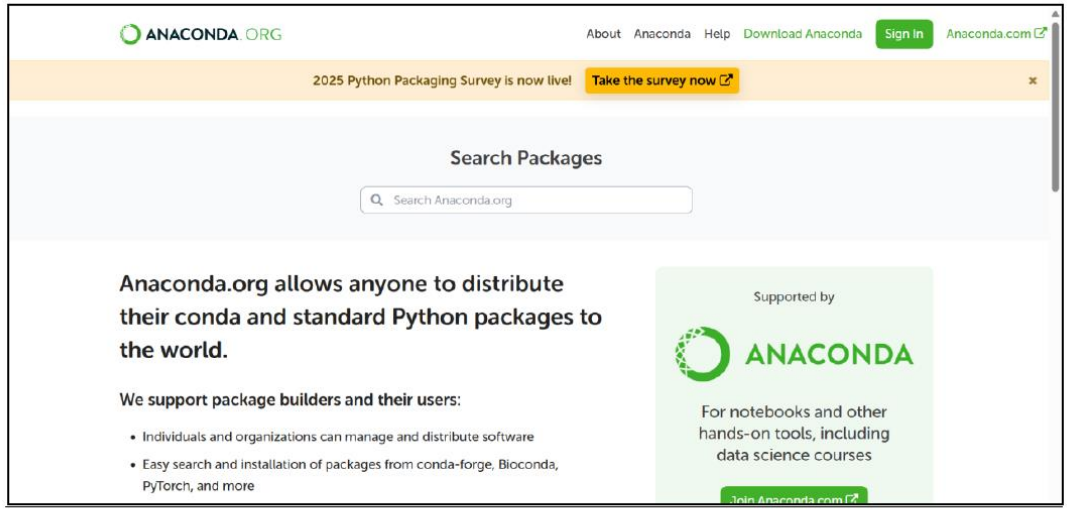
- However, we have developed an ABC Atlas Access Installation Guide (Windows), which describes how to use Anaconda and GitHub Desktop to install and Run ABC Atlas Access on a Windows machine
- (From what I have been told, usage on Mac and Unix is more straightforward)

Atlas Access Installation Guide (Windows) – Anaconda & GitHub Desktop

This guide goes through the steps of installing Anaconda (pages 1-5) and GitHub Desktop (page 6) on a Windows machine, for the purpose of accessing Allen Institute ABC Atlas Access Jupyter Notebooks (pages 7-11). Anaconda is an open-source library management system used to create and manage virtual environments (Anaconda is one of many options for this step). GitHub Desktop is used to access repositories on GitHub.

Installing Anaconda on Windows

Step 1: Go to http://www.anaconda.org and click "Download Anaconda."



Step 2: Choose if you want to register or not by clicking "Get Started" or "skip registration."

…

https://github.com/AllenInstitute/spoctt ➡ / ABCAtlas_UseCases / ABC_Atlas_Access_Installation_Guide_(Windows).pdf

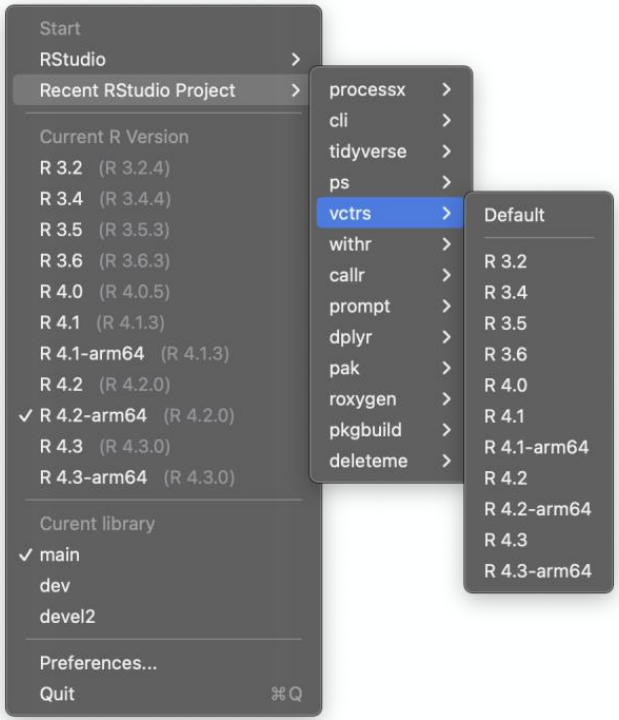
setting up your R environment/rig + renv

- This is also beyond the scope of this workshop…
- However, some guidelines:



rig

- Rig for R (and Rtools) versioning



https://github.com/r-lib/rig

+

renv

- Renv for R package management

+

GitHub

- GitHub for reproducible code
- *(Also good practice for python!)*

https://rstudio.github.io/renv

setting up your environment/docker

- **What is Docker:**
  - Docker packages software + all dependencies into a portable unit called an image, which you run as a container (a lightweight, isolated runtime).
  - Essentially, Docker replaces the rig+renv steps on the previous slide
- **When to use Docker:**
  - Some Allen Institute code (e.g., scrattch) require specific environments to run, which have been pre-built
- **How to use Docker:**
  - Pull an image from Docker Hub (e.g., https://hub.docker.com/r/alleninst/scrattch)
  - Load image in Docker Desktop (Windows) or use the terminal (any OS)
  - Start a container and then run R normally
- **Challenges with using Docker:**
  - Getting docker working is not user-friendly and often takes trial and error
  - (I probably can't help with this step today, unfortunately ☹).

ABC Atlas Access/

https://github.com/AllenInstitute/abc_atlas_access/

ABC atlas access/

- **ABC Atlas Access files and Jupyter notebooks provides a standard entry point into multiple single cell and spatial transcriptomics data sets at the Allen Institute**

- Organized as a website describing the data and showing the notebooks
  - https://github.com/AllenInstitute/abc_atlas_access/

- Includes a Python class AbcProjectCache that:
  - downloads data released through the ABC Atlas
  - tracks data release versions
  - tracks what files have been downloaded
  - verifies file integrity
  - provides an easier way of getting at data whether that be an anndata file, cell/gene metadata tables etc.

- Includes tutorial Jupyter notebooks that:
  - Walk users through how to use the cache object
  - Show example analyses that can be performed on the various data released in the ABC Atlas

https://alleninstitute.github.io/abc_atlas_access/intro.html

what's in the ABC Atlas/

mouse

- **Whole Mouse Brain** (scRNAseq and MERFISH data)
  - Yao et al., 2023, Zhang et al., 2023 – Nature
- **Consensus Whole Mouse Brain** (scRNAseq)
  - Yao et al., 2023, Langlieb et al., 2023 – Nature
- **Aging Mouse** (scRNAseq)
  - Jin et al., 2025 – Nature.
- **Developing Mouse Visual Cortex** (scRNAseq)
  - Gao et al., 2025 – Nature

human & primate

- **Whole Human Brain** (snRNAseq)
  - Siletti et al., 2023 – Science
- **Human Alzheimer's Disease** – MTG and DLPFC (SEA-AD) (snRNAseq and MERFISH data)
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  - (No publication—collaboration with MJFF/ASAP)
- **Cross-species HMBA Basal Ganglia** (snRNAseq and MERFISH data)
  - Johansen & Fu et al, 2026; Hewitt & Turner et al, 2026 - bioRxiv

https://alleninstitute.github.io/abc_atlas_access/intro.html

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- **Aging Mouse** (scRNAseq)
  - Jin et al., 2025 – Nature.
- **Developing Mouse Visual Cortex** (scRNAseq)
  - Gao et al., 2025 – Nature

human & primate

- **Whole Human Brain** (snRNAseq)
  - Siletti et al., 2023 – Science
- **Human Alzheimer's Disease** – Multiregion and caudate nucleus (SEA-AD) (snRNAseq)
  - Travaglini, Gabitto, Ding & et al., 2026 – bioRxiv
  - Kana et al 2026 - bioRXiv
- **Human Parkinson's Disease** (ASAP Human Postmortem BrainSeq) (snRNAseq)
  - (No publication—collaboration with MJFF/ASAP)
- **Cross-species HMBA Basal Ganglia** (snRNAseq and MERFISH data)
  - Johansen & Fu et al, 2026; Hewitt & Turner et al, 2026 - bioRxiv

For SEA-AD, coming soon:

- *The MTG data will be accessible via external Jupyter notebooks*
- *The Multiregion and caudate nucleus data will be in the ABC Atlas*

https://alleninstitute.github.io/abc_atlas_access/intro.html

new SEA-AD data sets available via ABC atlas access/

SEA-AD Multiregion taxonomy and snRNASeq analysis: clustering and annotations

Alzheimer's disease (AD) is characterized by the progressive accumulation of amyloid-beta (Aβ) plaques and hyperphosphorylated tau (pTau) tangles across brain regions. While these regions differ in architecture, function, and susceptibility to pathology, many share common cellular populations.

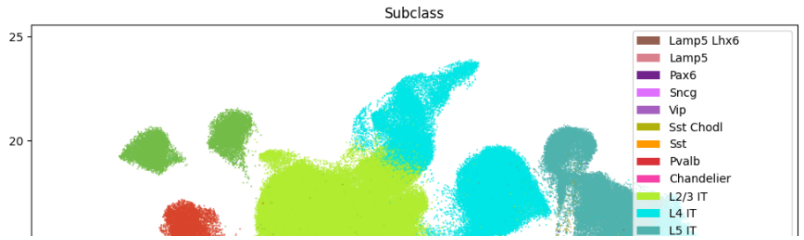
In [10]:
```
clinical = abc_cache.get_metadata_dataframe(
    directory='SEA-AD-Multiregion-10X',
    file_name='clinical'
).set_index('donor_label')
clinical
```

```
clinical.csv: 100%|██████████| 13.3k/13.3k [00:00<00:00, 136kMB/s]
```

Out[10]:

donor_label	Overall AD neuropathological Change	Thal	Braak	CERAD score	Overall CAA Score	Highest Lewy Body Disease	Total Microinfarcts (not observed grossly)	Total microinfarcts in screening sections	Atherosclerosis	Arteriol
H19.33.004	Not AD	Thal 0	Braak IV	Absent	Not identified	Not Identified (olfactory bulb not assessed)	1	1	Mild	
H20.33.001	Low	Thal 2	Braak IV	Sparse	Not identified	Not Identified (olfactory bulb not assessed)	0	0	Mild	

In [39]:
```
fig, ax = plot_umap(
    plot_cell_extended['x'],
    plot_cell_extended['y'],
    cc=plot_cell_extended['Subclass_color'],
    labels=plot_cell_extended['Subclass'],
    term_orders=plot_cell_extended['Subclass_order'],
    fig_width=12,
    fig_height=12
)
res = ax.set_title("Subclass")
plt.show()
```



SEA-AD CaH snRNASeq analysis: clustering and annotations

The SEA-AD CaH taxonomy represents is formed from a combination of taxons from the (HMBA-BG taxonomy)[HMBA-BG-taxonomy-CCN20250428.md] and the SEA-AD MTG Taxonomy. These 66 taxons are built into a new hierarchy with 16 subclasses, and 8 neighborhoods. Of 886,037 total nuclei, 686,439 pass the quality checks to be included in this analysis.

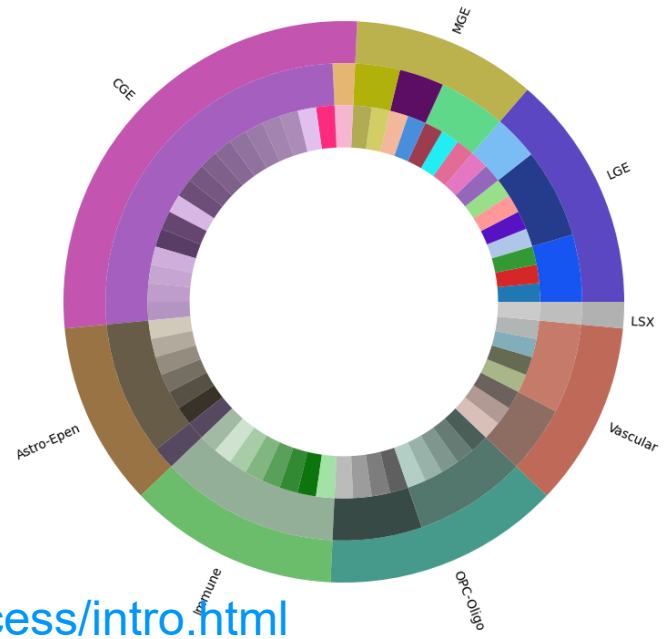
The notebook presented here demonstrates how to access and visualize taxonomy annotations, cell type metadata, donor information, brain region annotations, and AD-associated cellular changes from the SEA-AD CaH dataset. These examples focus on precomputed metadata accessible through the Allen Brain Cell Atlas Access package. Additional data, including links to raw sequencing, spatial transcriptomic, and neuropathology data, are available at SEA-AD.org.

Visualizing the SEA-AD CaH Taxonomy

Term sets: neighborhood, subclass, and supertype define the SEA-AD CaH taxonomy. We can visualize the taxonomy as a sunburst diagram that shows the single inheritance hierarchy through a series of rings, that are sliced for each annotation term. Each ring corresponds to a level in the hierarchy. We have ordered the rings so that the class level. Rings are divided based on their hierarchical relationship to the parent slice.

In [46]:
```
levels = ['Neighborhood', 'Subclass', 'Supertype']
df = {}

# Copy the term order of the parent into each of the level below it.
if term_with_counts.index.name != 'cluster_annotation_term_label':
    term_with_counts = term_with_counts.set_index('cluster_annotation_term_label')
```

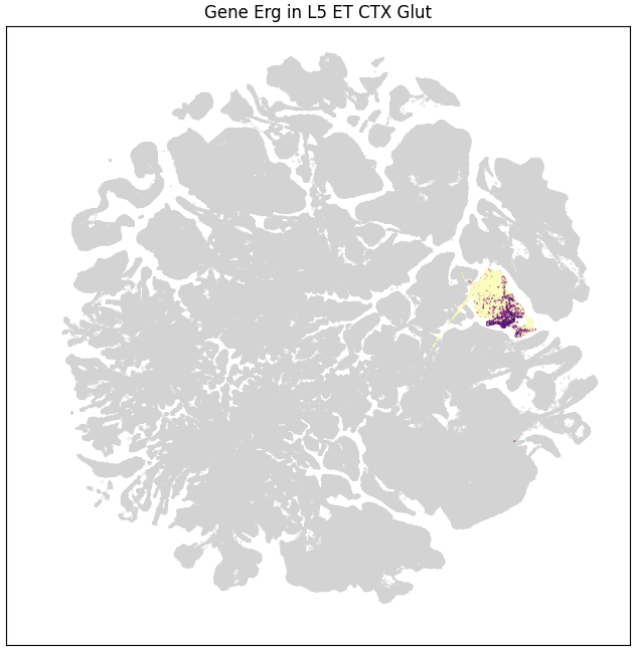


https://alleninstitute.github.io/abc_atlas_access/intro.html

What's in the Demo today

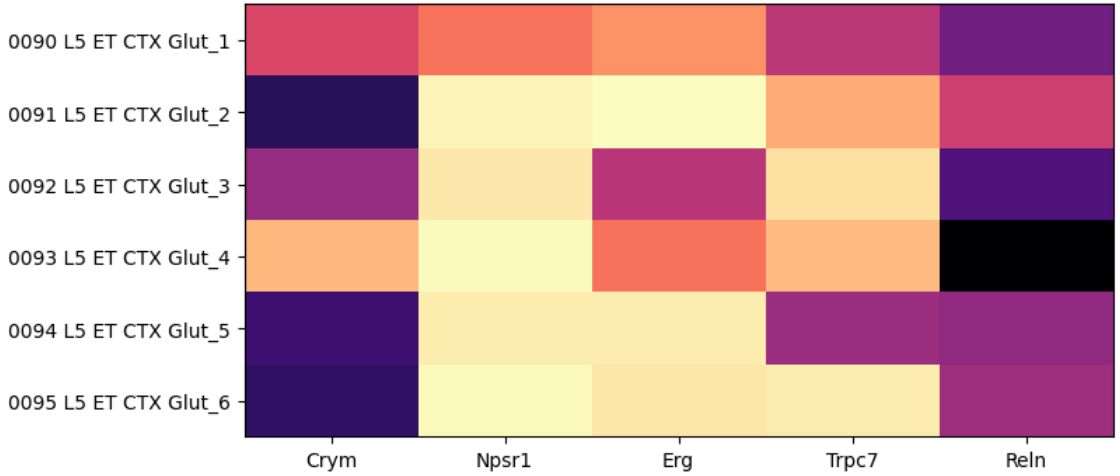
- Quick run through of the getting_started notebook.
  - Including how to setup the cache and download data.

- Processing of Whole Mouse Brain (WMB) 10X data
  - Extracting genes from the WMB-10X files



- Plotting celltypes and genes expressed in L5 ET CTX Glut in a UMAP

- Plotting gene expression in a heatmap ➔



https://alleninstitute.github.io/abc_atlas_access/intro.html

Helpful Links

- Python
  - Python/conda: https://docs.anaconda.com/anaconda/install/
  - Jupyter: https://jupyter.org/
  - Pandas: https://pandas.pydata.org/docs/
- abc_atlas_access:
  - Website
  - Github code repository
  - Notebooks
- Video tutorials
  - Webinar: https://www.youtube.com/watch?v=JyeyemlrDCw
  - Webinar Q&A: https://www.youtube.com/watch?v=VDZuPocFpnc
  - Workshop runthrough: https://www.youtube.com/watch?v=LfZCv7ZkACY&vl=en

- *(NOTE: these can all be found on brain-map.org… we'll also provide more links at the end)*

https://alleninstitute.github.io/abc_atlas_access/intro.html

Demo links

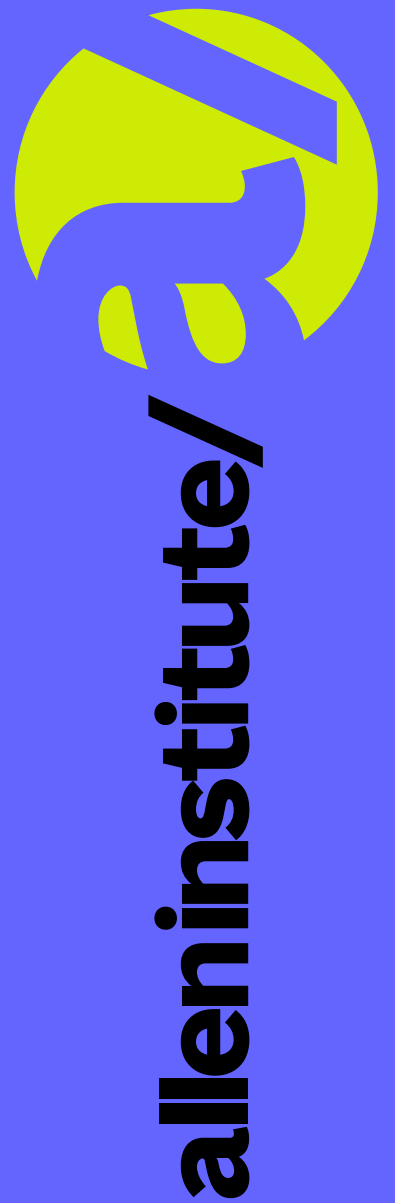
- Demo files on Github:
    - https://github.com/AllenInstitute/abc_atlas_access/tree/u/morriscb/taxonomyWorkshop/abc_atlas_workshop
- Demo files on Google Drive (w/ example genes)
    - https://drive.google.com/drive/folders/1MOJwh_AuwYwfA96GMInD_pxZMJK6Swfy?usp=drive_link

https://alleninstitute.github.io/abc_atlas_access/intro.html

any questions/

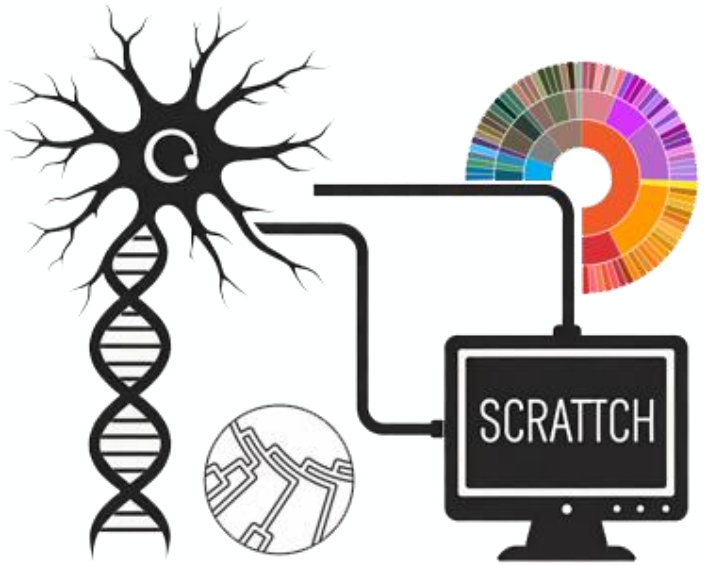


user data/transcriptomic analysis via scrattch



working with snRNA-seq data/

Single-**c**ell **R**NA-seq **a**nalysis for **t**ranscriptomic **t**ype **ch**aracterization (or "scrattch") is a suite of R and python scripts from the Allen Institute for Brain Science. The core R libraries are linked in the umbrella package called `scrattch`, which is modeled after the `tidyverse` package.



- Many cell type taxonomies discussed this workshop were developed using scrattch libraries at multiple analysis steps
- Patch-seq and spatial transcriptomics data sets were mapped to taxonomies using scrattch
- Lots of paper figures are generated using scrattch
- Scrattch was written by MANY folks at the Allen Institute, including Lucas Graybuck, Zizhen Yao, Nelson Johansen, Cindy van Velthoven, and Inkar Kapen

https://alleninstitute.github.io/scrattch/

the scrattch suite of R and python libraries/

Data preparation: file formats and schema

- `scrattch.taxonomy` - Taxonomy building scripts for RNA-seq based taxonomies following the Allen Institute (AIT) schema. A table of available AIT-formatted taxonomies **can be found here**.
- `scrattch.io` - [deprecated]. Library for file handling and data formatting, replaced by `scrattch.taxonomy` in 2024.

Data analysis: cell clustering and mapping (also called label transfer)

- `scrattch.hicat` - **H**ierarchical, **i**terative **c**lustering for **a**nalysis of **t**ranscriptomics
- `scrattch.bigcat` - Clustering analysis for extremely large single cell dataset
- `scrattch.mapping` - Generalized mapping scripts for single cell RNA-seq, Patch-seq, spatial transcriptomics, or related data types
- `scrattch.patchseq` - Functions for generating additional QC metrics and output files for patch-seq analysis

Data visualization

- `scrattch.vis` - Plotting functions for visualization of single cell RNA-seq data

Example data: small RNA-seq data sets

- `tasic2016data` - Data from Tasic, et al. (2016), which is used for demos
- `hodge2019data` - Data subset from Hodge, et al. (2019), which is used for demos
- A table of actual AIT-formatted taxonomies **can be found here**.

Python equivalent

scrattch_taxonomy_py

transcriptomic_clustering

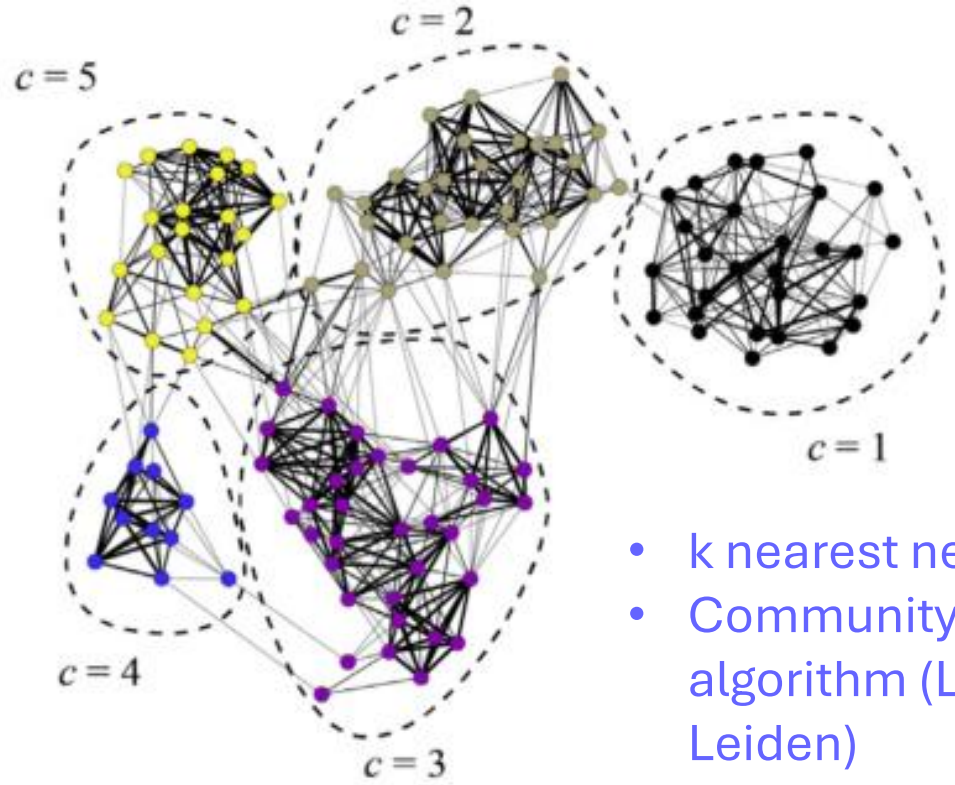
cell_type_mapper

- Scrattch includes parallel R and python packages for several topics
- **These packages require Docker** but you can do clustering without it

https://alleninstitute.github.io/scrattch/

iterative, graph-based clustering/

- Cells are first clustered at a coarse level to separate major cell classes (e.g., neurons, glia).
- Each cluster is then **recursively subdivided** into increasingly fine transcriptomic groups.
  - At each iteration, clustering is driven by genes that are most informative at that level of resolution.
- New subclusters are retained only if they are mutually separable by stringent differential gene expression criteria.



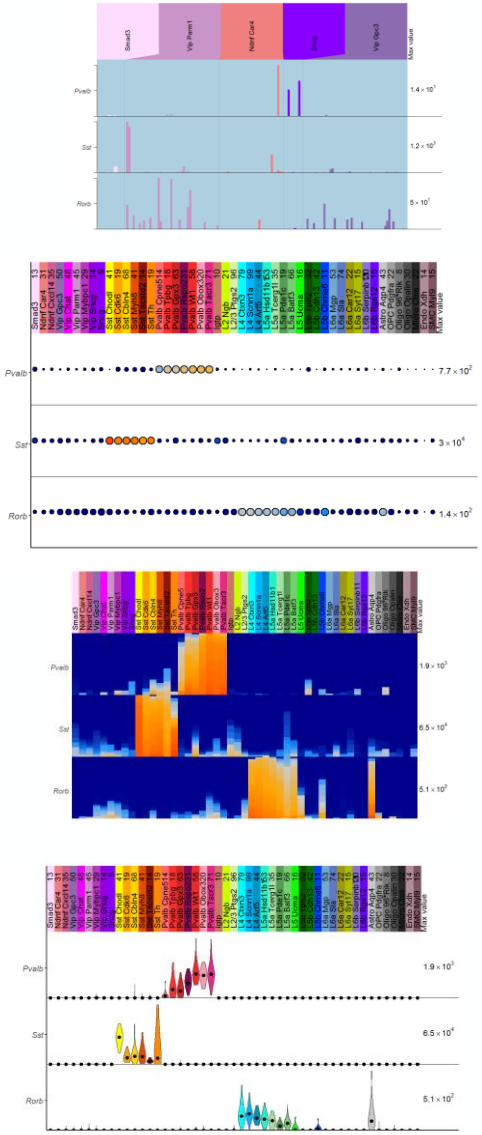
- k nearest neighbors
- Community detection algorithm (Louvain, Leiden)

- **Tutorials show you how do the clustering, whether or not you understand the details!**

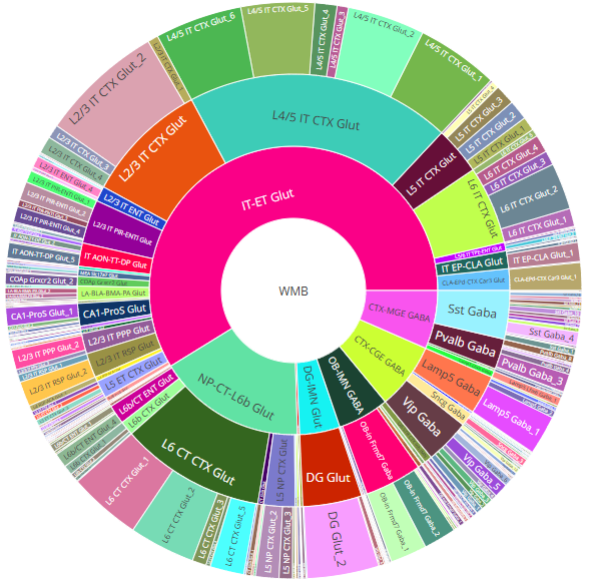
scrattch.hicat, scrattch.bigcat, transcriptomic_clustering

visualizing results/

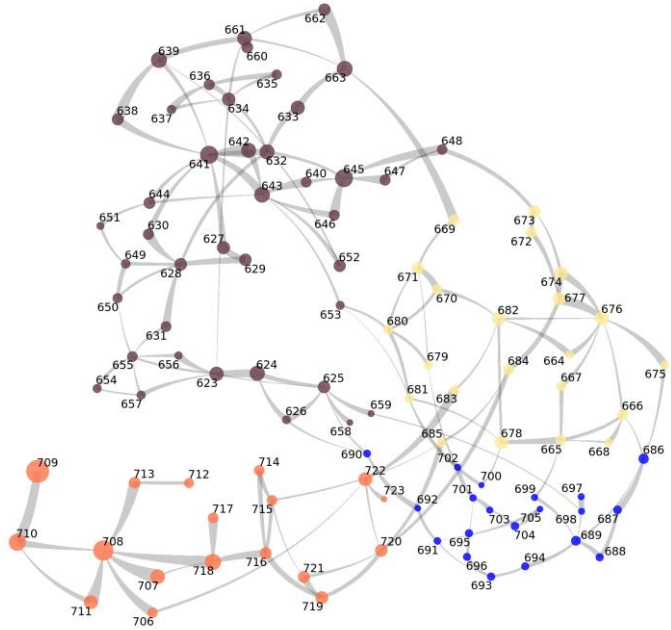
- **Gene summaries**



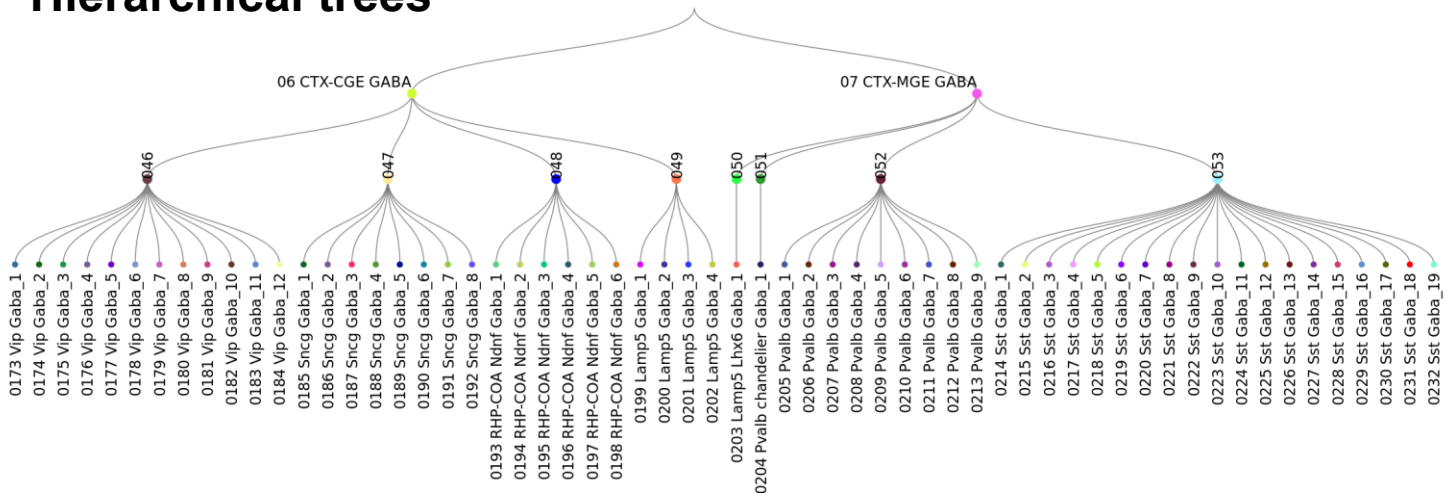
- **Sunburst plots**



- **Constellation diagrams**



- **Hierarchical trees**



scrattch.vis, scrattch.example

building a cell type taxonomy/

- **Cell type taxonomy**: all the data (optionally), metadata, clustering, annotations, and analyses stored in a tidy package, in a standardized format
  - We call that format “Allen Institute Taxonomy (AIT)”

scrattch.taxonomy, scrattch_taxonomy_py

building a cell type taxonomy/

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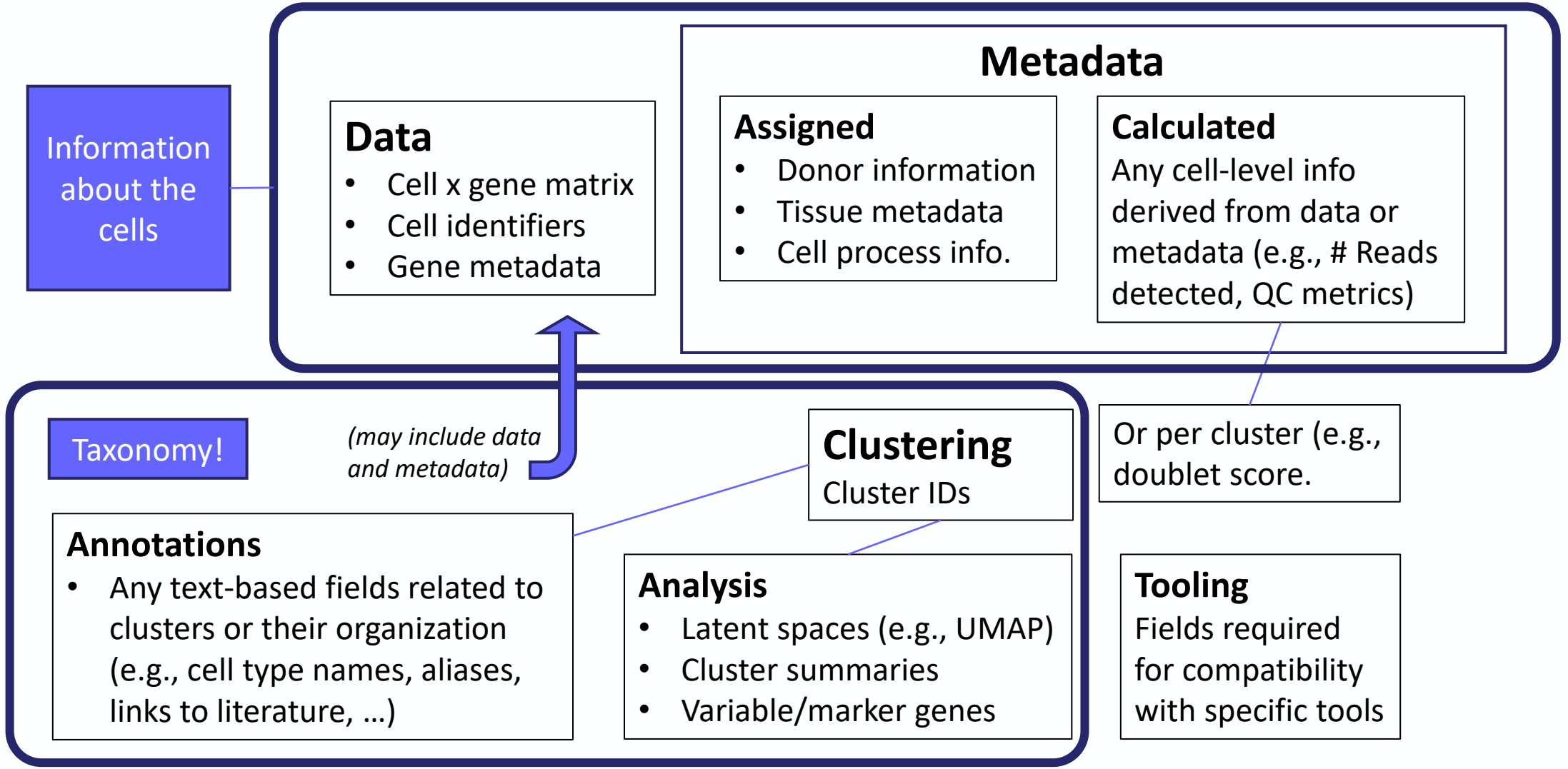
- **An .h5ad file** which encapsulates the essential components of a taxonomy required for downstream analysis including analysis, mapping and cell type annotation
  - Compatible with external schema and tools (e.g., CZI CELLxGENE)

- **A formalized schema** that includes controlled vocabulary for data and metadata fields in the .h5ad file
  - https://github.com/AllenInstitute/AllenInstituteTaxonomy

- **R (and python) code** to generate and work with .h5ad files following AIT format/schema
  - An environment (docker) for using the R code

scrattch.taxonomy, scrattch_taxonomy_py

contents of an AIT file/

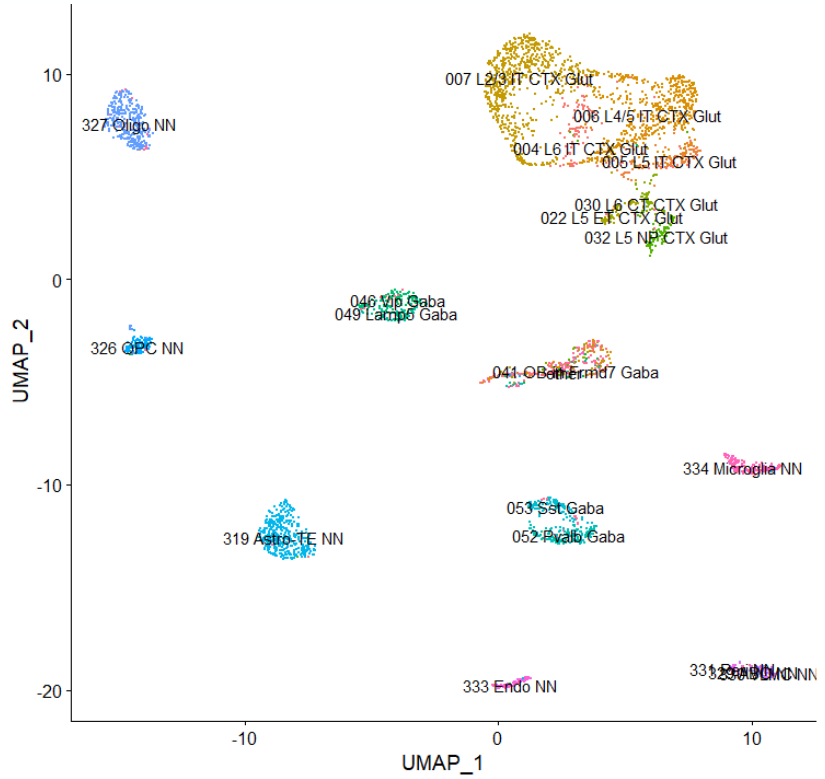


scrattch.taxonomy, scrattch_taxonomy_py

mapping query data to reference cell types/

- **Cell type mapping**: transferring cell type labels from a cell type taxonomy to user-provided (single cell or spatial) omics data.

Example output: Mouse M1



scrattch.mapping algorithms

- **Correlation mapping**: Map query cells directly to taxonomy clusters.
- **Tree Mapping**: Walk query cells down the taxonomy dendrogram.
- **Seurat Mapping**: Align taxonomy and query cells into an integrated latent space to annotate.
- **MapMyCells Hierarchical**: Walk query cells down a cell type hierarchy.
- **MapMyCells Flat**: Hierarchical mapping, but maps directly to leaf node of taxonomy.

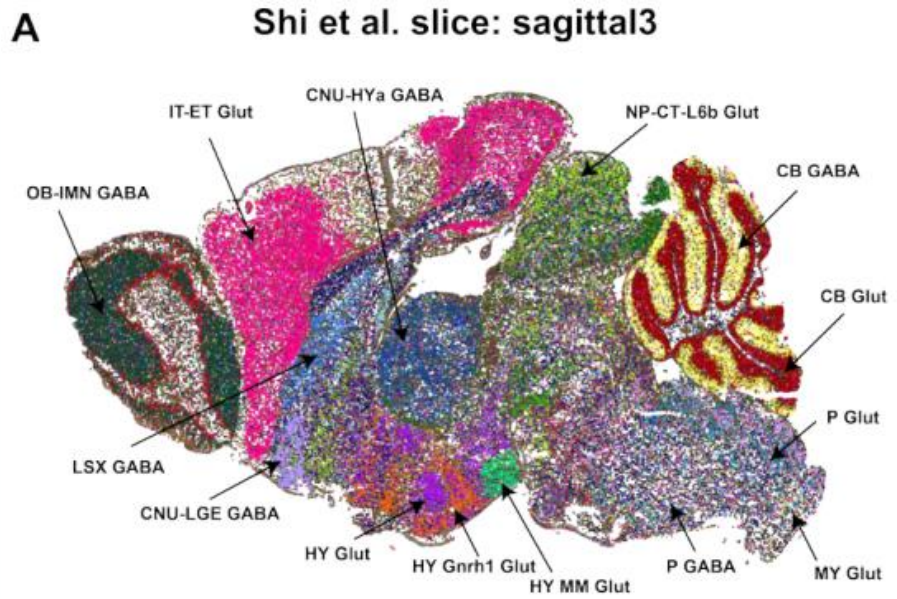
*

scrattch.mapping, cell_type_mapper*, MapMyCells*

cell_type_mapper/python version of MapMyCells

- The online version of MapMyCells is sufficient for many of the most common use cases
- scrattch.mapping extends this for use with R
- Both run cell_type_mapper on the back-end

- Jupyter notebooks exist for several use cases:
  - **Mapping to Allen Institute taxonomies on your own machine**: useful for pipeline implementation and for large files
  - **Mapping to a user-defined taxonomy**: useful for studying non-brain organ systems, other species, or new taxonomies
  - **Cross-species mapping**: this happens automatically in the on-line implementation but the notebooks show some examples
  - **Creating and mapping to a taxonomy defined from a subset of the Allen Institute's data:** useful when focused on a single brain region or with patch-seq data (to exclude glia)



Example mapping of data from an external study to mouse whole brain taxonomy.
From Daniel et al, 2026, bioRxiv

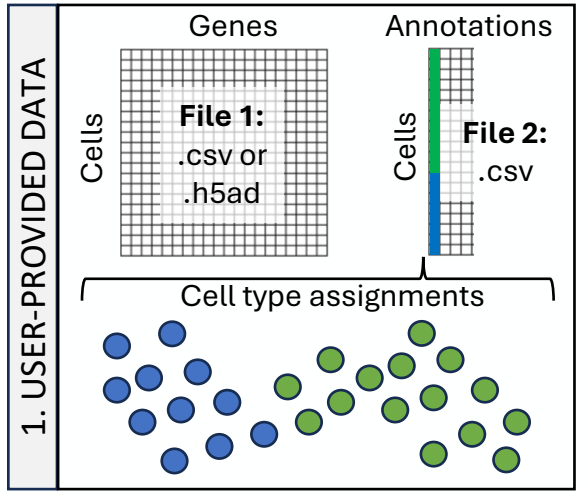
related tools/

- There are a few other tools related to single cell analysis which are listed on the scrattch landing page:
  - **mfishtools**: R and python packages for building combinatorial marker gene panels for spatial transcriptomics
  - **Cell HierARchy Gene Explorer (CHARGE):** Interactive web tool for identifying genes of potential biological interest in a cell type taxonomy
    - *Backend files can be built from any AIT file via the 'charge_taxonomy' repo*
  - **Annotation Comparison Explorer (ACE):** Web tool and R shiny app for comparison of annotations such as (i) cell type assignments, (ii) donor metadata (e.g., donor, sex, age), and (iii) cell metadata (e.g., anatomic location, QC metrics)

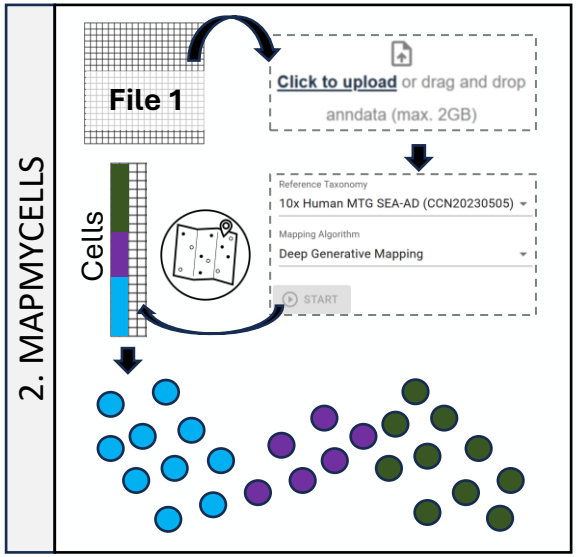
Use case: comparing mapping and clustering results

use case/aligning user cell type names

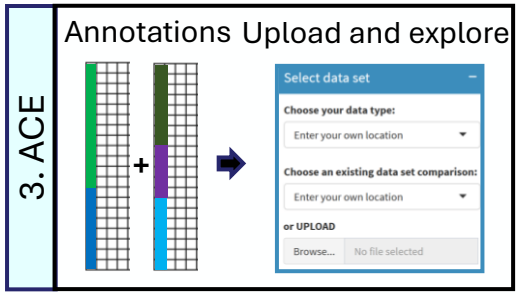
- **Set-up**: I have created my own cell type taxonomy and want to link it to a reference taxonomy to utilize Allen Brain Map tools and knowledge. How?



1. Get a single cell data set with assigned cell types (and other metadata)



2. Run MapMyCells on same dataset and download results



3. Join cluster and mapping tables and upload to ACE.

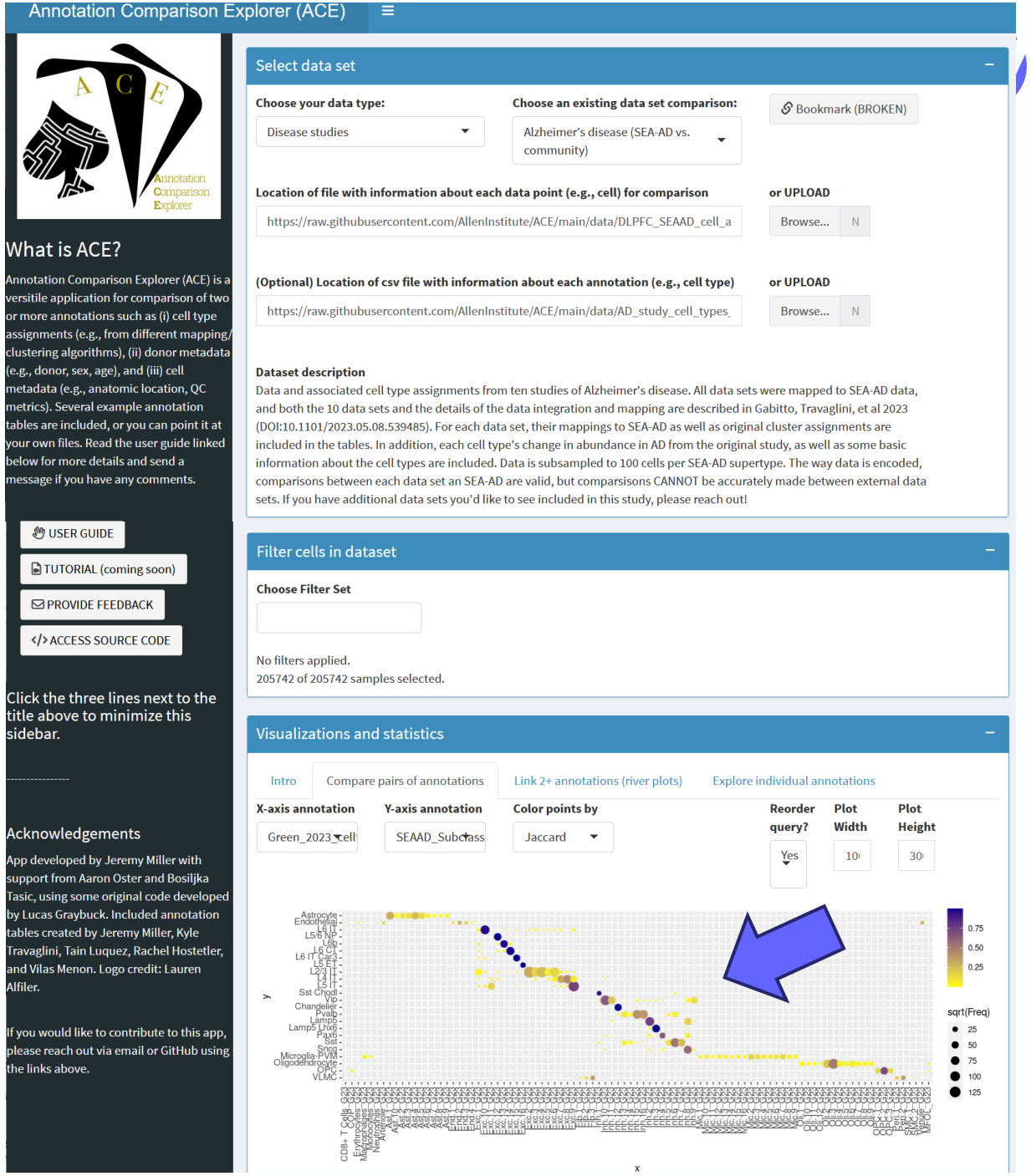
ACE: Annotation Comparison Explorer

- ACE is an interactive tool for this use case (and others)

- ACE must be run in three sequential steps separated on the app:

1. **Select a data set** for which annotations will be compared. These can be preselected or user provided.
2. **Filter cells** in dataset to include only annotations of interest (e.g., all cells, neurons, microglia)
3. **Visualizations and statistics** for doing the actual comparisons

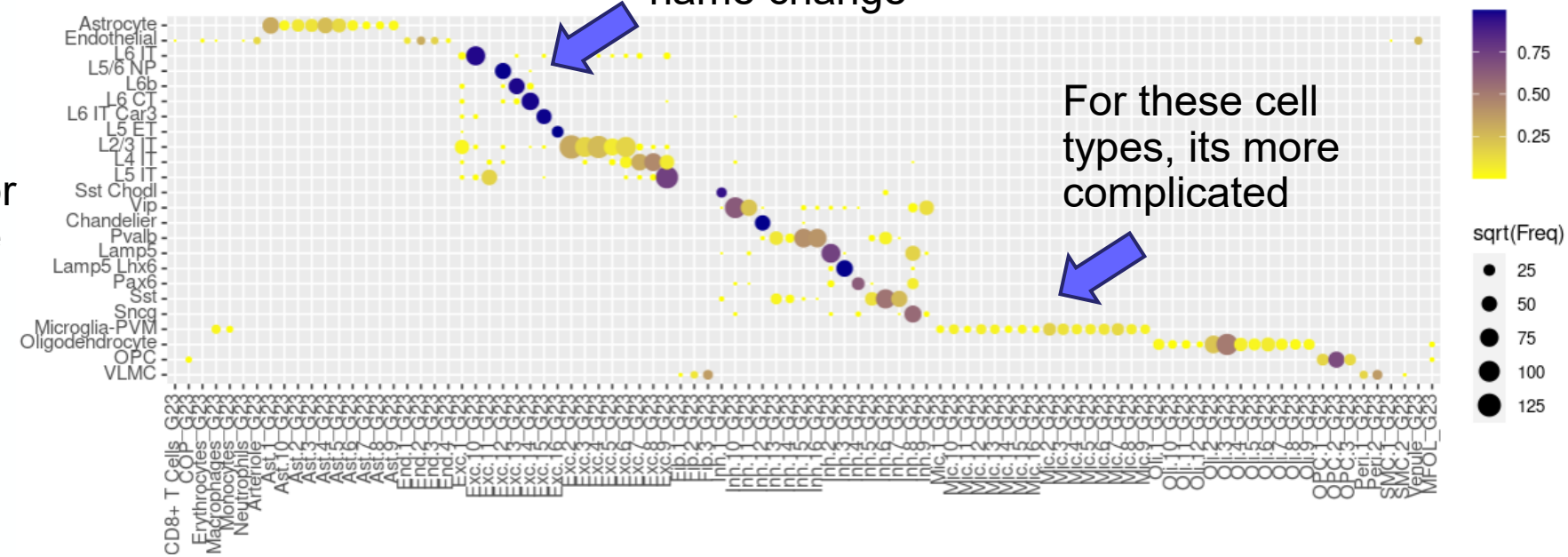
https://sea-ad.shinyapps.io/ACEapp/



example/ mapping Green et al 2023 data set

This is called a confusion matrix

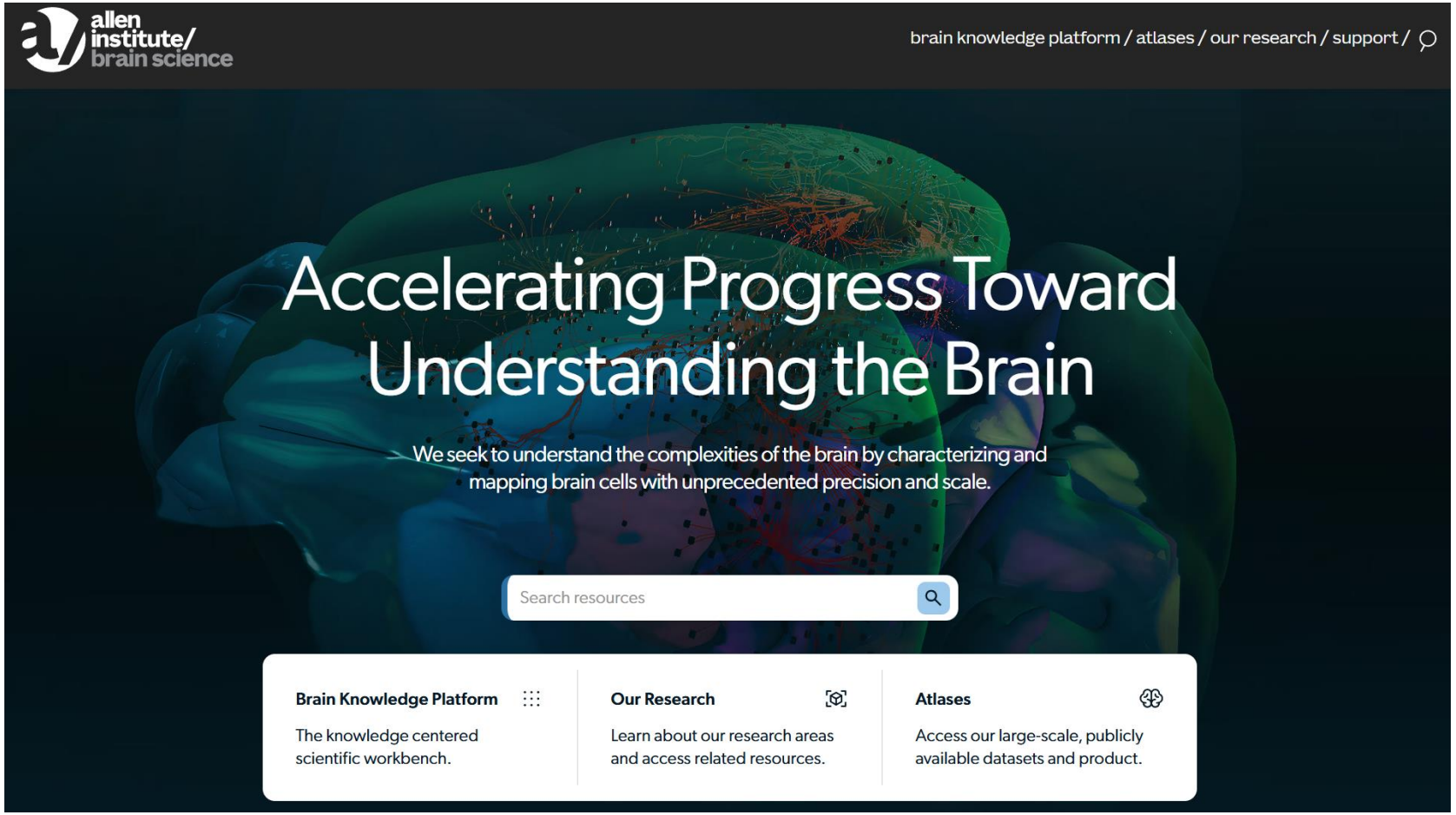
SEA-AD names for the same cells

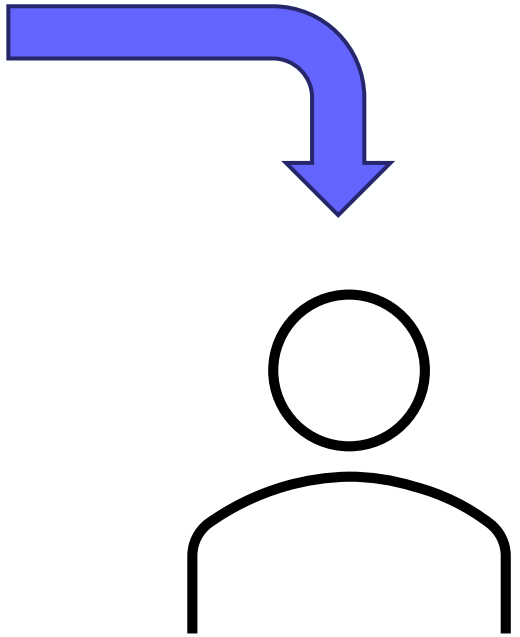


What Green et al 2023 called the cells

If we align on the same terminology, then any knowledge derived from one study can be immediately applied to other studies…

goal/bring our knowledge to your data





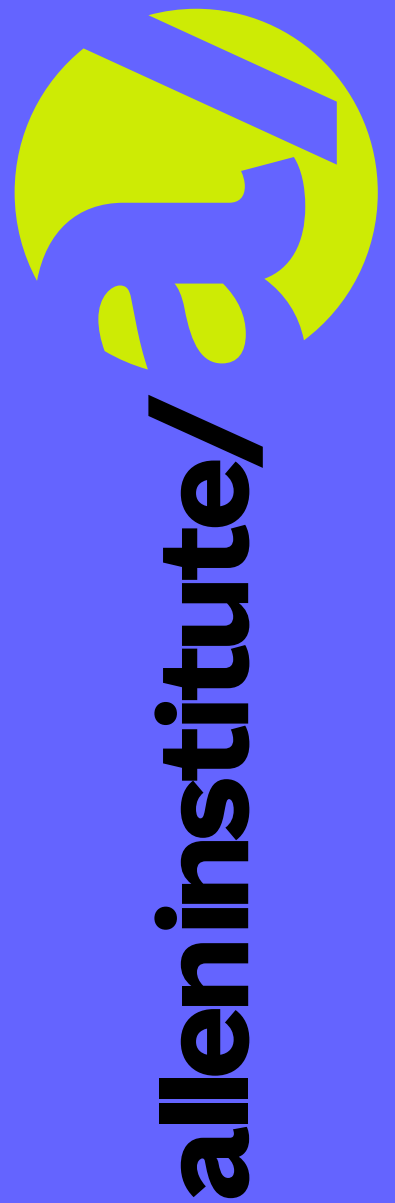
any questions/



break time/see you at 3:15



closing session/



agenda/

- **3:15-3:30 |** What's next in cell typing at the Allen Institute
  - *Coming soon on the ABC Atlas*
  - *Towards a cross-species whole brain atlas*

- **3:30-3:45 |** How to stay in touch
- **3:45-4:00 |** Feedback survey
  - *PLEASE fill out this survey! It helps us to fund and to improve future workshops.*

- **4:00-5:00 |** User exploration of Allen Institute tools
- **5:00-6:00 |** Networking reception

what's next?/new data, references, and capabilities

Share your feedback and ideas!

More Data

- **Connectivity:** EM, bar-seq, whole-brain lightsheet, expansion lightsheet
- **Developing** mouse: gene expression, spatial
- **Whole-brain human & NHP** spatial, single-cell, patch-seq
- **Disease**: SEA-AD, Parkinson's, and more
- **Behavior**: in vivo physiology from Neural Dynamics

Updated References

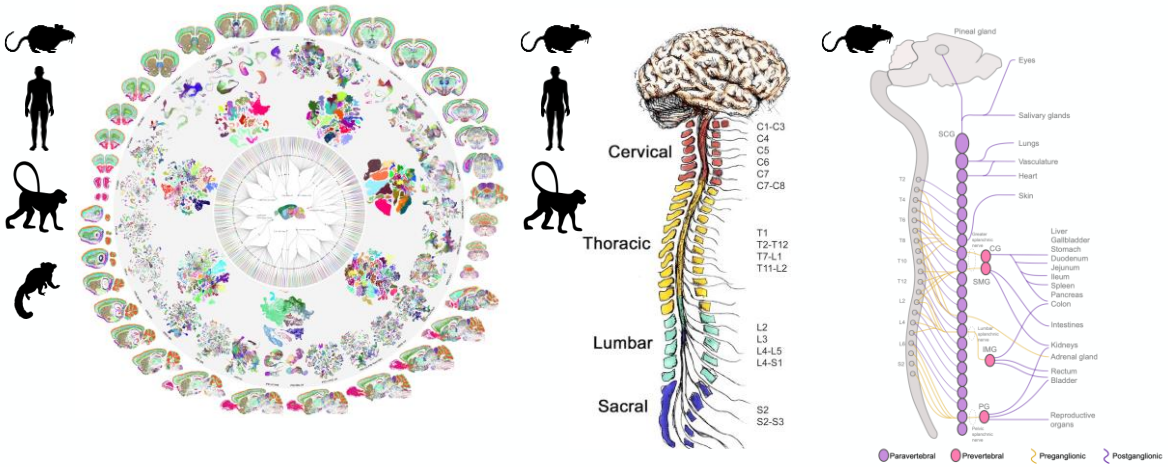
- Adult mouse **CCF v4**
- Whole-brain, **human-NHP-mouse consensus** cell types

More Capabilities

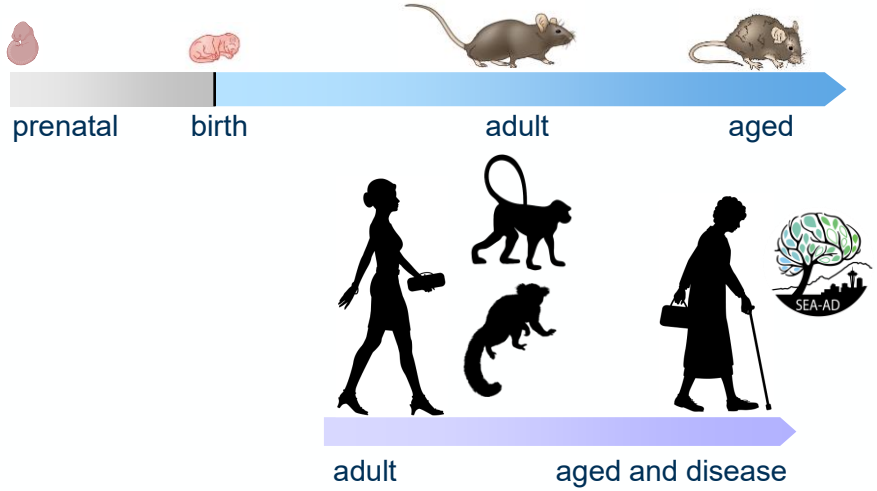
- **Visualization and discovery**: multi-modality heatmaps, differential search
- **Expanded MapMyCells** for multiple modalities
- Scalable **processing** pipelines
- Improved **programmatic** tools
- Analyze and disseminate ***your data*** with BKP

in-process foundational molecular cell type atlases/

Across the brain, spinal cord and peripheral nervous system

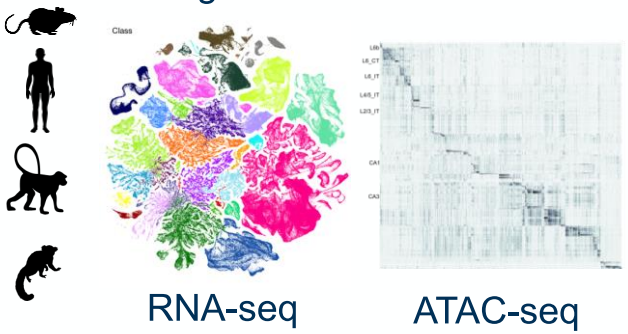


Across time and in disease



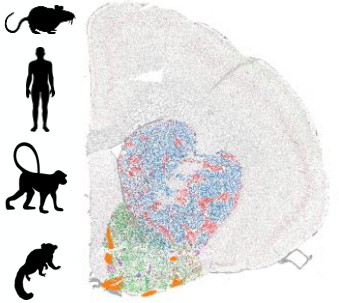
Across multiple data modalities and with spatial distributions

Single cell multiomics

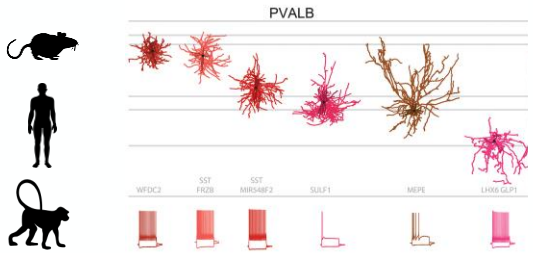


RNA-seq ATAC-seq

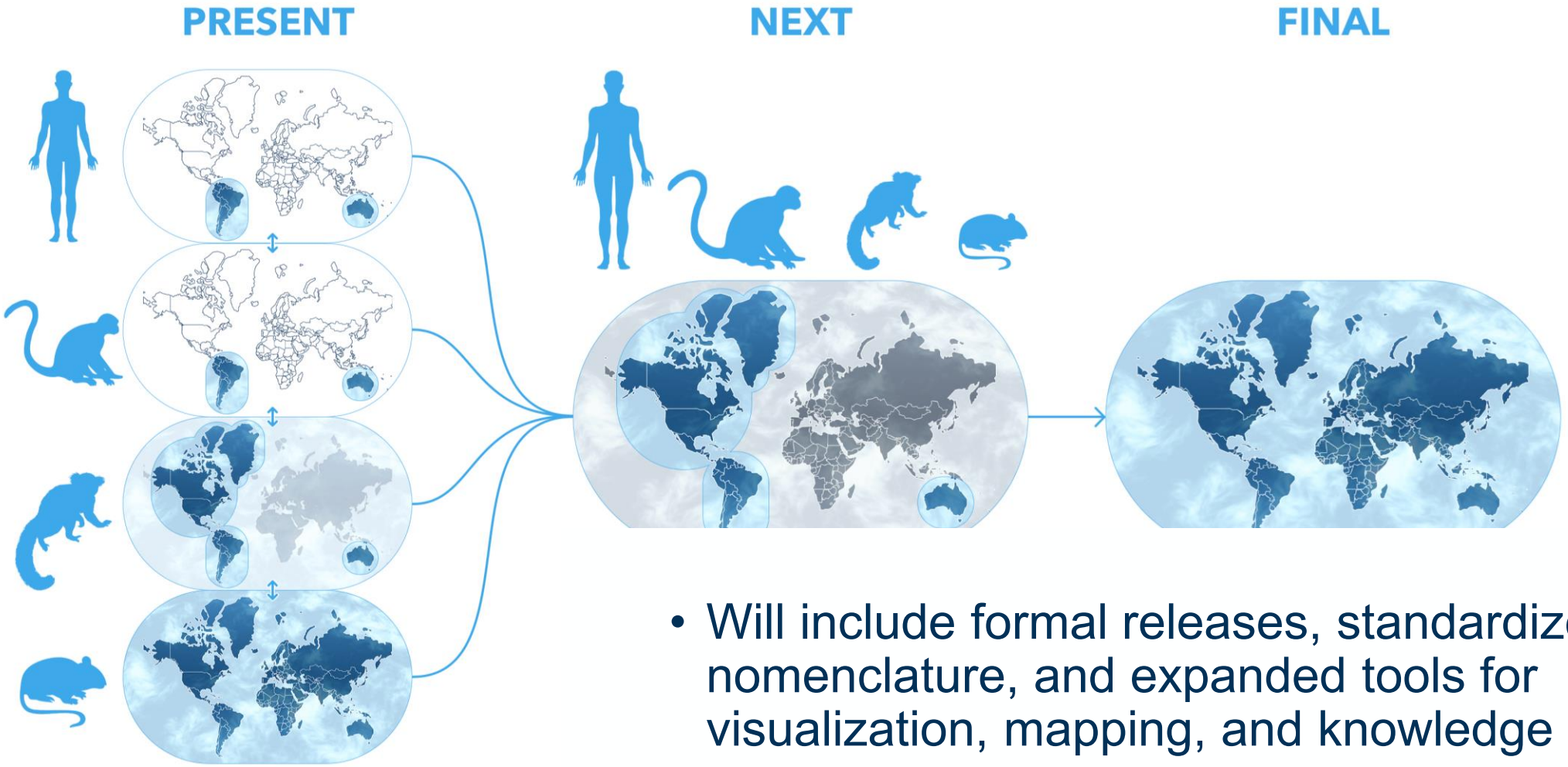
Spatial transcriptomics



Morphology and physiology

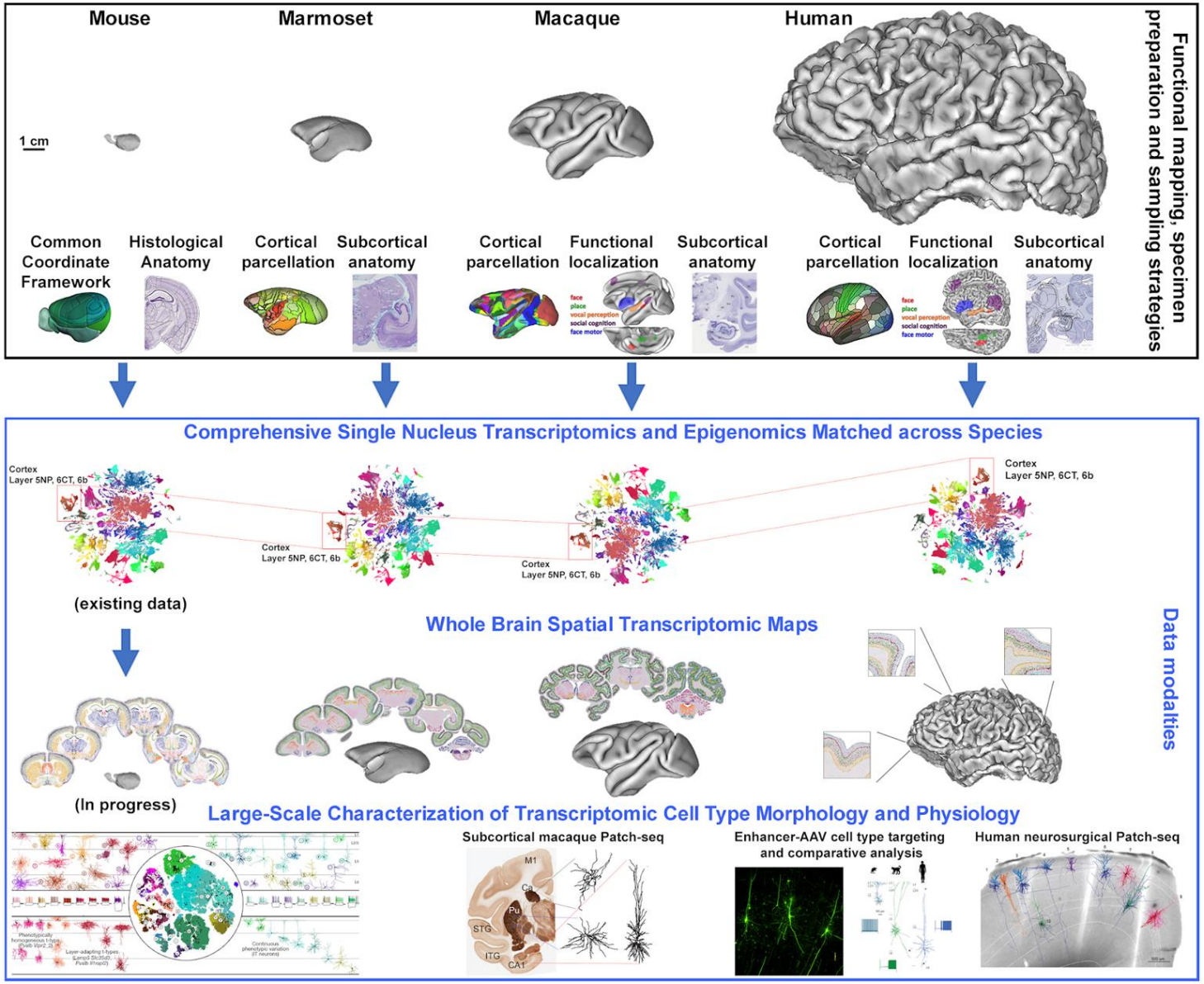


heading towards brain-wide cell atlases as foundational references similar to the human genome/



- Will include formal releases, standardized nomenclature, and expanded tools for visualization, mapping, and knowledge

the Human and Mammalian Brain Atlas project aims to get us close in the next couple of years/



Atlasing
Brain regions and coordinate spaces

Taxonomy
Cell type definition and characterization

and a new Brain Health Accelerator collaborative initiative will extend to characterizing (and treating) multiple diseases/



Broader neurodegenerative disease focus: Alzheimer's, Parkinson's, Lewy body, Huntington's, ALS

Broader multiomic profiling, on well characterized donor cohorts with available biofluids

Major focus on data modeling and AI, aiming to identify actionable **early biomarkers and therapeutic targets**

Goal: create a cell and circuit-targeted therapeutic for a clinical trial

summary/

- The Allen Institute has created multiple atlases of cell types across mammalian species in developing, adult, and aged brain
- Data, protocols, and related tools and resources are freely and publicly available for anyone to use (even you!)
- These tools can be used to address many brain research topics…
- …And we provide support!

need help after the workshop/

Scientific support

brain-map.org/support

Tutorials, videos, and other content aimed at making our open science tools usable

Educator support

alleninstitute.org/education

Support for teaching students using the Allen Brain Map

Ask a scientist

community.brain-map.org

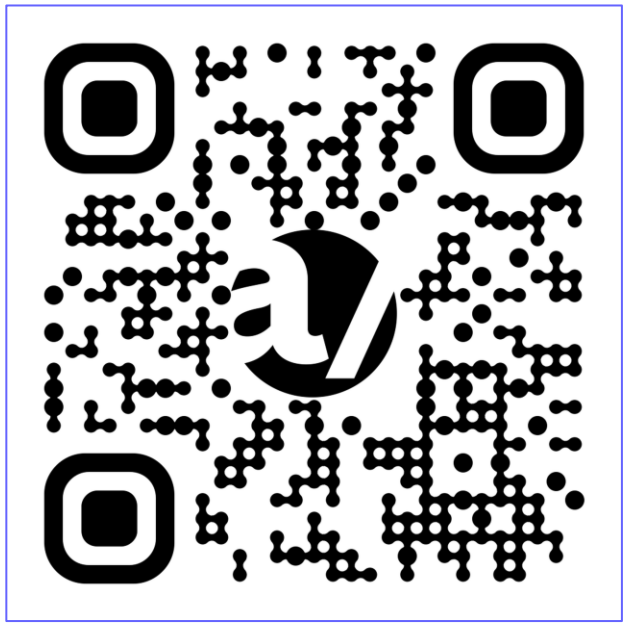
Ask a question or submit feedback in the Community Forum

rachel.hostetler@alleninstitute.org

Talk directly to an Allen Institute scientist

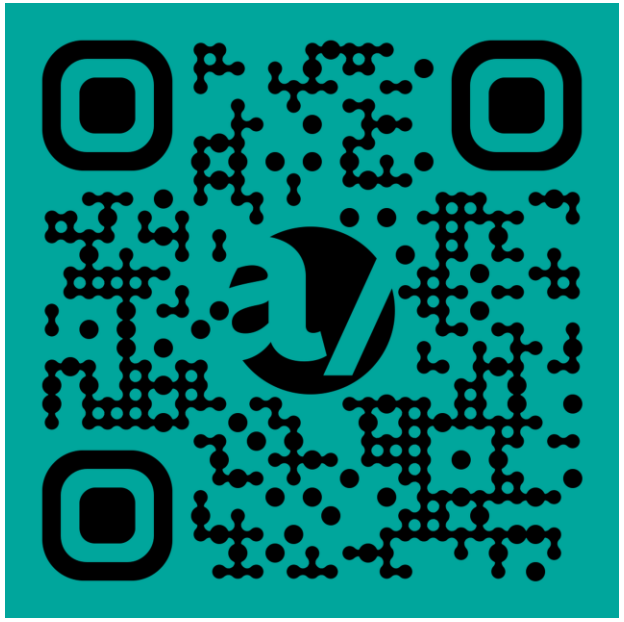
connect with us/

Follow us on LinkedIn

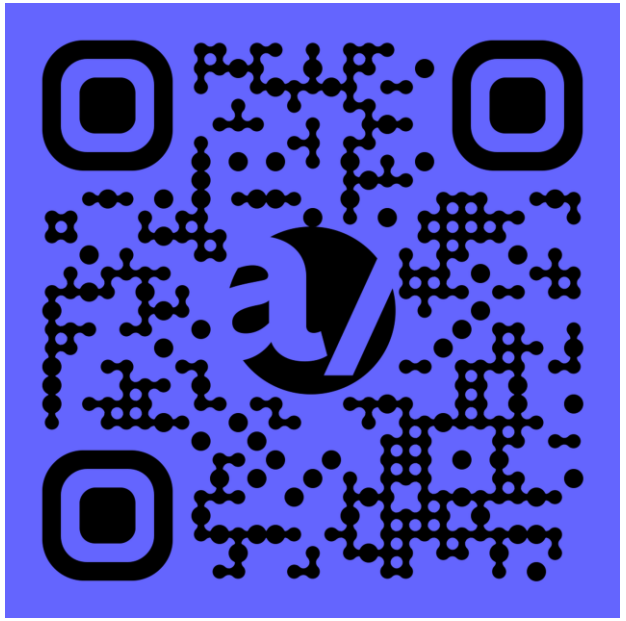


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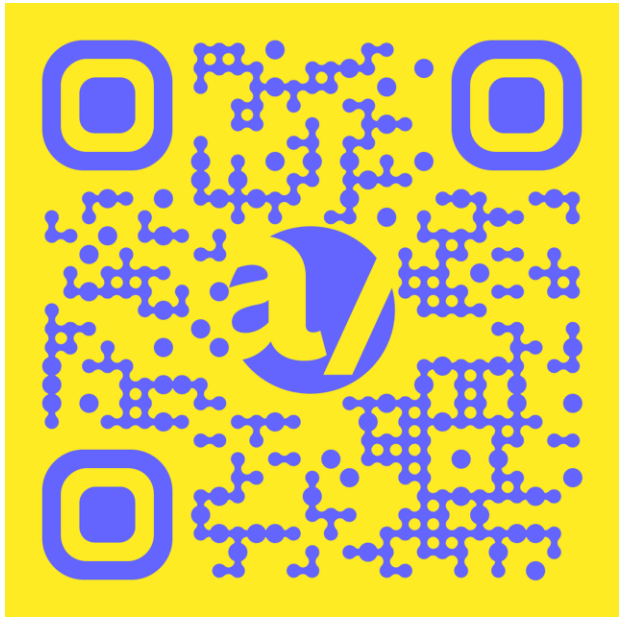
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Watch our YouTube videos



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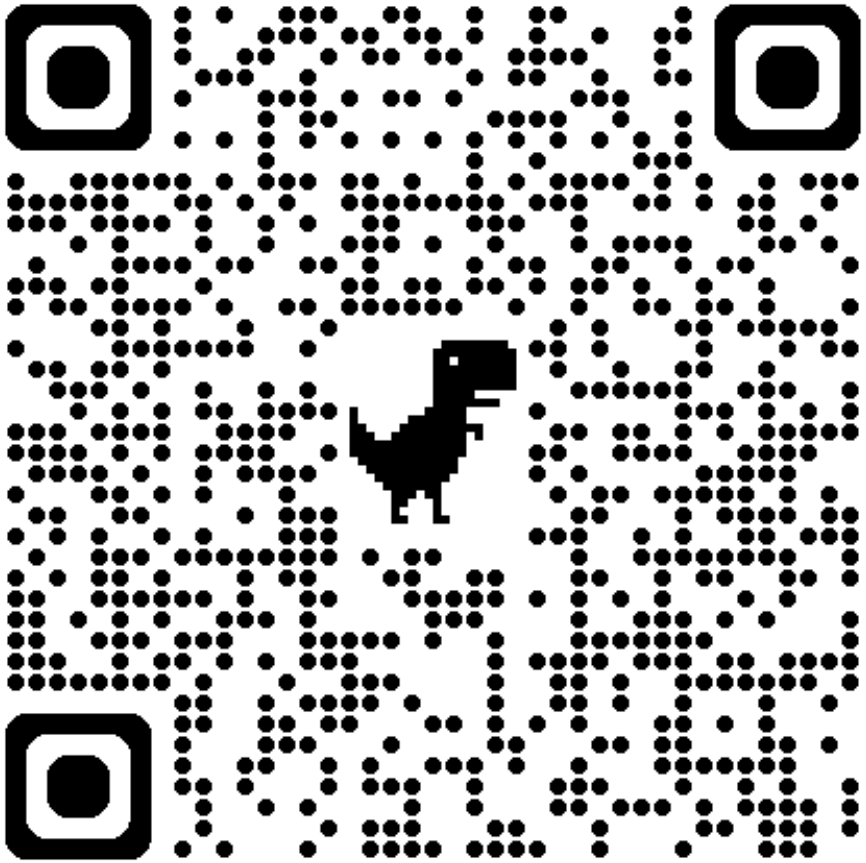
work with us/

- We hire straight out of graduate school – no post-doc required!
  - This is typically a Scientist I
  - The average “Scientist I” salary is about $80k-$100k USD
  - Each position will be clear if it supports Visa or not - it is position dependent
- Hiring is year-round, although sometimes more open spots during the start of the new year
- We often look for specific skill sets to fill open positions; the skill set depends on the position.

NOTE: The Allen Institute does not have any undergraduate internships available for international students, and as of now, we do not have any PhD-level internships either

All careers will be posted on:
alleninstitute.org/careers

Feedback survey



thankyou

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- **NINDS (1U24NS133077)**